

SHARP PLANAR QUANTITATIVE SAINT–VENANT STABILITY BY REDUCTION TO A GRAPH INSET

ABSTRACT. We prove the sharp planar quantitative Saint–Venant inequality $\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega)$, where $\mathcal{A}$ is the Fraenkel asymmetry and δ_T is the torsional (Saint–Venant) deficit, with an absolute constant C and no regularity assumption on Ω . The proof reduces a near-extremal Ω to a small-amplitude radial graph by a purely level-set construction governed by the two level deficits — the Cauchy–Schwarz/Serrin defect D_H and the isoperimetric defect D_I — and concludes with the perturbative stability of nearly spherical sets. In particular the argument uses neither the Brasco–De Philippis–Velichkov selection principle, nor mass transport, nor symmetry-type (Serrin / Magnanini–Poggesi) estimates. Two sharp ingredients drive the reduction: a fold-content bound free of the quantitative-isoperimetric square-root loss, obtained from a point-source flux identity and a thin-annulus cancellation; and the observation that excising the low-torsion boundary caps to manufacture a graph *decreases* the scale-invariant torsional deficit. The concluding nearly-spherical coercivity — the perturbative estimate that here replaces the selection principle — is proved self-containedly for graphs of small *amplitude* and *arbitrary slope*, through an exact energy identity and a slope-free annular Steklov lower bound; this is the regime the level-set reduction produces and the one in which classical (slope-controlled) nearly-spherical theorems do not apply. The reduction is carried out unconditionally except for two precisely isolated analytic points — the localisation of a harmonic boundary-layer term in the cap-cutting step, and a quadratic mode-one smallness of the graph profile at the optimal centre (replacing a barycentre normalisation) — on which the final estimate is conditional and which are stated as explicit scoped hypotheses where used; every other step is proved in full, the only external inputs being classical theorems quoted with precise references.

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1. INTRODUCTION

The Saint–Venant inequality and its quantitative form. Let $\Omega \subset \mathbb{R}^2$ be a bounded open set of finite perimeter with $|\Omega| = \pi$. The *torsion function* of Ω is the unique solution $u \in H_0^1(\Omega)$ of $-\Delta u = 1$ in Ω , $u = 0$ on $\partial\Omega$; its integral $T(\Omega) := \int_{\Omega} u$ is the *torsional rigidity*. The classical Saint–Venant conjecture, proved by Pólya and Szegő [10] via the Schwarz symmetrization of the torsion potential, asserts that the disc maximises torsional rigidity at fixed area:

$$T(\Omega) \leq T(B_1) = \frac{\pi}{8},$$

with equality if and only if Ω is a disc. A more precise, and substantially more difficult, question is to quantify stability: how large must Ω deviate from a disc when the deficit $\delta_T(\Omega) := \frac{\pi}{8} - T(\Omega)$ is small? The natural measure of deviation is the *Fraenkel asymmetry*

$$\mathcal{A}(\Omega) := \min_{x_0 \in \mathbb{R}^2} \frac{|\Omega \Delta B(x_0, r_{\Omega})|}{|\Omega|}, \quad r_{\Omega} := \sqrt{|\Omega|/\pi},$$

which is zero if and only if Ω is a disc. The sharp quantitative Saint–Venant inequality is the statement

$$\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega) \tag{1}$$

for some *absolute* constant C (independent of Ω , of $|\Omega|$, and of any regularity of $\partial\Omega$). The exponent 2 on the left and the exponent 1 on the right are both optimal: the ratio $\mathcal{A}(\Omega)^2/\delta_T(\Omega)$ is bounded and bounded away from zero for a one-parameter family of ellipses.

Talenti [12] proved a pointwise comparison between the decreasing rearrangement of u_{Ω} and the torsion function of the equimeasure disc, giving $T(\Omega) \leq T(B_1)$ and the quantitative comparison $\mu(v) \leq \mu_B(v)$ between the distribution functions. This Talenti comparison, which we state precisely as Theorem 3.8, is a key input in the present work. Talenti’s result also implies the profile identity $\int_0^{1/4} (\mu_B - \mu) dv = \delta_T$ (see Lemma 3.9), which localizes the deficit to a single integral of a non-negative integrand.

Prior work on quantitative stability. The Faber–Krahn problem for the first Dirichlet eigenvalue λ_1 is the structural analogue of the Saint–Venant problem, and the two share much of their quantitative theory. For the quantitative isoperimetric inequality the sharp form $\mathcal{A}(E) \leq C_F \delta_{\text{iso}}(E)^{1/2}$ (where $\delta_{\text{iso}}(E)$ denotes the isoperimetric deficit and C_F is an absolute constant; see Theorem 2.11) was established by Fusco, Maggi, and Pratelli [7]; this result, which we use as a black box, is the *only* external quantitative inequality the present paper imports.

For Saint–Venant/Faber–Krahn stability, Brasco and Pratelli [4] proved a quantitative stability result for both the torsional rigidity and the first eigenvalue, establishing (1) with a *non-sharp* exponent. The sharp inequality (1) — with exponent 2 and an absolute constant, valid with no regularity on $\partial\Omega$ — was established by Brasco, De Philippis, and Velichkov [3]. Their argument proceeds by a *selection principle*: from a near-extremal sequence one extracts a subsequence that, after appropriate normalisation, converges in a strong topology to a $C^{1,\gamma}$ domain; the stability of the latter is then proved by a second-variation analysis. The selection principle is a powerful and elegant tool, but it is non-constructive, non-quantitative in the convergence, and it relies on the regularity theory for minimizers of shape functionals in a fundamental way. A survey of quantitative spectral inequalities, including the state of the art up to 2017, is given by Brasco and De Philippis [2].

The main theorem. The main result of this paper is the following, proved in Section 9 as Theorem 9.2.

Theorem 1.1 (Main theorem; see Theorem 9.2). *There exists an absolute constant C such that every bounded open set $\Omega \subset \mathbb{R}^2$ with $|\Omega| = \pi$ satisfies*

$$\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega).$$

This is the same conclusion as [3], but obtained by an entirely different route. The proof requires no selection principle, no mass transport, and no symmetry-type argument (Serrin overdetermined problem / Magnanini–Poggesi estimates). The only external inputs are: classical function-space and geometric-measure-theory tools (the coarea formula, the planar isoperimetric inequality, the Talenti comparison, and the sharp quantitative isoperimetric inequality of [7]), all quoted with precise references in Section 2.

Strategy and what is new. The argument replaces the selection–regularity–second-variation architecture of [3] by a purely level-set construction driven by the two *level deficits*:

$$\begin{aligned} D_H(v) &:= \mu(v)(-\mu'(v)) - P(\{u > v\})^2 \geq 0, \\ D_I(v) &:= P(\{u > v\})^2 - 4\pi\mu(v) \geq 0, \end{aligned}$$

where $\mu(v) = |\{u > v\}|$ is the distribution function and $P(\{u > v\})$ is the perimeter of the superlevel set. The central identity (Theorem 3.4, proved in Section 3)

$$\int_0^m (D_H(v) + D_I(v)) dv = 4\pi \delta_T(\Omega)$$

encodes the entire deficit in a non-negative integral over the level parameter. All the estimates in the paper are consequences of this identity and the classical Talenti comparison.

The two sharp ingredients. Two novel observations make the reduction exact:

- (1) *Fold-content bound with no square-root loss.* At the optimal inset level $\hat{v} \sim \sqrt{\delta_T}$, the connected component $\tilde{E}$ of $\{u > \hat{v}\}$ of largest measure is trapped in a thin annulus $B_{\rho_i}(z) \subset \tilde{E} \subset B_{\rho_e}(z)$ of width $\eta = \rho_e - \rho_i \lesssim \delta_T^{1/8}$ (established in Section 5). The total angular measure of rays along which $\tilde{E}$ is *not* a radial graph is bounded, via the point-source flux identity and the cancellation $|I_{\tilde{E}} - 2\pi\hat{r}| \leq a\eta/(\rho_i\hat{r})$ (where $a = |\tilde{E} \Delta B_{\hat{r}}(z)|$), by a quantity of order $\text{Exc} + a\eta$ — without the factor of $\sqrt{\delta_{\text{iso}}}$ that the naive application of the quantitative isoperimetric inequality [7] would introduce. This is the content of Proposition 6.3 and Corollary 6.5 in Section 6.
- (2) *Cutting caps decreases the deficit.* To manufacture a genuine radial graph G from $\tilde{E}$ one must excise the “caps” (the portions of $\tilde{E}$ that cause each ray to intersect $\partial\tilde{E}$ more than once). A potential concern is that this excision increases the scale-invariant deficit $\delta_*(G) := (T(B_G) - T(G))/T(B_G)$. In fact it does not: the equal-area comparison ball shrinks faster than torsion is lost, so δ_* *decreases*. This is Lemma 7.4 in Section 7, proved via the Green-function representation of torsional rigidity and a barrier estimate in the collar.

With $\tau := \sqrt{\delta_T}$, the two observations combine to produce a continuous radial graph G with amplitude $\lesssim \delta_T^{1/8}$, scale-invariant deficit $\delta_*(G) \leq C\delta_*(\Omega)$, and symmetric difference $|\Omega \Delta G| \leq C\sqrt{\delta_T}$

(the precise reduction is Theorem 9.1 in Section 9). The perturbative stability of nearly spherical sets (Theorem 8.1 in Section 8) then gives $\mathcal{A}(G)^2 \leq C_G \delta_*(G)$, and the triangle inequality for the symmetric difference transfers this to $\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega)$.

Organization of the paper. Section 2 collects the classical prerequisites: the theory of sets of finite perimeter (De Giorgi–Federer structure theorem, coarea formula, differentiation of the distribution function), the planar isoperimetric inequality and the sharp quantitative version [7], the interior regularity and analyticity of the torsion function [8], and Sard’s theorem.

Section 3 introduces the two level deficits D_H and D_I , proves their non-negativity, establishes the central level-set deficit identity $\int_0^m (D_H + D_I) dv = 4\pi \delta_T$ (Theorem 3.4), and derives the Talenti comparison and the profile identity.

Section 4 carries out the level-set construction: it proves deficit transfer to any superlevel set, selects the good level $\hat{v}$ by an averaging argument, establishes the closeness of $\{u > \hat{v}\}$ to Ω , and isolates the inset $\tilde{E}$ as the dominant connected component.

Section 5 proves that a connected hole-free finite-perimeter set with controlled isoperimetric deficit is trapped in a thin concentric annulus, using the angular-perimeter coarea identity and a crossing-number argument.

Section 6 bounds the fold content of $\tilde{E}$ — the obstruction to being a radial graph — via the point-source flux identities and the thin-annulus cancellation, achieving the sharp estimate without the quantitative-isoperimetric square-root loss.

Section 7 proves that excising the caps to manufacture a radial graph decreases the scale-invariant torsion deficit, via a Green-function comparison and a barrier estimate.

Section 8 establishes the perturbative stability of nearly spherical sets for the torsion functional, proving that any radial graph of small amplitude satisfies $\mathcal{A}^2 \leq C_G \delta_*$, with explicit absolute constants, by a second-order expansion of the torsion deficit in Fourier modes.

Section 9 assembles all pieces and optimises the level parameter at $\tau = \sqrt{\delta_T}$. The reduction is complete except for two precisely isolated analytic points, on which the conclusion is conditional: the localisation of the harmonic boundary-layer term in the cap-cutting estimate (98) (the residual flux of the torsion function through the interior cap floors, Remark 7.2), and the quadratic mode-one smallness (154) of the graph profile at the optimal centre (Remark 9.3), which plays the role of a barycentre normalisation without recentring the set. Both are stated explicitly as scoped hypotheses where they are used; every other step — including the trapping annulus, the hole-filling, the sharp fold content, the measurable-graph realisation, and the nearly-spherical coercivity — is proved unconditionally.

Notation. Throughout the paper, $\Omega \subset \mathbb{R}^2$ is a bounded open set with $|\Omega| = \pi$. The unit disc is B_1 ; $B(x_0, r)$ denotes the open disc of radius r centred at x_0 . The symbol C denotes a generic positive absolute constant whose value may change from line to line; named constants such as C_G , η_G , A_0 , δ_* are fixed at their first occurrence. All *absolute* constants are independent of Ω and of any geometric or regularity data of $\partial\Omega$. The notation $f \lesssim g$ means $f \leq Cg$ for an absolute C .

2. PRELIMINARIES

Throughout this paper $\Omega \subset \mathbb{R}^2$ denotes a bounded open set, $|\cdot|$ denotes Lebesgue measure on $\mathbb{R}^2$, $\mathcal{H}^1$ denotes one-dimensional Hausdorff measure, and $B_r(x) \subset \mathbb{R}^2$ is the open disc of radius r centred at x ; we write $B_r = B_r(0)$. The fixed normalisation $|\Omega| = \pi$ is in force from Section 1 onward but is not required for the statements in this section unless explicitly noted. We collect here the definitions and classical results the rest of the paper relies on. Standard results are stated precisely with full hypotheses and cited to published sources.

2.1. Sets of finite perimeter. We follow the conventions of [9].

Definition 2.1 (Perimeter). Let $E \subset \mathbb{R}^2$ be Lebesgue measurable and let $A \subset \mathbb{R}^2$ be open. The *perimeter of E in A* is

$$P(E; A) := \sup \left\{ \int_E \operatorname{div} \varphi \, dx : \varphi \in C_c^1(A; \mathbb{R}^2), \|\varphi\|_{L^\infty} \leq 1 \right\}.$$

We set $P(E) := P(E; \mathbb{R}^2)$. The set E is of *finite perimeter* (in A) if $P(E; A) < \infty$.

If E has locally finite perimeter, the distributional gradient $\mu_E := -D\mathbf{1}_E$ is an $\mathbb{R}^2$ -valued Radon measure with total variation $|\mu_E|(A) = P(E; A)$ [9, Proposition 12.1, Definition 12.1].

Definition 2.2 (Reduced boundary). For E of locally finite perimeter, the *reduced boundary* $\partial^* E$ is the set of $x \in \operatorname{supp} |\mu_E|$ at which

$$\nu_E(x) := \lim_{r \rightarrow 0^+} \frac{\mu_E(B_r(x))}{|\mu_E|(B_r(x))}$$

exists and satisfies $|\nu_E(x)| = 1$. The vector $\nu_E(x)$ is the *measure-theoretic outer unit normal* to E .

Theorem 2.3 (De Giorgi structure theorem). *Let $E \subset \mathbb{R}^2$ have locally finite perimeter. Then $\partial^* E$ is countably $\mathcal{H}^1$ -rectifiable, $\mu_E = \nu_E \mathcal{H}^1 \llcorner \partial^* E$, and for every Borel set A ,*

$$P(E; A) = \mathcal{H}^1(\partial^* E \cap A).$$

Proof. See [9, Theorem 15.9, Corollary 16.1] or [5, Section 5.7, Theorem 5.15]. $\square$

Definition 2.4 (Essential boundary). For a measurable set $E \subset \mathbb{R}^2$, its *essential boundary* is $\partial^e E := \mathbb{R}^2 \setminus (E^{(0)} \cup E^{(1)})$, where $E^{(t)} := \{x : \lim_{r \rightarrow 0} |E \cap B_r(x)| / |B_r(x)| = t\}$.

Theorem 2.5 (Federer). *If $E \subset \mathbb{R}^2$ has locally finite perimeter then $\partial^* E \subseteq E^{(1/2)} \subseteq \partial^e E$ and $\mathcal{H}^1(\partial^e E \setminus \partial^* E) = 0$. In particular, $P(E; A) = \mathcal{H}^1(\partial^e E \cap A)$ for every Borel set A .*

Proof. See [9, Theorem 15.5, Theorem 16.2] or [5, Theorem 5.16]. $\square$

2.2. The coarea formula and differentiation of the distribution function.

Theorem 2.6 (Coarea formula). *Let $\Omega \subset \mathbb{R}^2$ be open and let $u \in W^{1,1}(\Omega)$. Then for every nonnegative Borel function $g : \Omega \rightarrow [0, \infty]$,*

$$\int_{\Omega} g(x) |\nabla u(x)| \, dx = \int_{-\infty}^{\infty} \left(\int_{\{u=t\}} g \, d\mathcal{H}^1 \right) dt. \quad (2)$$

Moreover, for Lebesgue-almost every $t \in \mathbb{R}$ the superlevel set $\{u > t\}$ has finite perimeter in Ω , the level set $\{u = t\}$ is $\mathcal{H}^1$ -rectifiable with $\mathcal{H}^1(\{u = t\}) < \infty$, and

$$P(\{u > t\}; \Omega) = \mathcal{H}^1(\{u = t\}) \quad \text{for a.e. } t.$$

Proof. The coarea formula (2) for $W^{1,1}$ functions is [9, Theorem 13.1]; see also [5, Section 3.4, Theorem 3.13]. The perimeter identity and rectifiability of almost every level set follow from the BV-coarea formula [1, Theorem 3.40]: since $\int_{\mathbb{R}} P(\{u > t\}; \Omega) dt = \int_{\Omega} |\nabla u| dx < \infty$, the set $\{u > t\}$ has finite perimeter for a.e. t . $\square$

The following corollary, which we call the *differentiation lemma* for the distribution function, is used repeatedly in the level-set analysis.

Lemma 2.7 (Differentiation of the distribution function). *Let $\Omega \subset \mathbb{R}^2$ be bounded and let $u \in W^{1,1}(\Omega) \cap C^1(\Omega)$ with $u \geq 0$ and*

$$|\{x \in \Omega : \nabla u(x) = 0\}| = 0.$$

Define $\mu(v) := |\{u > v\}|$ for $v \geq 0$. Then μ is non-increasing and absolutely continuous, and for Lebesgue-almost every $v > 0$,

$$-\mu'(v) = \int_{\{u=v\}} \frac{1}{|\nabla u|} d\mathcal{H}^1. \quad (3)$$

In particular, the Cauchy-Schwarz inequality applied to the pair $(1, |\nabla u|^{1/2})$ on $\{u = v\}$ gives

$$P(\{u > v\})^2 = \mathcal{H}^1(\{u = v\})^2 \leq \left(\int_{\{u=v\}} |\nabla u| d\mathcal{H}^1 \right) (-\mu'(v)) \quad \text{for a.e. } v. \quad (4)$$

Proof. The coarea formula (2) with $g = |\nabla u|^{-1} \mathbf{1}_{\{\nabla u \neq 0\}}$ gives, for $0 \leq s < t$,

$$|\{s < u \leq t\} \cap \{\nabla u \neq 0\}| = \int_s^t \left(\int_{\{u=v\} \cap \{\nabla u \neq 0\}} \frac{1}{|\nabla u|} d\mathcal{H}^1 \right) dv.$$

The critical set has zero Lebesgue measure by hypothesis, so the left side is $|\{s < u \leq t\}| = \mu(s) - \mu(t)$. By Sard's theorem (Theorem 2.17 below), $\{u = v\} \cap \{\nabla u = 0\}$ is empty for a.e. v ; for those v the inner integral may be written over the whole level $\{u = v\}$. Thus μ is the tail integral of an L^1_{loc} function, is absolutely continuous, and $-\mu'(v) = \int_{\{u=v\}} |\nabla u|^{-1} d\mathcal{H}^1$ for a.e. v .

For (4): by Cauchy-Schwarz on $\{u = v\}$,

$$\mathcal{H}^1(\{u = v\})^2 = \left(\int_{\{u=v\}} 1 d\mathcal{H}^1 \right)^2 \leq \left(\int_{\{u=v\}} |\nabla u| d\mathcal{H}^1 \right) \left(\int_{\{u=v\}} \frac{1}{|\nabla u|} d\mathcal{H}^1 \right) = \left(\int_{\{u=v\}} |\nabla u| d\mathcal{H}^1 \right) (-\mu'(v)),$$

and by Theorem 2.6, $\mathcal{H}^1(\{u = v\}) = P(\{u > v\})$ for a.e. v . $\square$

Remark 2.8. The integral $\int_{\{u=v\}} |\nabla u| d\mathcal{H}^1$ is the *flux* of the torsion function across the level set. When $-\Delta u = 1$ in $\{u > v\}$ with $u = v$ on $\partial\{u > v\}$ (for a.e. regular v), integration of $-\Delta u = 1$ over $\{u > v\}$ and Green's first identity give $\int_{\{u=v\}} |\nabla u| d\mathcal{H}^1 = |\{u > v\}| = \mu(v)$. Hence (4) becomes $P(\{u > v\})^2 \leq \mu(v)(-\mu'(v))$ for a.e. v ; the deficit from equality is $D_H(v) := \mu(v)(-\mu'(v)) - P(\{u > v\})^2 \geq 0$. This is developed in Section 3.

2.3. Isoperimetric inequalities.

Theorem 2.9 (Planar isoperimetric inequality). *For every set of finite perimeter $E \subset \mathbb{R}^2$ with $|E| < \infty$,*

$$P(E)^2 \geq 4\pi |E|, \quad (5)$$

with equality if and only if E is equivalent (up to a Lebesgue-null set) to a disc $B_r(x)$.

Proof. This is [9, Theorem 14.1]; see also [5, Section 5.6, Theorem 5.6]. $\square$

Theorem 2.10 (Relative isoperimetric inequality in a disc). *There is an absolute constant $C_{\text{rel}} > 0$ such that for every set of finite perimeter $E \subset \mathbb{R}^2$, every $x \in \mathbb{R}^2$, and every $r > 0$,*

$$\min\{|E \cap B_r(x)|, |B_r(x) \setminus E|\} \leq C_{\text{rel}} P(E; B_r(x))^2. \quad (6)$$

The constant C_{rel} is independent of E , x , and r .

Proof. By the relative isoperimetric inequality for convex domains, applied to the unit disc $B_1 \subset \mathbb{R}^2$, there is a constant $C_{\text{rel}} > 0$ such that for every set of finite perimeter E , $\min\{|E \cap B_1|, |B_1 \setminus E|\} \leq C_{\text{rel}} P(E; B_1)^2$; see [9, Theorem 12.37] (stated in $\mathbb{R}^n$, here with $n = 2$, $K = B_1$) or [5, Theorem 5.9]. The general case at radius r follows by the substitution $\tilde{E} := \{y/r : y \in E\}$: since $|\tilde{E} \cap B_1| = r^{-2}|E \cap B_r(x)|$ and $P(\tilde{E}; B_1) = r^{-1}P(E; B_r(x))$, the unit-disc inequality for $\tilde{E}$ multiplied by r^2 yields (6) with the same constant C_{rel} . $\square$

2.4. The sharp quantitative isoperimetric inequality. For a set of finite perimeter $E \subset \mathbb{R}^2$ with $0 < |E| < \infty$ we define the *Fraenkel asymmetry*

$$\mathcal{A}(E) := \min_{x_0 \in \mathbb{R}^2} \frac{|E \Delta B_{r_E}(x_0)|}{|E|}, \quad r_E := \sqrt{|E|/\pi}, \quad (7)$$

and the *isoperimetric deficit*

$$\delta_{\text{iso}}(E) := \frac{P(E)}{2\sqrt{\pi|E|}} - 1 \geq 0, \quad (8)$$

where $2\sqrt{\pi|E|}$ is the perimeter of a disc of area $|E|$. Both quantities are dimensionless, translation- and scaling-invariant, and nonnegative, vanishing precisely when E is a disc.

Theorem 2.11 (Fusco–Maggi–Pratelli). *There is an absolute constant $C_F > 0$ such that for every set of finite perimeter $E \subset \mathbb{R}^2$ with $0 < |E| < \infty$,*

$$\mathcal{A}(E) \leq C_F \delta_{\text{iso}}(E)^{1/2}. \quad (9)$$

The exponent $\frac{1}{2}$ is optimal: it cannot be replaced by any strictly larger power, as shown by elongated ellipses with small eccentricity. The value of C_F is not explicit in [7]; it is finite and absolute (depending on neither E nor any geometric parameter of Ω).

Proof. This is Theorem 1.1 of [7] (stated there in $\mathbb{R}^n$; the planar case $n = 2$ is the specialisation $C_F := C(2)$). The theorem is proved in [7] using symmetrisation and a selection-principle argument; we use it as a black box. $\square$

Remark 2.12 (Relation between δ_{iso} and the additive deficit). The additive isoperimetric deficit $D_I(E) := P(E)^2 - 4\pi|E| \geq 0$ (see Definition 3.2) satisfies

$$\delta_{\text{iso}}(E) = \sqrt{1 + \frac{D_I(E)}{4\pi|E|}} - 1 = \frac{D_I(E)/(4\pi|E|)}{\sqrt{1 + D_I(E)/(4\pi|E|)} + 1} \leq \frac{D_I(E)}{8\pi|E|},$$

since $\sqrt{1+s} + 1 \geq 2$ for $s \geq 0$. Hence $\delta_{\text{iso}}(E)^{1/2} \leq (D_I(E)/(8\pi|E|))^{1/2}$, and (9) gives

$$\mathcal{A}(E)^2 \leq C_F^2 \delta_{\text{iso}}(E) \leq \frac{C_F^2}{8\pi} \frac{D_I(E)}{|E|}.$$

We will apply this estimate in Section 9 to the inset $\tilde{E}$.

2.5. Interior regularity of the torsion function. Let $u \in H_0^1(\Omega)$ be the unique weak solution of the torsion problem

$$-\Delta u = 1 \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega, \quad (10)$$

i.e., $\int_{\Omega} \nabla u \cdot \nabla \varphi \, dx = \int_{\Omega} \varphi \, dx$ for all $\varphi \in H_0^1(\Omega)$. Existence and uniqueness follow from the Lax–Milgram theorem; $u \geq 0$ by the weak maximum principle [8, Theorem 8.1].

Theorem 2.13 (Interior smoothness and analyticity of u). *Let $\Omega \subset \mathbb{R}^2$ be a bounded open set and let $u \in H_0^1(\Omega)$ be the weak solution of (10). Then $u \in C^\infty(\Omega)$. Moreover, u is real-analytic in Ω .*

Proof. Since the right-hand side of (10) is $f \equiv 1 \in C^\infty(\mathbb{R}^2)$ and the operator $-\Delta$ has smooth (constant) coefficients, interior Schauder estimates bootstrap to give $u \in C_{\text{loc}}^{k,\alpha}(\Omega)$ for every $k \geq 0$ and every $\alpha \in (0, 1)$; see [8, Theorem 6.17, Corollary 6.18]. Hence $u \in C^\infty(\Omega)$.

Real-analyticity follows from the general fact that solutions of linear elliptic equations with real-analytic coefficients and real-analytic right-hand side are real-analytic in the interior: the Laplacian has constant (hence analytic) coefficients and $f \equiv 1$ is analytic, so u is real-analytic in Ω ; this is a classical result of Morrey–Nirenberg, discussed in [8, Chapter 6, §6.8 and the historical notes]. $\square$

Theorem 2.14 (Interior gradient and Hessian estimates). *There is an absolute constant $C > 0$ with the following property. Let $\Omega \subset \mathbb{R}^2$ be a bounded open set, let u satisfy (10) weakly, let $x \in \Omega$, and set $d := \text{dist}(x, \partial\Omega)$. Then*

$$|\nabla u(x)| \leq C \left(\frac{\|u\|_{L^\infty(B_d(x))}}{d} + d \right), \quad |D^2 u(x)| \leq C \left(\frac{\|u\|_{L^\infty(B_d(x))}}{d^2} + 1 \right). \quad (11)$$

Proof. Apply the interior Schauder estimate of [8, Theorem 6.2] to u on $B_d(x)$ where $-\Delta u = 1$. The scale-invariant form of the estimate gives

$$d \|\nabla u\|_{L^\infty(B_{d/2}(x))} + d^2 \|D^2 u\|_{L^\infty(B_{d/2}(x))} \leq C (\|u\|_{L^\infty(B_d(x))} + d^2),$$

with C depending only on the dimension $n = 2$. Dividing by d and d^2 respectively yields (11) (the sup over $B_{d/2}(x)$ bounds the pointwise value at x). $\square$

Lemma 2.15 (Critical set of the torsion function has zero area). *Let $\Omega \subset \mathbb{R}^2$ be a bounded open set and let u solve (10). Then*

$$|\{x \in \Omega : \nabla u(x) = 0\}| = 0.$$

Consequently the torsion function satisfies the hypothesis of Lemma 2.7.

Proof. By Theorem 2.13, u is real-analytic in Ω . The open set Ω has at most countably many connected components. On any component U , the two first derivatives of u cannot both vanish identically: if they did, u would be constant on U , contradicting $-\Delta u = 1$ there. Hence for each component U at least one derivative, say $\partial_j u$, is a non-trivial real-analytic function on U , and

$$\{\nabla u = 0\} \cap U \subset \{\partial_j u = 0\}.$$

The zero set of a non-trivial real-analytic function on a connected open subset of $\mathbb{R}^2$ has Lebesgue measure zero. Indeed, if such a zero set had positive measure, a density point and the Taylor expansion at that point would force all derivatives to vanish there, and the analytic identity theorem would force the function to vanish identically. Thus $|\{\nabla u = 0\} \cap U| = 0$ on every component U , and the countable union over components has measure zero. $\square$

Lemma 2.16 (L^∞ bound). *Let $\Omega \subset \mathbb{R}^2$ be bounded and let u solve (10). Then $0 \leq u(x) \leq \frac{1}{4}(\text{diam } \Omega)^2$ for all $x \in \Omega$. In particular $m := \sup_\Omega u \leq \frac{1}{4}(\text{diam } \Omega)^2 < \infty$ (the supremum is taken over the open set Ω ; no continuity of u up to $\partial\Omega$ is assumed or asserted).*

Proof. The lower bound $u \geq 0$ follows from the weak maximum principle [8, Theorem 8.1]. For the upper bound, let $R = \text{diam } \Omega$ and pick any $x_0 \in \Omega$; then $\Omega \subset B_R(x_0)$. The function $w(x) := (R^2 - |x - x_0|^2)/4$ satisfies $-\Delta w = 1$ in $\mathbb{R}^2$ and $w \geq 0$ on Ω . Hence $h := w - u \in H^1(\Omega)$ satisfies $-\Delta h = 0$ weakly in Ω and $h = w - u \geq -u$, so $h^- := \max\{-h, 0\} = (u - w)^+ \leq u^+ = u$ has zero boundary trace, $(u - w)^+ \in H_0^1(\Omega)$ (as $u \in H_0^1$ and $w \geq 0$). By the weak maximum principle [8, Theorem 8.1] applied to the harmonic h , $h \geq 0$ a.e. in Ω , i.e. $u \leq w \leq R^2/4$ in Ω . No continuity of u up to $\partial\Omega$ is used. $\square$

2.6. Sard's theorem and regular values of the torsion function.

Theorem 2.17 (Sard). *Let $U \subseteq \mathbb{R}^d$ be open and $f \in C^k(U; \mathbb{R}^m)$ with $k \geq \max\{1, d - m + 1\}$. Then the set of critical values of f ,*

$$f(\{x \in U : \text{rank } Df(x) < m\}),$$

has Lebesgue measure zero in $\mathbb{R}^m$. In particular, if $f \in C^\infty(U)$ and $m = 1$, the set $f(\{\nabla f = 0\}) \subset \mathbb{R}$ is a Lebesgue-null set.

Proof. See [9, Theorem A.6] for the statement in the context used here; the original source is [11]. $\square$

Lemma 2.18 (Almost every level of u is regular). *Let $\Omega \subset \mathbb{R}^2$ be a bounded open set and let u solve (10). Then for Lebesgue-almost every $v \in (0, m)$ the level v is a regular value of u in Ω : that is, $\nabla u(x) \neq 0$ at every point $x \in \{u = v\} \cap \Omega$. For every such v ,*

- (i) *the level set $\{u = v\} \cap \Omega$ is a smooth, properly embedded 1-dimensional submanifold of Ω (a disjoint union of smooth curves); if moreover $u \in C(\bar{\Omega})$ with $u = 0$ on $\partial\Omega$, it is a finite union of smooth closed curves, compactly contained in Ω ;*
- (ii) *the superlevel set $\{u > v\}$ is a finite-perimeter set with $P(\{u > v\}) = \mathcal{H}^1(\{u = v\})$;*
- (iii) *the differentiation formula (3) holds at v , with $1/|\nabla u|$ finite and smooth on $\{u = v\}$.*

Proof. By Theorem 2.13, $u \in C^\infty(\Omega)$ (indeed real-analytic). Applying Theorem 2.17 with $d = 2$, $m = 1$: the set of critical values $u(\{\nabla u = 0\} \cap \Omega) \subset \mathbb{R}$ has Lebesgue measure zero. Hence for a.e. v , every point of $\{u = v\} \cap \Omega$ is a regular point. For such v :

(i): At a regular value v the implicit function theorem gives $\{u = v\} \cap \Omega$ as a smooth, properly embedded 1-dimensional submanifold of the open set Ω (without boundary in Ω), i.e. a disjoint union of smooth curves, each either a closed loop in Ω or a properly embedded line escaping every compact subset of Ω . This is the honest *interior* statement and uses no boundary regularity. If, in addition, $u \in C(\overline{\Omega})$ with $u = 0$ on $\partial\Omega$ (which holds whenever $\partial\Omega$ is, say, continuous), then for $v > 0$ the level set $\{u = v\}$ is compactly contained in Ω and so is a *finite* disjoint union of smooth closed curves. The two genuinely interior consequences (ii) and (iii) below, proved from the coarea formula alone, hold with no boundary regularity whatsoever.

(ii): By Theorem 2.6, $P(\{u > v\}) = \mathcal{H}^1(\{u = v\})$ for a.e. v .

(iii): For a.e. v , (3) holds and $|\nabla u| > 0$ on $\{u = v\}$, so $1/|\nabla u|$ is finite and smooth there by Theorem 2.13. $\square$

3. THE LEVEL DEFICITS AND THE VOLUME PROFILE

This section is the analytic heart of the paper. We package the Saint-Venant deficit

$$\delta_T = \delta_T(\Omega) = \frac{\pi}{8} - T(\Omega) \geq 0$$

of a normalised domain $|\Omega| = \pi$ into the level sets of its torsion function u . Recall (Section 2) that $u \in H_0^1(\Omega) \cap C^\infty(\Omega)$ solves $-\Delta u = 1$ in Ω , $u = 0$ on $\partial\Omega$, that

$$T(\Omega) := \int_{\Omega} u \, dx = \int_{\Omega} |\nabla u|^2 \, dx, \quad m := \max_{\Omega} u,$$

the second equality holding because testing $-\Delta u = 1$ against $u \in H_0^1(\Omega)$ gives $\int_{\Omega} |\nabla u|^2 = \int_{\Omega} u$. We define two non-negative defects of each level set; their integral over all levels turns out to be exactly $4\pi\delta_T$. We then use Talenti's comparison to pin the volume profile $\mu(v) := |\{u > v\}|$, obtaining $m \leq \frac{1}{4}$, an exact profile identity, and the pointwise bound $-\mu' \geq 4\pi$. These facts feed directly into the inset construction of Section 4.

Throughout, a value $v \in (0, m)$ is called *regular* if $\nabla u \neq 0$ on $\{u = v\}$; by Lemma 2.18 almost every $v \in (0, m)$ is regular, and for such v the level set $\{u = v\}$ is a smooth one-dimensional submanifold of Ω and the coarea quotient $1/|\nabla u|$ is finite and smooth on it. We write

$$E_v := \{x \in \Omega : u(x) > v\}, \quad \mu(v) := |E_v|, \quad P(v) := P(E_v) = \mathcal{H}^1(\{u = v\})$$

for the super-level set, its area, and its perimeter (the last equality holding for regular v , the level set being a smooth curve).

3.1. The flux and coarea identities. The two structural identities below are the only place where the equation $-\Delta u = 1$ enters this section; everything afterwards is geometry of the level sets. The first (flux) identity quantifies the source; the second (coarea) is the differentiation formula for μ . We prove both.

Lemma 3.1 (Flux and coarea identities). *For a.e. $v \in (0, m)$,*

$$\int_{\{u=v\}} |\nabla u| \, d\mathcal{H}^1 = \mu(v), \tag{12}$$

and for a.e. $v \in (0, m)$ the function μ is differentiable with

$$-\mu'(v) = \int_{\{u=v\}} \frac{1}{|\nabla u|} d\mathcal{H}^1. \quad (13)$$

Moreover μ is absolutely continuous and nonincreasing on $(0, m)$, with $\mu(0^+) = |\Omega| = \pi$ and $\mu(m^-) = 0$.

Proof. Flux (12). We derive this directly from the weak formulation, so that no boundary regularity of Ω is needed. Recall $u \in H_0^1(\Omega)$ solves

$$\int_{\Omega} \nabla u \cdot \nabla \varphi dx = \int_{\Omega} \varphi dx \quad \text{for all } \varphi \in H_0^1(\Omega). \quad (14)$$

Fix $0 < v < m$ and, for small $h > 0$ with $v + h < m$, take the Lipschitz cut-off

$$\psi_h(x) := \min\left\{1, \frac{1}{h}(u(x) - v)_+\right\} \in [0, 1].$$

Since $t \mapsto \min\{1, \frac{1}{h}(t - v)_+\}$ is Lipschitz and vanishes for $t \leq v$ (in particular at $t = 0$), and $u \in H_0^1(\Omega)$, the composition ψ_h lies in $H_0^1(\Omega)$ and is an admissible test function in (14), with

$$\nabla \psi_h = \frac{1}{h} \nabla u \mathbf{1}_{\{v < u < v+h\}} \quad \text{a.e.}$$

Inserting $\varphi = \psi_h$ into (14),

$$\frac{1}{h} \int_{\{v < u < v+h\}} |\nabla u|^2 dx = \int_{\Omega} \nabla u \cdot \nabla \psi_h dx = \int_{\Omega} \psi_h dx.$$

We pass to the limit $h \downarrow 0$ on each side. On the left, the coarea formula (Theorem 2.6) gives

$$\frac{1}{h} \int_{\{v < u < v+h\}} |\nabla u|^2 dx = \frac{1}{h} \int_v^{v+h} \left(\int_{\{u=t\}} |\nabla u| d\mathcal{H}^1 \right) dt \xrightarrow{h \downarrow 0} \int_{\{u=v\}} |\nabla u| d\mathcal{H}^1$$

for a.e. v , namely at every Lebesgue point of the $L^1(0, m)$ function $t \mapsto F(t) := \int_{\{u=t\}} |\nabla u| d\mathcal{H}^1$ (that $F \in L^1$ follows from $\int_0^m F dt = \int_{\{0 < u < m\}} |\nabla u|^2 dx \leq \int_{\Omega} |\nabla u|^2 < \infty$ by coarea). On the right, $0 \leq \psi_h \leq 1$ and $\psi_h \uparrow \mathbf{1}_{\{u > v\}}$ pointwise as $h \downarrow 0$, so by monotone (or dominated, since $|\Omega| < \infty$) convergence

$$\int_{\Omega} \psi_h dx \xrightarrow{h \downarrow 0} \int_{\Omega} \mathbf{1}_{\{u > v\}} dx = |\{u > v\}| = \mu(v).$$

Equating the two limits yields (12) for a.e. $v \in (0, m)$.

Coarea (13) and absolute continuity. This is the differentiation of the distribution function, Lemma 2.7: for a.e. v , μ is differentiable with $-\mu'(v) = \int_{\{u=v\}} |\nabla u|^{-1} d\mathcal{H}^1$. We record the absolute-continuity statement explicitly, as it is used repeatedly. Set $\Phi(v) := \int_{\{u=v\}} |\nabla u|^{-1} d\mathcal{H}^1 \geq 0$ for those v where this is finite. The coarea formula (Theorem 2.6) with $g = \mathbf{1}_{\{u > 0\}} |\nabla u|^{-1} \mathbf{1}_{\{\nabla u \neq 0\}}$ gives

$$\int_0^m \Phi(\tau) d\tau = \int_{\{0 < u < m\} \cap \{\nabla u \neq 0\}} 1 dx = |\Omega| = \pi < \infty,$$

where the middle equality uses Lemma 2.15. Hence $\Phi \in L^1(0, m)$. For $0 < v_1 < v_2 < m$ the coarea formula applied with $g = \mathbf{1}_{\{v_1 < u \leq v_2\}} |\nabla u|^{-1} \mathbf{1}_{\{\nabla u \neq 0\}}$ yields

$$\mu(v_1) - \mu(v_2) = |\{v_1 < u \leq v_2\}| = \int_{v_1}^{v_2} \Phi(\tau) d\tau.$$

Thus $\mu(v) = \int_v^m \Phi(\tau) d\tau$ for $v \in (0, m)$; being the tail integral of an L^1 function, μ is absolutely continuous and nonincreasing on $(0, m)$, with $\mu' = -\Phi$ a.e., which is (13). Finally $\mu(0^+) = \int_0^m \Phi = |\Omega| = \pi$ (equivalently $|\{u > 0\}| = |\Omega|$ since $u > 0$ in Ω by the strong minimum principle), and $\mu(m^-) = \int_m^m \Phi = 0$. $\square$

3.2. The two level deficits.

Definition 3.2 (The two level deficits). For a regular value $v \in (0, m)$ we define the *Cauchy–Schwarz* (or Hölder/Serrin) defect and the *isoperimetric* defect of the level set $\{u = v\}$ by

$$D_H(v) := \mu(v) (-\mu'(v)) - P(v)^2, \quad D_I(v) := P(v)^2 - 4\pi \mu(v).$$

The names anticipate Lemma 3.3: D_H vanishes precisely when the flux is equidistributed (constant $|\nabla u|$ on the level curve), and D_I vanishes precisely when the level set is a disk.

Lemma 3.3 (Non-negativity of the deficits). *For a.e. $v \in (0, m)$ one has*

$$D_H(v) \geq 0, \quad D_I(v) \geq 0.$$

Moreover $D_I(v) = 0$ if and only if E_v is (equivalent to) a disk, and, for regular v , $D_H(v) = 0$ if and only if $|\nabla u|$ is constant $\mathcal{H}^1$ -almost everywhere on $\{u = v\}$.

Proof. $D_I \geq 0$. This is the planar isoperimetric inequality (Theorem 2.9) applied to the finite-perimeter set E_v : $P(E_v)^2 \geq 4\pi|E_v|$, i.e. $P(v)^2 \geq 4\pi\mu(v)$, with equality if and only if E_v is a disk up to a Lebesgue-null modification.

$D_H \geq 0$. Fix a regular value v at which (12) and (13) hold (a.e. v , by Lemmas 2.18 and 3.1); then $\Sigma = \{u = v\}$ is a smooth embedded curve on which $|\nabla u| > 0$ and $P(v) = \mathcal{H}^1(\Sigma) < \infty$. Apply the Cauchy–Schwarz inequality on the finite measure $\mathcal{H}^1 \llcorner \Sigma$ to the functions $|\nabla u|^{1/2}$ and $|\nabla u|^{-1/2}$:

$$P(v)^2 = \left(\int_{\Sigma} 1 d\mathcal{H}^1 \right)^2 = \left(\int_{\Sigma} |\nabla u|^{1/2} \cdot |\nabla u|^{-1/2} d\mathcal{H}^1 \right)^2 \leq \left(\int_{\Sigma} |\nabla u| d\mathcal{H}^1 \right) \left(\int_{\Sigma} \frac{1}{|\nabla u|} d\mathcal{H}^1 \right).$$

By (12) the first factor is $\mu(v)$, and by (13) the second factor is $-\mu'(v)$. Hence $P(v)^2 \leq \mu(v)(-\mu'(v))$, i.e. $D_H(v) \geq 0$. Equality in Cauchy–Schwarz holds if and only if $|\nabla u|^{1/2}$ and $|\nabla u|^{-1/2}$ are proportional $\mathcal{H}^1$ -a.e. on Σ , i.e. if and only if $|\nabla u|$ is $\mathcal{H}^1$ -a.e. constant on $\{u = v\}$. $\square$

3.3. The level-set deficit identity. We now prove the central identity and, as the mandate requires, derive the constant 4π independently of any prior computation, directly from coarea and integration by parts (the Saint–Venant route).

Theorem 3.4 (Level-set deficit identity). *With the notation above,*

$$\int_0^m (D_H(v) + D_I(v)) dv = 4\pi \delta_T(\Omega). \quad (15)$$

Proof. The proof has three steps: a pointwise telescoping, two integrations, and the assembly with the explicit value of the constant.

Step 1: telescoping. For a.e. $v \in (0, m)$ the squared-perimeter terms in Definition 3.2 cancel:

$$D_H(v) + D_I(v) = \left(\mu(v)(-\mu'(v)) - P(v)^2 \right) + \left(P(v)^2 - 4\pi\mu(v) \right) = \mu(v)(-\mu'(v)) - 4\pi\mu(v). \quad (16)$$

The perimeter has disappeared entirely; what remains involves only μ and μ' .

Step 2: the first term, by integration by parts. By Lemma 3.1, μ is absolutely continuous on $(0, m)$ with $\mu' \in L^1$, $\mu(0^+) = \pi$ and $\mu(m^-) = 0$. Consequently $v \mapsto \frac{1}{2}\mu(v)^2$ is absolutely continuous with derivative $\mu(v)\mu'(v) \in L^1(0, m)$ (a product of a bounded absolutely continuous function and an L^1 function, with the Leibniz rule valid because μ is bounded and absolutely continuous). Hence

$$\int_0^m \mu(v)(-\mu'(v)) dv = -\left[\frac{1}{2}\mu(v)^2\right]_{v=0^+}^{v=m^-} = \frac{1}{2}\mu(0^+)^2 - \frac{1}{2}\mu(m^-)^2 = \frac{1}{2}|\Omega|^2 = \frac{\pi^2}{2}. \quad (17)$$

Step 3: the second term, by the layer-cake formula. Since $u > 0$ in Ω and $\max u = m$,

$$\int_0^m 4\pi \mu(v) dv = 4\pi \int_0^\infty |\{u > v\}| dv = 4\pi \int_\Omega u dx = 4\pi T(\Omega), \quad (18)$$

the middle equality being Cavalieri's (layer-cake) formula for the nonnegative function u , valid since $\mu(v) = 0$ for $v \geq m$.

Assembly and the constant. Integrating (16) over $(0, m)$ and inserting (17)–(18),

$$\int_0^m (D_H + D_I) dv = \frac{\pi^2}{2} - 4\pi T(\Omega).$$

It remains to identify $\frac{\pi^2}{2} - 4\pi T(\Omega)$ with $4\pi\delta_T$. We do *not* invoke any earlier formula for $T(B_1)$; instead we read off the constant from Step 2. Indeed

$$\frac{\pi^2}{2} = \frac{|\Omega|^2}{2} = 4\pi \cdot \frac{|\Omega|^2}{8\pi} = 4\pi \cdot \frac{\pi^2}{8\pi} = 4\pi \cdot \frac{\pi}{8},$$

so that

$$\int_0^m (D_H + D_I) dv = 4\pi \left(\frac{\pi}{8} - T(\Omega) \right) = 4\pi \delta_T(\Omega),$$

by the definition $\delta_T = \frac{\pi}{8} - T(\Omega)$. This proves (15).

We pause to make the provenance of the constant transparent, since it is the keystone of the paper. The factor 4π in D_I is the planar isoperimetric constant ($P^2 \geq 4\pi A$). The factor $\frac{1}{2}|\Omega|^2$ in Step 2 is purely the integration of $\mu(-\mu')$, owing nothing to the equation beyond the boundary values $\mu(0^+) = |\Omega|$, $\mu(m^-) = 0$. These two independent inputs combine into the deficit because of the elementary algebraic identity $\frac{1}{2}A^2 = 4\pi \cdot A^2/(8\pi)$ for any area A ; with $A = |\Omega| = \pi$ this reads $\frac{1}{2}|\Omega|^2 = 4\pi \cdot \frac{\pi}{8}$, which is the algebra displayed above. The number $\frac{\pi}{8}$ so produced is exactly the disk torsion $T(B_1)$ (computed independently in Lemma 3.9 below), so the right-hand side is $4\pi(T(B_1) - T(\Omega))$. Thus the right-hand side is $4\pi(T(B_1) - T(\Omega))$, the Saint-Venant deficit scaled by 4π , with no hidden normalisation. $\square$

Remark 3.5 (The identity as an exact, scale-aware statement). Identity (15) expresses the (global) Saint-Venant deficit as the total over all levels of two pointwise non-negative geometric defects of the level sets: D_H , the failure of $|\nabla u|$ to be constant on $\{u = v\}$; and D_I , the failure of E_v to be a disk. Both vanish identically (for all v) if and only if every level set is a disk on which $|\nabla u|$ is constant, i.e. if and only if Ω is a disk and $u = u_B$; this is Serrin's overdetermined characterisation, and it is the rigidity behind (15). The quantitative content is the converse, exploited from Section 4 on: small δ_T forces both defects to be small *on most levels*.

3.4. Scale invariance of the identity. For the inset of Section 4 we will apply the identity not to Ω but to its super-level sets, which are not area-normalised. We therefore record the unnormalised form

of (15) and its exact homogeneity. Let $E \subset \mathbb{R}^2$ be any bounded open set of finite measure which is regular for the Dirichlet problem, with torsion function w_E ($-\Delta w_E = 1$ in E , $w_E \in H_0^1(E)$), torsion $T(E) = \int_E w_E = \int_E |\nabla w_E|^2$, maximum $m_E = \max w_E$, and level quantities $\mu_E(v) = |\{w_E > v\}|$, $P_E(v) = P(\{w_E > v\})$,

$$D_H^E(v) := \mu_E(v)(-\mu'_E(v)) - P_E(v)^2, \quad D_I^E(v) := P_E(v)^2 - 4\pi\mu_E(v).$$

Proposition 3.6 (Unnormalised identity and degree-4 homogeneity). *For any such E ,*

$$\int_0^{m_E} (D_H^E(v) + D_I^E(v)) dv = \frac{1}{2}|E|^2 - 4\pi T(E) = 4\pi(T(B_E) - T(E)), \quad T(B_E) := \frac{|E|^2}{8\pi}, \quad (19)$$

where B_E is any disk with $|B_E| = |E|$. Both sides are homogeneous of degree 4 under dilations: writing $sE = \{sx : x \in E\}$ for $s > 0$, one has

$$\int_0^{m_{sE}} (D_H^{sE} + D_I^{sE}) dv = s^4 \int_0^{m_E} (D_H^E + D_I^E) dv, \quad T(B_{sE}) - T(sE) = s^4(T(B_E) - T(E)). \quad (20)$$

Consequently the scale-invariant deficit $\delta_*(E) := (T(B_E) - T(E))/T(B_E) \in [0, 1]$ satisfies

$$\delta_*(E) = \frac{T(B_E) - T(E)}{T(B_E)} = \frac{\frac{1}{4\pi} \int_0^{m_E} (D_H^E + D_I^E) dv}{|E|^2/(8\pi)} = \frac{2}{|E|^2} \int_0^{m_E} (D_H^E + D_I^E) dv, \quad (21)$$

and $\delta_*(sE) = \delta_*(E)$ for all $s > 0$.

Proof. Identity (19). The proof of Lemma 3.1 and Theorem 3.4 used only $-\Delta w_E = 1$, $w_E \in H_0^1(E)$, interior smoothness, and Sard's theorem, all of which hold for E verbatim (interior regularity and analyticity of the torsion function depend only on the interior equation, not on boundary regularity; Theorem 2.13). Repeating the three steps gives

$$\int_0^{m_E} (D_H^E + D_I^E) dv = \frac{1}{2}\mu_E(0^+)^2 - 4\pi T(E) = \frac{1}{2}|E|^2 - 4\pi T(E),$$

since $\mu_E(0^+) = |\{w_E > 0\}| = |E|$ ($w_E > 0$ in E). The last equality in (19) is the algebraic identity $\frac{1}{2}|E|^2 = 4\pi \cdot |E|^2/(8\pi) = 4\pi T(B_E)$.

Homogeneity (20). The torsion function transforms under dilation by $w_{sE}(x) = s^2 w_E(x/s)$: indeed w_{sE} so defined vanishes on $\partial(sE)$ and $-\Delta w_{sE}(x) = -s^2 \cdot s^{-2}(\Delta w_E)(x/s) = 1$, and by uniqueness it is the torsion function of sE . Hence $m_{sE} = s^2 m_E$, and

$$T(sE) = \int_{sE} w_{sE} dx = \int_{sE} s^2 w_E(x/s) dx \stackrel{x=sy}{=} s^2 \int_E s^2 w_E(y) dy = s^4 T(E),$$

while $|sE|^2 = s^4|E|^2$; both terms on the right of (19) scale by s^4 , giving the second equality in (20), and the first follows from it via (19). (Directly: for a level $v = s^2 v'$ one has $\{w_{sE} > v\} = s\{w_E > v'\}$, so $\mu_{sE}(s^2 v') = s^2 \mu_E(v')$, $P_{sE}(s^2 v') = s P_E(v')$, $\mu'_{sE}(s^2 v') = \mu'_E(v')$, whence $D_H^{sE}(s^2 v') = s^2 D_H^E(v')$ and $D_I^{sE}(s^2 v') = s^2 D_I^E(v')$; the change of variables $v = s^2 v'$, $dv = s^2 dv'$ then produces the factor $s^2 \cdot s^2 = s^4$.)

Scale invariance of δ_ .* From (19), $T(B_E) - T(E) = \frac{1}{4\pi} \int_0^{m_E} (D_H^E + D_I^E) dv$ and $T(B_E) = |E|^2/(8\pi)$, so dividing gives $\delta_*(E) = \frac{2}{|E|^2} \int_0^{m_E} (D_H^E + D_I^E) dv$, which is (21). By (20) numerator and denominator $T(B_E)$ both scale by s^4 , so $\delta_*(sE) = \delta_*(E)$. $\square$

Remark 3.7 (Consistency with the normalisation). Applied to $E = \Omega$ with $|\Omega| = \pi$, (19) returns (15): $\frac{1}{2}\pi^2 - 4\pi T(\Omega) = 4\pi\delta_T$, and $T(B_\Omega) = \pi^2/(8\pi) = \pi/8 = T(B_1)$. The two normalisations are related by $\delta_*(\Omega) = \delta_T/T(B_1) = \frac{8}{\pi}\delta_T$, in agreement with the global notation.

3.5. Talenti's comparison and the volume profile. To convert “small on most levels” into quantitative geometry we need the size of the level sets, i.e. the profile $\mu(v)$. This is furnished by Talenti's pointwise comparison with the model disk. Recall the unit-disk torsion function $u_{B_1}(x) = \frac{1}{4}(1 - |x|^2)$ and its distribution function

$$\mu_B(v) := |\{u_{B_1} > v\}| = \pi(1 - 4v)_+, \quad v \geq 0,$$

where $t_+ = \max\{t, 0\}$.

Theorem 3.8 (Talent comparison for the profile). *Let u be the torsion function of Ω with $|\Omega| = \pi$. Then*

$$\mu(v) = |\{u > v\}| \leq \mu_B(v) = \pi(1 - 4v)_+ \quad \text{for all } v \geq 0. \quad (22)$$

Proof. We invoke Talenti's comparison theorem [12, Theorem 1] in the form proved there: if $u \in H_0^1(\Omega)$ solves $-\Delta u = f$ with $f \geq 0$, and u^* denotes the Schwarz (radially symmetric decreasing) rearrangement of u , while $\tilde{u}$ solves $-\Delta \tilde{u} = f^*$ in the ball Ω^* centred at the origin with $|\Omega^*| = |\Omega|$ and $\tilde{u} \in H_0^1(\Omega^*)$, then

$$u^*(x) \leq \tilde{u}(x) \quad \text{for all } x \in \Omega^*.$$

We apply this with $f \equiv 1$. Then $f^* \equiv 1$, $\Omega^* = B_1$ (since $|\Omega| = \pi = |B_1|$), and the comparison solution is the disk torsion function $\tilde{u} = u_{B_1}$, because $u_{B_1} = \frac{1}{4}(1 - |x|^2)$ solves $-\Delta u_{B_1} = 1$ in B_1 with zero boundary data and is its own rearrangement. Thus $u^* \leq u_{B_1}$ pointwise on B_1 .

Schwarz rearrangement is equimeasurable: $|\{u > v\}| = |\{u^* > v\}|$ for every v . Since $u^* \leq u_{B_1}$, we have $\{u^* > v\} \subseteq \{u_{B_1} > v\}$, hence $|\{u^* > v\}| \leq |\{u_{B_1} > v\}|$. Combining,

$$\mu(v) = |\{u > v\}| = |\{u^* > v\}| \leq |\{u_{B_1} > v\}| = \mu_B(v) = \pi(1 - 4v)_+,$$

where the last equality is $\{u_{B_1} > v\} = \{|x|^2 < 1 - 4v\}$, of area $\pi(1 - 4v)_+$. $\square$

Lemma 3.9 (Exact volume profile and the cap on the maximum). *The Saint-Venant deficit equals the L^1 gap between the disk profile and the domain profile over $(0, \frac{1}{4})$:*

$$\int_0^{1/4} (\mu_B(v) - \mu(v)) dv = \delta_T(\Omega), \quad (23)$$

the integrand $\mu_B(v) - \mu(v) \geq 0$ being nonnegative for every v . In particular the maximum is capped,

$$m = \max_\Omega u \leq \frac{1}{4}. \quad (24)$$

Proof. The maximum cap (24). Suppose, for contradiction, that $m > \frac{1}{4}$. Pick v with $\frac{1}{4} < v < m$. Then $\{u > v\} \neq \emptyset$ and is open (as u is continuous), so $\mu(v) = |\{u > v\}| > 0$; but $v > \frac{1}{4}$ gives $\mu_B(v) = \pi(1 - 4v)_+ = 0$, contradicting (22). Hence $m \leq \frac{1}{4}$. In particular $\mu(v) = 0$ for $v \geq \frac{1}{4} \geq m$.

The profile identity (23). Nonnegativity of the integrand on all of $(0, \frac{1}{4})$ is exactly (22). By the layer-cake formula and $m \leq \frac{1}{4}$,

$$T(\Omega) = \int_\Omega u dx = \int_0^\infty \mu(v) dv = \int_0^{1/4} \mu(v) dv,$$

the last step using $\mu(v) = 0$ for $v \geq \frac{1}{4}$. Likewise, by the same formula for the disk torsion function u_{B_1} ,

$$T(B_1) = \int_{B_1} u_{B_1} dx = \int_0^\infty \mu_B(v) dv = \int_0^{1/4} \pi(1 - 4v) dv = \pi \left[v - 2v^2 \right]_0^{1/4} = \pi \left(\frac{1}{4} - \frac{1}{8} \right) = \frac{\pi}{8}.$$

Subtracting,

$$\int_0^{1/4} (\mu_B(v) - \mu(v)) dv = T(B_1) - T(\Omega) = \frac{\pi}{8} - T(\Omega) = \delta_T(\Omega).$$

This is (23). (The computation also reconfirms independently that $T(B_1) = \pi/8$, hence that the deficit is well defined and non-negative: $\delta_T = \int_0^{1/4} (\mu_B - \mu) \geq 0$.) $\square$

Remark 3.10 (Two complementary decompositions of δ_T). We have written δ_T in two ways: as an integral over levels of the *shape* defects $D_H + D_I$ (Theorem 3.4, weight 4π), and as an integral over levels of the *volume* gap $\mu_B - \mu$ (Lemma 3.9, weight 1). The inset construction in Section 4 uses both: the first transfers the torsion deficit to a single level set, the second certifies that this level set captures almost all of the area of Ω .

3.6. A pointwise lower bound on $-\mu'$. The final consequence we extract is a clean pointwise estimate, used to quantify how fast the volume profile decreases.

Lemma 3.11 (Lower bound on the profile slope). *For a.e. $v \in (0, m)$,*

$$-\mu'(v) \geq 4\pi. \quad (25)$$

Proof. Fix $v \in (0, m)$ a regular value at which (13) holds (a.e. v). By Lemma 3.3, $D_H(v) \geq 0$ and $D_I(v) \geq 0$, i.e.

$$\mu(v)(-\mu'(v)) \geq P(v)^2 \quad \text{and} \quad P(v)^2 \geq 4\pi\mu(v).$$

Chaining these,

$$\mu(v)(-\mu'(v)) \geq P(v)^2 \geq 4\pi\mu(v).$$

Since $v < m$, the super-level set $E_v = \{u > v\}$ is nonempty and open, so $\mu(v) > 0$; dividing by $\mu(v) > 0$ gives $-\mu'(v) \geq 4\pi$. $\square$

Remark 3.12 (Sharpness and the model). All four statements of this section are sharp and are simultaneously saturated by the disk: for u_{B_1} one has $\mu = \mu_B$, $-\mu'_B(v) = 4\pi$ on $(0, \frac{1}{4})$, $D_H \equiv D_I \equiv 0$, $\delta_T = 0$, and $m = \frac{1}{4}$. Thus (25) is an equality for the model, and the level identity (15) reads $0 = 0$. The quantitative theory of the remaining sections measures the deviation from this rigid configuration.

4. THE INSET: REGULARITY, DEFICIT TRANSFER, LEVEL SELECTION

Throughout this section $\Omega \subset \mathbb{R}^2$ is open with $|\Omega| = \pi$, $u \in H_0^1(\Omega)$ is its torsion function ($-\Delta u = 1$ in Ω , $u = 0$ on $\partial\Omega$), $m := \max_\Omega u$, and $\delta_T = \delta_T(\Omega) = \frac{\pi}{8} - T(\Omega) \geq 0$ is the Saint-Venant deficit. We write $E_v := \{u > v\}$ for the open super-level set, $\mu(v) := |E_v|$, $P(v) := P(E_v)$, and recall the two level deficits of Definition 3.2,

$$D_H(v) = \mu(v)(-\mu'(v)) - P(v)^2 \geq 0, \quad D_I(v) = P(v)^2 - 4\pi\mu(v) \geq 0$$

(Lemma 3.3), the level-set deficit identity $\int_0^m (D_H + D_I) dv = 4\pi\delta_T$ (Theorem 3.4), the Talenti profile $\mu_B(v) := \pi(1 - 4v)_+$ with $\mu \leq \mu_B$ and the profile identity $\int_0^{1/4} (\mu_B - \mu) dv = \delta_T$ (Theorem 3.8, Lemma 3.9), the bound $-\mu'(v) \geq 4\pi$ a.e. (Lemma 3.11), and the flux/coarea identities $\int_{\{u=v\}} |\nabla u| d\mathcal{H}^1 = \mu(v)$, $-\mu'(v) = \int_{\{u=v\}} |\nabla u|^{-1} d\mathcal{H}^1$ for a.e. regular v (Theorem 2.6, Lemma 2.7). By Sard's theorem (Theorem 2.17) and Lemma 2.18 the non-regular values of u form a Lebesgue-null set, and for every regular $v \in (0, m)$ the set E_v has finite perimeter with $\mathcal{H}^1$ -rectifiable boundary $\{u = v\}$ (Theorem 2.13). Absolute constants are denoted C (value may change) or named $(A_0, \delta_\star)$; $\delta_\star > 0$ is a small absolute constant, fixed at the end of the section. We set $\tau := \sqrt{\delta_T}$ only at the very end; until then τ is a free parameter in $(0, \frac{1}{16})$.

4.1. Standing measure-theoretic convention and interior containment. The torsion function u is regular only in the *interior* of Ω : by Theorem 2.13, $u \in C^\infty(\Omega)$ (indeed real-analytic) and $u > 0$ in Ω , but for a general bounded open Ω we do *not* assume $u \in C(\overline{\Omega})$, nor that $u = 0$ holds pointwise on $\partial\Omega$, nor that the super-level sets $E_v = \{u > v\}$ are compactly contained in Ω . Indeed they need not be: for the punctured disc $\Omega = B_1 \setminus \{0\}$ (area π) the torsion function is $u = u_{B_1} = (1 - |x|^2)/4$ (the removed point $\{0\}$ is polar, hence invisible to H_0^1), and every super-level set $\{u \geq t\} = \overline{B_{\sqrt{1-4t}}}$ accumulates at the boundary point $0 \in \partial\Omega$, so its closure is *not* a compact subset of Ω . The naive statement “ $\overline{E_v}$ is compactly contained in Ω ” is therefore false in general, and we adopt throughout the only correct, measure-theoretic, stance: all quantities of the paper — $|\Omega|$, $T(\Omega)$, the asymmetry $\mathcal{A}$, the level sets E_v , their measures $\mu(v)$ and perimeters $P(v)$ — depend on Ω only through its measure-theoretic structure, and a *polar* (capacity-zero) subset of $\partial\Omega$ may be added to or removed from Ω without changing any of them. We make this precise and isolate the containment property we actually use.

Lemma 4.1 (Interior containment of super-level sets, modulo a polar set). *Let $\Omega \subset \mathbb{R}^2$ be bounded open with torsion function u , and let $t > 0$. Then:*

- (i) *for a.e. $t > 0$ the set $E_t = \{u > t\}$ has finite perimeter, $P(E_t) = \mathcal{H}^1(\{u = t\})$, and E_t is an open subset of Ω that coincides with its own measure-theoretic interior up to an $\mathcal{H}^1$ -null set;*
- (ii) *for every regular $t > 0$ and every connected component E of E_t , the relative boundary of E in Ω lies in the level set,*

$$\partial E \cap \Omega \subset \{u = t\} \subset \Omega,$$

an $\mathcal{H}^1$ -rectifiable curve contained in the open set Ω and carrying the full perimeter, $P(E) = \mathcal{H}^1(\partial^ E) = \mathcal{H}^1(\partial E \cap \Omega)$; the set E is bounded (so $\overline{E} \subset \mathbb{R}^2$ is compact), and the $\mathbb{R}^2$ -closure $\overline{E}$ may meet $\partial\Omega$, but $\overline{E} \cap \partial\Omega$ carries no perimeter of E : $\mathcal{H}^1(\partial^* E \cap \partial\Omega) = 0$. (This is the precise, measure-theoretic substitute for “ $\overline{E}$ compactly contained in Ω ”, which fails for the punctured disc.)*

- (iii) *consequently, replacing Ω by $\Omega \cup Z$ or $\Omega \setminus Z$ for any polar set Z (capacity zero, hence Lebesgue-null and not charged by the perimeter measure of any finite-perimeter set, [5, Thm. 4.5, §4.7]) leaves u , $T(\Omega)$, every E_t , $\mu(t)$, $P(t)$, δ_T and $\mathcal{A}(\Omega)$ unchanged. In particular the symmetric differences $|\Omega \Delta E_t|$ and $|\Omega \Delta G|$ used downstream are insensitive to such Z .*

Proof. (i). By the coarea formula (Theorem 2.6) applied to $u \in H^1(\Omega) \cap C^\infty(\Omega)$, $\int_0^m P(E_t) dt = \int_\Omega |\nabla u| < \infty$, so E_t has finite perimeter for a.e. t , with $P(E_t) = \mathcal{H}^1(\{u = t\})$ at every regular t

(Lemma 2.18). The set E_t is open as a super-level set of the continuous function u on the open set Ω ; for a regular value $\nabla u \neq 0$ on $\{u = t\}$, so $\{u = t\}$ is an $\mathcal{H}^1$ -rectifiable C^1 curve, of zero Lebesgue measure, whence E_t equals its measure-theoretic interior up to that null curve.

(ii). Fix a regular t and a component E of E_t . E is open (super-level set of the continuous u) and bounded ($E \subset \Omega$, Ω bounded), so $\overline{E} \subset \mathbb{R}^2$ is compact. The relative boundary $\partial E \cap \Omega$ consists of points $q \in \Omega$ that are limits of points of E (where $u > t$) and of points of $\Omega \setminus E$; by continuity of u on the open set Ω , $u(q) = t$, so $\partial E \cap \Omega \subset \{u = t\}$, which at the regular value t is an $\mathcal{H}^1$ -rectifiable C^1 curve in Ω along which $\nabla u \neq 0$ (Lemma 2.18). The reduced boundary satisfies $\partial^* E \subset \partial E$, and the part of $\partial^* E$ inside Ω is $\mathcal{H}^1$ -equivalent to $\{u = t\} \cap \partial E$ (Theorem 2.6, $P(E) = \mathcal{H}^1(\{u = t\} \cap \partial E)$). It remains to see that $\overline{E}$ may touch $\partial\Omega$ only off the reduced boundary. Indeed $\partial^* E \subset \partial E$, and the part of ∂E on $\partial\Omega$ is disjoint from $\Omega \supset \{u = t\}$; since $\partial^* E \cap \Omega \subset \{u = t\} \subset \Omega$ already carries the full perimeter $P(E) = \mathcal{H}^1(\partial^* E)$, the remainder $\partial^* E \cap \partial\Omega$ has $\mathcal{H}^1$ -measure zero. (For the punctured disc, $\overline{E} = \overline{B_{\sqrt{1-4t}}}$ touches $\partial\Omega$ at the polar puncture 0, which is a single point, hence $\mathcal{H}^1$ -null and not a reduced-boundary point.) This proves (ii).

(iii). A polar set $Z \subset \mathbb{R}^2$ has zero 2-capacity, hence zero Lebesgue measure and is removable for H_0^1 : the torsion function of Ω and of $\Omega \cup Z$ coincide ($H_0^1(\Omega) = H_0^1(\Omega \cup Z)$ when Z is polar of finite measure, since capacity-zero sets are negligible for H^1 traces, [5, §4.7]). All listed quantities are integrals against Lebesgue measure of u or indicators of finite-perimeter sets, or perimeters thereof; Lebesgue measure and the perimeter measure (which is absolutely continuous with respect to $\mathcal{H}^1 \llcorner_{\partial^* E}$ and assigns no mass to the $\mathcal{H}^1$ -null, indeed Lebesgue-null, polar set Z , [5, §5.7], [9, Thm. 16.2]) both ignore Z . The claim follows. $\square$

Remark 4.2 (The role of Lemma 4.1). Lemma 4.1 is the honest substitute for the (false in general) assertion that $\overline{E_v}$ is compactly contained in Ω . What the later sections genuinely need is only: (a) E_v and each of its components are open subsets of Ω on which u is real-analytic with $\nabla u \neq 0$ along $\{u = v\}$ at regular v (so ∂E is an $\mathcal{H}^1$ -rectifiable curve in the open set Ω , with $P(E) = \mathcal{H}^1(\partial^* E)$); and (b) the measure-theoretic insensitivity to polar sets, (iii), which guarantees that all symmetric differences and deficits are well posed. In Section 5 we work with the *measure-theoretic representative* of $\tilde{E}$ (equivalently the finely-open representative): the topological boundary $\partial \tilde{E}$ there is the boundary of this representative, which differs from the reduced and essential boundaries only on an $\mathcal{H}^1$ -null set and lies in the open set Ω . No regularity of $\partial\Omega$ is used anywhere.

4.2. No interior holes. We now record a topological regularity property of the super-level sets, which is a direct consequence of the superharmonicity of u ($-\Delta u = 1 \geq 0$) and the minimum principle. We use only the *interior* regularity of Theorem 2.13 — $u \in C^\infty(\Omega)$, $u > 0$ in Ω — together with Lemma 4.1; in particular we do *not* assume $u \in C(\overline{\Omega})$ or any regularity of $\partial\Omega$. The phrase “ $\overline{K} \cap \partial\Omega \neq \emptyset$ ” below means that K , a relatively closed bounded subset of Ω , is *not* compactly contained in Ω : its closure in $\mathbb{R}^2$ meets $\partial\Omega$ (equivalently, K has a limit point on $\partial\Omega$).

Lemma 4.3 (No interior holes). *For every $v \in (0, m)$ the following hold.*

- (i) *Every connected component K of the open set $\{x \in \Omega : u(x) < v\}$ has $\overline{K} \cap \partial\Omega \neq \emptyset$.*
- (ii) *For every regular value v , each connected component E of $E_v = \{u > v\}$ has no interior holes: every bounded connected component H of $\mathbb{R}^2 \setminus \overline{E}$ satisfies $\overline{H} \not\subset \Omega$; in particular $\mathbb{R}^2 \setminus \overline{E}$ is connected up to sets touching $\partial\Omega$, and the inset defined below is simply connected modulo $\partial\Omega$.*

Proof. The entire argument takes place in the open set Ω , where u is smooth (Theorem 2.13); no value of u on $\partial\Omega$ and no continuity of u up to $\partial\Omega$ is invoked. The negations we contradict are exactly the statements that the relevant components are *compactly contained in* Ω (closure in $\mathbb{R}^2$ a compact subset of Ω), the case Lemma 4.1(ii) shows cannot occur for super-level components; here we prove the analogous exclusion for the complementary sub-level islands and for holes.

(i). Suppose, for contradiction, that some component K of $\{u < v\}$ satisfies $\overline{K} \Subset \Omega$, i.e. $\overline{K}$ is a compact subset of Ω not meeting $\partial\Omega$. We claim $u \equiv v$ on ∂K . Let $x_0 \in \partial K$. Since $\overline{K} \subset \Omega$ we have $x_0 \in \Omega$, so u is continuous at x_0 (interior continuity, Theorem 2.13; no boundary regularity is used). If $u(x_0) < v$ then by continuity $u < v$ on a ball around x_0 , which would be a connected subset of $\{u < v\}$ meeting K , hence contained in K , placing x_0 in the *interior* of K — contradicting $x_0 \in \partial K$. If $u(x_0) > v$ then by continuity $u > v$ near x_0 , so a punctured neighbourhood of x_0 is disjoint from $\{u < v\} \supset K$, contradicting $x_0 \in \partial K$ (as K accumulates at x_0). Hence $u(x_0) = v$, proving $u = v$ on ∂K .

Now u is continuous on the compact set $\overline{K}$ and attains its minimum there; since $u < v$ at any interior point of K and $u = v$ on ∂K , the minimum is attained at an interior point and equals $\min_{\overline{K}} u < v = \min_{\partial K} u$. But $-\Delta u = 1 \geq 0$ in Ω makes u superharmonic, and the (weak) minimum principle for superharmonic functions in C^2 [8, Theorem 3.1] gives $\min_{\overline{K}} u = \min_{\partial K} u$. This contradiction proves (i).

(ii). Let v be regular and E a component of E_v , and suppose H is a bounded component of $\mathbb{R}^2 \setminus \overline{E}$ with $\overline{H} \subset \Omega$. Then H is an open subset of Ω with $\overline{H} \Subset \Omega$, and we consider the two open sets $W^- := H \cap \{u < v\}$ and $W^+ := H \cap \{u > v\}$, both contained in H , hence both with closures contained in $\overline{H} \Subset \Omega$.

We first note that every connected component G of W^- (resp. of W^+) is in fact a connected component of $\{u < v\}$ (resp. of $\{u > v\}$), and is contained in H . Indeed, G is connected and contained in the open set $\{u \neq v\}$, so it lies in a unique component $\hat{G}$ of $\{u < v\}$ (resp. $\{u > v\}$). If $\hat{G}$ left H it would, being connected and meeting H , have to cross ∂H ; but $\partial H \subset \overline{E}$ and on $\overline{E}$ one has $u \geq v$, while points of ∂H with $u > v$ lie in the open set E (a component of E_v , hence open) which is disjoint from $\hat{G}$ when $\hat{G} \subset W^-$, and equals a different component of E_v disjoint from E when $\hat{G} \subset W^+$ (as $\hat{G} \cap E \subset W^+ \cap E \subset H \cap E = \emptyset$). In either case $\hat{G}$ cannot meet ∂H , so $\hat{G} \subset H$ and $\hat{G} = G$. Thus each component of W^- is a connected component of $\{u < v\}$ whose closure lies in $\overline{H} \Subset \Omega$ and so does not meet $\partial\Omega$; by (i) this forces $W^- = \emptyset$. Each connected component E' of W^+ is a connected component of $E_v = \{u > v\}$ with $\overline{E'} \Subset \Omega$; on it $u = v$ on $\partial E'$ and $u > v$ inside, so u attains an interior maximum strictly above its boundary value. But u is superharmonic ($-\Delta u = 1 \geq 0$), so $-u$ is subharmonic and satisfies the maximum principle [8, Theorem 3.1]: $\max_{\overline{E'}} u = \max_{\partial E'} u = v$, contradicting $u > v$ inside. Hence $W^+ = \emptyset$ as well.

Therefore $u \equiv v$ on the open set H , which has positive Lebesgue measure; but v is a regular value, so $\{u = v\}$ is an $\mathcal{H}^1$ -rectifiable curve, of zero Lebesgue measure. This contradicts $|H| > 0$, so no such H exists. $\square$

Remark 4.4 (Measure-theoretic reading; no boundary regularity). Part (ii) uses both the minimum principle (via (i), excluding holes that are sub-level islands) and the maximum principle (excluding super-level islands separated from E), and only interior regularity of u (Theorem 2.13); it is

consistent with the standing convention of Section 4.1. The statement “ $\overline{H} \not\subset \Omega$ ” is read in the sense of Lemma 4.1(ii): a putative hole H that is compactly contained in Ω is excluded, while a putative hole that accumulates only on $\partial\Omega$ (as the centre of the punctured disc would, were $\{0\}$ removed) is not a genuine interior hole and carries no perimeter of E . This does *not* by itself prove that the full exterior $\mathbb{R}^2 \setminus \overline{\tilde{E}}$ is connected; that stronger topological input is supplied unconditionally by the hole-filling construction Lemma 4.12 of Section 4.8, which fills any boundary-touching holes at a cost quadratic in the isoperimetric deficit. The consequences used unconditionally here are connectedness of $\tilde{E}$, exclusion of compactly contained holes, and the measure-theoretic fact $|\{u = \hat{v}\}| = 0$ used repeatedly to identify symmetric differences. A reader content with these may keep (i) and the line “ $u \equiv v$ on H with $|H| > 0$ contradicts $|\{u = v\}| = 0$ ” as the whole of (ii).

4.3. Deficit transfer to super-level sets. The torsion function of any super-level set E_v is $u - v$, and the level-set deficit identity, being homogeneous of degree four, transfers the scale-invariant deficit from Ω to E_v as a tail of the positive-integrand identity of Theorem 3.4.

Lemma 4.5 (Deficit transfer). *For every regular $v \in (0, m)$,*

$$\delta_*(E_v) \leq \frac{\pi^2}{\mu(v)^2} \delta_*(\Omega). \quad (26)$$

Proof. On the open set E_v the relevant function is $w := (u - v)_+ \upharpoonright_{E_v} = u - v$. This is the torsion function of E_v in the weak Sobolev sense. Indeed, by the capacity characterisation of zero traces, the quasi-continuous representative of $(u - v)_+$ vanishes outside E_v up to capacity zero, hence $(u - v)_+ \in H_0^1(E_v)$ [5, §4.7]. For every $\varphi \in C_c^\infty(E_v)$,

$$\int_{E_v} \nabla w \cdot \nabla \varphi = \int_{\Omega} \nabla u \cdot \nabla \varphi = \int_{\Omega} \varphi = \int_{E_v} \varphi.$$

Thus $-\Delta w = 1$ in E_v with zero Sobolev trace, and uniqueness in $H_0^1(E_v)$ gives the claim. Therefore $T(E_v) = \int_{E_v} w = \int_{E_v} (u - v)$. Its super-level sets are $\{w > s\} = \{u > v + s\} = E_{v+s}$ for $s \in (0, m - v)$, so the perimeter and distribution functions of w at height s coincide with those of u at height $v + s$; in particular the two level deficits of w at height s equal $D_H(v + s)$ and $D_I(v + s)$.

We now apply the derivation of the level-set deficit identity (Theorem 3.4, eq. (15)) to w on E_v ; the derivation there uses only the flux/coarea identities, the absolute continuity of the distribution function, and the layer-cake formula, none of which requires the area to equal π . For a domain D of area $|D|$ with torsion function having maximum m_D and level deficits D_H^D, D_I^D it yields

$$\int_0^{m_D} (D_H^D + D_I^D) ds = \frac{1}{2}|D|^2 - 4\pi T(D) = 4\pi \left(\frac{|D|^2}{8\pi} - T(D) \right) = 4\pi (T(B_D) - T(D)),$$

where B_D is the disc of area $|D|$, $T(B_D) = |D|^2/(8\pi)$; the middle equality is exactly the computation in the proof of Theorem 3.4 with $\frac{1}{2}|D|^2$ in place of $\frac{1}{2}\pi^2$. Taking $D = E_v$, $m_{E_v} = m - v$, and using the identification of the level deficits of w with $D_H(v + \cdot), D_I(v + \cdot)$,

$$4\pi (T(B_{E_v}) - T(E_v)) = \int_0^{m-v} (D_H(v+s) + D_I(v+s)) ds = \int_v^m (D_H + D_I) dv' \leq \int_0^m (D_H + D_I) dv' = 4\pi \delta_T,$$

the inequality by non-negativity of the integrand (Lemma 3.3). Hence $T(B_{E_v}) - T(E_v) \leq \delta_T$. Dividing by $T(B_{E_v}) = \mu(v)^2/(8\pi)$ and using $\delta_*(\Omega) = \frac{8}{\pi}\delta_T$, i.e. $\delta_T = \frac{\pi}{8}\delta_*(\Omega)$,

$$\delta_*(E_v) = \frac{T(B_{E_v}) - T(E_v)}{T(B_{E_v})} \leq \frac{\delta_T}{\mu(v)^2/(8\pi)} = \frac{8\pi\delta_T}{\mu(v)^2} = \frac{8\pi}{\mu(v)^2} \cdot \frac{\pi}{8} \delta_*(\Omega) = \frac{\pi^2}{\mu(v)^2} \delta_*(\Omega). \quad \square$$

The content of (26) is that the *scale-invariant* torsion deficit of every super-level set is comparable to that of Ω (a tail of a positive integral), even at levels near $\partial\Omega$ where the single-level deficits $D_H(v), D_I(v)$ are individually inflated. This is the mechanism that carries the small torsion deficit onto the inset.

4.4. Selection of a good level.

Lemma 4.6 (Good level). *Let $A_0 := 16\pi$. For every $\tau \in (0, \frac{1}{16})$ there is a regular value $\hat{v} \in (\tau, 2\tau)$ of u with*

$$D_H(\hat{v}) + D_I(\hat{v}) \leq A_0 \frac{\delta_T}{\tau}, \quad \mu_B(\hat{v}) - \mu(\hat{v}) \leq A_0 \frac{\delta_T}{\tau}. \quad (27)$$

Proof. Set $I := (\tau, 2\tau)$, of length $|I| = \tau$, and note $I \subset (0, \frac{1}{8}) \subset (0, \frac{1}{4})$. By Theorem 3.4 and non-negativity, $\int_I (D_H + D_I) dv \leq \int_0^m (D_H + D_I) dv = 4\pi\delta_T$. By Chebyshev's inequality,

$$|\{v \in I : D_H(v) + D_I(v) > A_0\delta_T/\tau\}| \leq \frac{\int_I (D_H + D_I)}{A_0\delta_T/\tau} \leq \frac{4\pi\delta_T}{A_0\delta_T/\tau} = \frac{4\pi\tau}{A_0}.$$

By Lemma 3.9, $\mu_B - \mu \geq 0$ and $\int_I (\mu_B - \mu) dv \leq \int_0^{1/4} (\mu_B - \mu) dv = \delta_T$, so again by Chebyshev,

$$|\{v \in I : \mu_B(v) - \mu(v) > A_0\delta_T/\tau\}| \leq \frac{\delta_T}{A_0\delta_T/\tau} = \frac{\tau}{A_0}.$$

With $A_0 = 16\pi$, the union of these two “bad” sets has measure at most $\frac{4\pi\tau}{A_0} + \frac{\tau}{A_0} = \frac{\tau}{4} + \frac{\tau}{16\pi} < \tau = |I|$. Adding the null set of non-regular values (Theorem 2.17, Lemma 2.18) does not change this. Hence the set of regular $\hat{v} \in I$ satisfying both bounds in (27) has positive measure, and in particular is nonempty. $\square$

From now on $\hat{v} \in (\tau, 2\tau)$ denotes a good regular level as in Lemma 4.6, and we write $\hat{v} < \frac{1}{4}$ (since $\hat{v} < 2\tau < \frac{1}{8}$), so that $\mu_B(\hat{v}) = \pi(1 - 4\hat{v})$.

4.5. Closeness of the super-level set.

Lemma 4.7 (Closeness). *At the level $\hat{v}$ of Lemma 4.6,*

$$|\Omega \Delta \{u > \hat{v}\}| = \pi - \mu(\hat{v}) \leq 8\pi\tau + A_0 \frac{\delta_T}{\tau}. \quad (28)$$

Proof. Since $E_{\hat{v}} = \{u > \hat{v}\} \subset \Omega$, the symmetric difference is $\Omega \setminus E_{\hat{v}} = \{x \in \Omega : u(x) \leq \hat{v}\}$, of measure $\pi - \mu(\hat{v})$ (the level set $\{u = \hat{v}\}$ is Lebesgue-null). Split

$$\pi - \mu(\hat{v}) = (\pi - \mu_B(\hat{v})) + (\mu_B(\hat{v}) - \mu(\hat{v})).$$

The first term is $\pi - \pi(1 - 4\hat{v}) = 4\pi\hat{v} \leq 4\pi(2\tau) = 8\pi\tau$, using $\hat{v} < 2\tau$; the second is $\leq A_0\delta_T/\tau$ by (27). Adding gives (28). $\square$

We now fix $\delta_\star \in (0, \frac{1}{256}]$ small enough that, whenever $\delta_T \leq \delta_\star$ and $\tau \in [\sqrt{\delta_T}, \frac{1}{16})$ with $\delta_T/\tau \leq \sqrt{\delta_T}$ (e.g. at the optimal choice $\tau = \sqrt{\delta_T}$, where the right side of (28) equals $(8\pi + A_0)\sqrt{\delta_T}$), one has

$$8\pi\tau + A_0 \frac{\delta_T}{\tau} \leq \frac{\pi}{2}, \quad \text{hence} \quad \mu(\hat{v}) \geq \frac{\pi}{2}. \quad (29)$$

Concretely, at $\tau = \sqrt{\delta_T}$ this requires $(8\pi + A_0)\sqrt{\delta_\star} \leq \pi/2$, i.e. $\delta_\star \leq \pi^2/(4(8\pi + A_0)^2)$; with $A_0 = 16\pi$ this is $\delta_\star \leq 1/(2304)$, comfortably compatible with $\delta_\star \leq \frac{1}{256}$ after taking the smaller value. We keep (29) as a standing hypothesis for the remainder of the section (it holds at the optimised scale $\tau = \sqrt{\delta_T}$); the value of $\hat{v}, \tau$ is otherwise left symbolic and optimised in Theorem 9.1.

4.6. One connected inset: the spillover is higher order. The super-level set $\{u > \hat{v}\}$ may have several components, but the total mass of all but the largest is quadratically small in $D_I(\hat{v})$, by a multi-component refinement of the isoperimetric inequality.

Lemma 4.8 (Spillover). *Assume (29). Let the connected components of $\{u > \hat{v}\}$ have areas $A_1 \geq A_2 \geq \dots > 0$, let $\tilde{E}$ be the component of area A_1 , and set $S := \sum_{i \geq 2} A_i = \mu(\hat{v}) - A_1$ for the total spillover. Then $A_1 \geq \pi/4$ and*

$$|\{u > \hat{v}\} \setminus \tilde{E}| = S \leq \frac{D_I(\hat{v})^2}{16\pi^3} \leq \frac{A_0^2}{16\pi^3} \frac{\delta_T^2}{\tau^2}. \quad (30)$$

Proof. Multi-component isoperimetric inequality. Perimeter is additive over the (essentially disjoint) components and the planar isoperimetric inequality (Theorem 2.9) applies to each, so $P(\hat{v}) = \sum_i P_i \geq \sum_i 2\sqrt{\pi A_i}$ with P_i the perimeter of the i -th component. Squaring and using $(\sum_i \sqrt{A_i})^2 = \sum_i A_i + 2 \sum_{i < j} \sqrt{A_i A_j}$,

$$D_I(\hat{v}) = P(\hat{v})^2 - 4\pi \sum_i A_i \geq 4\pi \left[\left(\sum_i \sqrt{A_i} \right)^2 - \sum_i A_i \right] = 8\pi \sum_{i < j} \sqrt{A_i A_j}. \quad (31)$$

Lower bound on A_1 . Since $A_j \leq A_1$ we have $\sqrt{A_j} = A_j/\sqrt{A_j} \geq A_j/\sqrt{A_1}$ for $j \geq 2$, hence $\sum_{j \geq 2} \sqrt{A_j} \geq S/\sqrt{A_1}$ and

$$\sum_i \sqrt{A_i} \geq \sqrt{A_1} + \frac{S}{\sqrt{A_1}} = \frac{A_1 + S}{\sqrt{A_1}} = \frac{\mu(\hat{v})}{\sqrt{A_1}}.$$

Combining with (31) (in the form $D_I \geq 4\pi[(\sum \sqrt{A_i})^2 - \mu(\hat{v})]$),

$$D_I(\hat{v}) \geq 4\pi \left(\frac{\mu(\hat{v})^2}{A_1} - \mu(\hat{v}) \right) = 4\pi \mu(\hat{v}) \frac{\mu(\hat{v}) - A_1}{A_1} = 4\pi \mu(\hat{v}) \frac{S}{A_1}. \quad (32)$$

Suppose, for contradiction, that $A_1 < \pi/4$. By (29), $\mu(\hat{v}) \geq \pi/2 > A_1$, so there are at least two components and $S = \mu(\hat{v}) - A_1 > \pi/2 - \pi/4 = \pi/4$. Then (32) gives

$$D_I(\hat{v}) > 4\pi \cdot \frac{\pi}{2} \cdot \frac{S}{\pi/4} = 8\pi S > 8\pi \cdot \frac{\pi}{4} = 2\pi^2.$$

But $D_I(\hat{v}) \leq A_0 \delta_T / \tau \leq 8\pi\tau + A_0 \delta_T / \tau \leq \pi/2 < 2\pi^2$ by (27) and (29), a contradiction. Hence $A_1 \geq \pi/4$.

Spillover bound. From (31),

$$D_I(\hat{v}) \geq 8\pi \sum_{i < j} \sqrt{A_i A_j} \geq 8\pi \sqrt{A_1} \sum_{j \geq 2} \sqrt{A_j} \geq 8\pi \sqrt{A_1} \sqrt{\sum_{j \geq 2} A_j} = 8\pi \sqrt{A_1} S,$$

where the middle step keeps only the pairs $(1, j)$ and the last step uses $\sum_{j \geq 2} \sqrt{A_j} \geq \sqrt{\sum_{j \geq 2} A_j}$ (superadditivity of $t \mapsto \sqrt{t}$ on non-negative terms). Squaring and using $A_1 \geq \pi/4$,

$$S \leq \frac{D_I(\hat{v})^2}{64\pi^2 A_1} \leq \frac{D_I(\hat{v})^2}{64\pi^2 \cdot (\pi/4)} = \frac{D_I(\hat{v})^2}{16\pi^3}.$$

The final bound in (30) follows from $D_I(\hat{v}) \leq A_0 \delta_T / \tau$ (Lemma 4.6). $\square$

Definition 4.9 (The inset). Assume $\delta_T \leq \delta_*$, τ subject to (29), and let $\hat{v} \in (\tau, 2\tau)$ be a good regular level (Lemma 4.6). The *inset* is the connected component $\tilde{E}$ of $\{u > \hat{v}\}$ of largest measure, i.e. the component of area A_1 in Lemma 4.8. We write

$$\mu_E := |\tilde{E}| = A_1 = \mu(\hat{v}) - S \geq \frac{\pi}{4}, \quad \hat{r} := \sqrt{\mu_E/\pi}, \quad \delta_{\text{iso}} := \delta_{\text{iso}}(\tilde{E}) = \frac{P(\tilde{E})}{2\sqrt{\pi\mu_E}} - 1,$$

and $\text{Exc} := P(\tilde{E}) - 2\pi\hat{r} = 2\pi\hat{r}\delta_{\text{iso}}$.

By Lemma 4.3 (ii), $\tilde{E}$ is connected and has no interior hole *compactly contained in* Ω . The stronger *global* exterior-connectedness ($\mathbb{R}^2 \setminus \tilde{E}$ connected) required by Sections 5 and 6 is supplied unconditionally by passing to the hole-filled inset of Lemma 4.12 below (Section 4.8); from Section 5 onward the geometric reduction is run on that saturation, which inherits all the controls of Proposition 4.10 at the same scale.

4.7. Controls on the inset. We collect the three controls of the inset used in the subsequent sections.

Proposition 4.10 (Inset controls). *Under Definition 4.9 there is an absolute constant C such that*

$$\delta_{\text{iso}}(\tilde{E}) \leq \frac{D_I(\hat{v}) + 4\pi S}{8\pi\mu_E} \leq C \frac{\delta_T}{\tau}, \quad (33)$$

$$\delta_*(\tilde{E}) \leq \frac{\pi^2}{\mu_E^2} \delta_*(\Omega) + \frac{S^2}{\mu_E^2} \leq C \delta_*(\Omega), \quad (34)$$

$$|\Omega \triangle \tilde{E}| \leq 8\pi\tau + A_0 \frac{\delta_T}{\tau} + \frac{A_0^2}{16\pi^3} \frac{\delta_T^2}{\tau^2}. \quad (35)$$

In particular, with the optimal choice $\tau = \sqrt{\delta_T}$ of Theorem 9.1, $\delta_{\text{iso}}(\tilde{E}) \leq C\sqrt{\delta_T}$, $\delta_(\tilde{E}) \leq C\delta_*(\Omega)$ and $|\Omega \triangle \tilde{E}| \leq C\sqrt{\delta_T}$.*

Proof. (33). Since $\partial\tilde{E} \subset \{u = \hat{v}\}$ and perimeter is additive over components, $P(\tilde{E}) \leq P(\hat{v})$, hence

$$D_I(\tilde{E}) := P(\tilde{E})^2 - 4\pi\mu_E \leq P(\hat{v})^2 - 4\pi\mu_E = (4\pi\mu(\hat{v}) + D_I(\hat{v})) - 4\pi\mu_E = D_I(\hat{v}) + 4\pi S,$$

using $P(\hat{v})^2 = 4\pi\mu(\hat{v}) + D_I(\hat{v})$ and $\mu(\hat{v}) - \mu_E = S$. From the definition of δ_{iso} , $(1 + \delta_{\text{iso}}(\tilde{E}))^2 = P(\tilde{E})^2/(4\pi\mu_E) = 1 + D_I(\tilde{E})/(4\pi\mu_E)$, so by $\sqrt{1+t} - 1 \leq t/2$,

$$\delta_{\text{iso}}(\tilde{E}) \leq \frac{D_I(\tilde{E})}{8\pi\mu_E} \leq \frac{D_I(\hat{v}) + 4\pi S}{8\pi\mu_E}.$$

By (30), $4\pi S \leq 4\pi D_I(\hat{v})^2/(16\pi^3) = D_I(\hat{v})^2/(4\pi^2)$; and $D_I(\hat{v}) \leq A_0 \delta_T / \tau \leq \pi/2$ by (27)–(29), so $4\pi S \leq D_I(\hat{v}) \cdot \frac{D_I(\hat{v})}{4\pi^2} \leq \frac{1}{8\pi} D_I(\hat{v}) \leq D_I(\hat{v})$. Hence $D_I(\hat{v}) + 4\pi S \leq 2D_I(\hat{v}) \leq 2A_0 \delta_T / \tau$, and with

$$\mu_E \geq \pi/4,$$

$$\delta_{\text{iso}}(\tilde{E}) \leq \frac{2A_0\delta_T/\tau}{8\pi \cdot (\pi/4)} = \frac{A_0}{\pi^2} \frac{\delta_T}{\tau} =: C \frac{\delta_T}{\tau}.$$

(34). We bound $\delta_*(\tilde{E})$ directly, using only the deficit transfer of Lemma 4.5 and the exact additivity of torsion over connected components; no appeal to Lemma 7.4 is needed (see Remark 4.14). Let $C_2, C_3, \dots$ be the spillover components of $E_{\hat{v}}$, of areas $A_2, A_3, \dots$ with $\sum_{i \geq 2} A_i = S$. On each component C of $E_{\hat{v}} = \{u > \hat{v}\}$, the restriction of $(u - \hat{v})_+$ to C belongs to $H_0^1(C)$. More explicitly, the Sobolev space on an open set decomposes over its connected components: if $E = \bigcup_i C_i$ is the disjoint union of open components and $w \in H_0^1(E)$, then $w_i := w \mathbf{1}_{C_i}$ belongs to $H_0^1(C_i)$ and $\nabla w_i = \mathbf{1}_{C_i} \nabla w$ a.e.; this follows from the quasi-continuous trace characterisation of H_0^1 and the locality of the Sobolev gradient. Applying this to $w = (u - \hat{v})_+ \in H_0^1(E_{\hat{v}})$ gives the claim. Testing against $C_c^\infty(C)$ gives $-\Delta(u - \hat{v}) = 1$ weakly in C , and uniqueness makes this restriction the torsion function of C ; hence $T(C) = \int_C (u - \hat{v})$. As $u - \hat{v}$ is shared and the components are pairwise disjoint, torsion is *exactly additive*:

$$T(E_{\hat{v}}) = \int_{E_{\hat{v}}} (u - \hat{v}) = \int_{\tilde{E}} (u - \hat{v}) + \sum_{i \geq 2} \int_{C_i} (u - \hat{v}) = T(\tilde{E}) + \sum_{i \geq 2} T(C_i). \quad (36)$$

By the Saint-Venant inequality applied to each spillover component (Proposition 3.6, $T(B_E) - T(E) \geq 0$, with $T(B_{C_i}) = A_i^2/(8\pi)$),

$$\sum_{i \geq 2} T(C_i) \leq \sum_{i \geq 2} \frac{A_i^2}{8\pi} \leq \frac{1}{8\pi} \left(\sum_{i \geq 2} A_i \right)^2 = \frac{S^2}{8\pi}, \quad (37)$$

the middle inequality because $\sum A_i^2 \leq (\sum A_i)^2$ for non-negative terms. Combining (36)–(37) with $\mu_E \leq \mu(\hat{v})$,

$$T(B_{\tilde{E}}) - T(\tilde{E}) = \frac{\mu_E^2}{8\pi} - T(E_{\hat{v}}) + \sum_{i \geq 2} T(C_i) \leq \frac{\mu_E^2 - \mu(\hat{v})^2}{8\pi} + (T(B_{E_{\hat{v}}}) - T(E_{\hat{v}})) + \frac{S^2}{8\pi} \leq \delta_T + \frac{S^2}{8\pi},$$

where we used $T(B_{\tilde{E}}) = \mu_E^2/(8\pi)$, $T(B_{E_{\hat{v}}}) = \mu(\hat{v})^2/(8\pi)$, $\mu_E^2 - \mu(\hat{v})^2 \leq 0$, and the bound $T(B_{E_{\hat{v}}}) - T(E_{\hat{v}}) \leq \delta_T$ established in the proof of Lemma 4.5. Dividing by $T(B_{\tilde{E}}) = \mu_E^2/(8\pi)$ and using $\delta_T = \frac{\pi}{8} \delta_*(\Omega)$,

$$\delta_*(\tilde{E}) \leq \frac{8\pi\delta_T}{\mu_E^2} + \frac{S^2}{\mu_E^2} = \frac{\pi^2}{\mu_E^2} \delta_*(\Omega) + \frac{S^2}{\mu_E^2}. \quad (38)$$

This is the first inequality in (34) up to the higher-order term S^2/μ_E^2 . The latter is negligible: by (30), $S \leq A_0^2 \delta_T^2 / (16\pi^3 \tau^2)$, and with the standing hypothesis (29) (in particular $A_0 \delta_T / \tau \leq \pi/2$, i.e. $\delta_T / \tau \leq 1/32$) one has $S \leq \frac{A_0^2}{16\pi^3} (\delta_T / \tau) \cdot \delta_T \leq \frac{A_0^2}{16\pi^3 \cdot 32} \delta_T \leq \delta_T$; hence, with $\mu_E \geq \pi/4$,

$$\frac{S^2}{\mu_E^2} \leq \frac{S \cdot \delta_T}{(\pi/4)^2} \leq \frac{16}{\pi^2} \delta_T^2 \leq \frac{16}{\pi^2} \delta_T \cdot \frac{\pi}{8} \delta_*(\Omega) = \frac{2}{\pi} \delta_T \delta_*(\Omega) \leq \delta_*(\Omega),$$

the last step for $\delta_T \leq \delta_* \leq \pi/2$. Thus, with $\mu_E \geq \pi/4$, (38) gives $\delta_*(\tilde{E}) \leq \frac{\pi^2}{\mu_E^2} \delta_*(\Omega) + \delta_*(\Omega) \leq (16 + 1) \delta_*(\Omega) \leq C \delta_*(\Omega)$, which is (34).

(35). By Lemma 4.7 and Lemma 4.8,

$$|\Omega \Delta \tilde{E}| = (\pi - \mu(\hat{v})) + S \leq \left(8\pi\tau + A_0 \frac{\delta_T}{\tau} \right) + \frac{A_0^2}{16\pi^3} \frac{\delta_T^2}{\tau^2}.$$

For the final assertions, set $\tau = \sqrt{\delta_T}$: then $\delta_T/\tau = \sqrt{\delta_T}$, so $\delta_{\text{iso}}(\tilde{E}) \leq C\sqrt{\delta_T}$ by (33); the spillover term $\delta_T^2/\tau^2 = \delta_T$ is higher order, so $|\Omega \triangle \tilde{E}| \leq (8\pi + A_0)\sqrt{\delta_T} + C\delta_T \leq C\sqrt{\delta_T}$; and (34) is independent of τ . $\square$

4.8. Hole-filling: global exterior connectedness at no cost. Lemma 4.3 excludes complementary components of $\tilde{E}$ that are *compactly contained* in Ω , but it does not by itself rule out a bounded complementary component whose closure reaches $\partial\Omega$: such a component evades the interior minimum principle, which is available only away from $\partial\Omega$. The downstream geometry of Sections 5 and 6, however, needs the *global* topological fact that the unbounded complement of the set is its only complement — the exterior-connectedness convention of Lemma 5.2. We supply this unconditionally by passing to the *saturation* (hole-filling) of $\tilde{E}$, the standard device of filling in all bounded complementary components. The decisive quantitative point is that, by an isoperimetric argument identical in spirit to the spillover bound Lemma 4.8, the filled area is bounded by the *square* of the isoperimetric deficit, hence is of the same higher order as the spillover S ; all inset scales are therefore preserved.

Definition 4.11 (Saturation of the inset). The *hole-filled inset* is

$$E^h := \tilde{E} \cup \bigcup \left\{ K : K \text{ a bounded connected component of } \mathbb{R}^2 \setminus \overline{\tilde{E}} \right\}, \quad (39)$$

the union of $\tilde{E}$ with all bounded complementary components of its $\mathbb{R}^2$ -closure. We write $H := |E^h \setminus \tilde{E}| \geq 0$ for the filled area.

Lemma 4.12 (Hole-filling). *Under Definition 4.9 the hole-filled inset E^h of Definition 4.11 satisfies the following.*

- (a) E^h is bounded, open and connected; its complement $\mathbb{R}^2 \setminus \overline{E^h}$ is connected and unbounded (it is exactly the unbounded component of $\mathbb{R}^2 \setminus \overline{\tilde{E}}$), and $\partial E^h \subset \partial \tilde{E}$. Thus E^h is globally hole-free and satisfies the exterior-connectedness convention Lemma 5.2 ($\mathbb{R}^2 \setminus \overline{E^h}$ connected, with $\partial(\mathbb{R}^2 \setminus \overline{E^h}) = \partial E^h$) unconditionally, with no hypothesis on $\partial\Omega$.
- (b) $\tilde{E} \subset E^h$, $|E^h| = \mu_E + H$, and the perimeter does not increase:

$$P(E^h) \leq P(\tilde{E}). \quad (40)$$

- (c) (Key bound.) With $D_I(\tilde{E}) = P(\tilde{E})^2 - 4\pi\mu_E \geq 0$,

$$H \leq \frac{D_I(\tilde{E})^2}{64\pi^2\mu_E} \leq C D_I(\tilde{E})^2. \quad (41)$$

- (d) At the inset scales of Proposition 4.10 ($D_I(\tilde{E}) \leq C\delta_T/\tau$, $\mu_E \geq \pi/4$) one has $H \leq C(\delta_T/\tau)^2$, so at $\tau = \sqrt{\delta_T}$, $H \leq C\delta_T$ (higher order, the order of the spillover S). Consequently all the

inset controls transfer to E^h , with the same rates: writing $\hat{r}^h := \sqrt{|E^h|/\pi}$,

$$|\Omega \Delta E^h| \leq |\Omega \Delta \tilde{E}| + H \leq C \sqrt{\delta_T}, \quad \hat{r}^h = \hat{r} + O(H) = \hat{r} + O(\delta_T); \quad (42)$$

$$\delta_{\text{iso}}(E^h) \leq \delta_{\text{iso}}(\tilde{E}) \leq C \frac{\delta_T}{\tau}, \quad \text{Exc}(E^h) \leq \text{Exc}(\tilde{E}) + C H \leq C \frac{\delta_T}{\tau}; \quad (43)$$

$$a(E^h) \leq a(\tilde{E}) + C H \leq a(\tilde{E}) + C \delta_T, \quad (\text{so } a(E^h) \leq C(\frac{\delta_T}{\tau})^{1/2}); \quad (44)$$

$$\delta_*(E^h) \leq \delta_*(\tilde{E}) + \frac{C H}{\mu_E} \leq \delta_*(\tilde{E}) + C \delta_T \leq C \delta_*(\Omega), \quad (45)$$

where $a(F) := |F \Delta \hat{B}_{\hat{r}_F}(z_F)|$ is the ball distance of F at its optimal centre and $\hat{r}_F := \sqrt{|F|/\pi}$.

Proof. (a). The open set $\mathbb{R}^2 \setminus \tilde{E}$ has, as the complement of a bounded set, exactly one unbounded connected component U_∞ (nonempty since $\tilde{E}$ is bounded) and at most countably many bounded components $\{K_i\}_{i \in I}$, all open and disjoint; by (39), $E^h = \tilde{E} \cup \bigcup_i K_i$, so that as sets

$$E^h = \mathbb{R}^2 \setminus (\partial \tilde{E} \cup U_\infty), \quad \text{equivalently} \quad \overline{E^h} = \mathbb{R}^2 \setminus U_\infty.$$

We first record the elementary topological facts. Each component K of $\mathbb{R}^2 \setminus \tilde{E}$ (bounded or not) is open with $\partial K \subset \partial \tilde{E}$: a point $x \in \partial K$ lies neither in the open set K nor in the open set $\mathbb{R}^2 \setminus \tilde{E} \setminus K$ (a union of other components), so $x \in \tilde{E}$; and $x \notin \tilde{E}$, since $\tilde{E}$ open with $x \in \overline{K}$ would put a neighbourhood of x inside $\tilde{E}$, disjoint from K . Hence $x \in \partial \tilde{E}$. In particular $\partial U_\infty \subset \partial \tilde{E}$ and $\partial K_i \subset \partial \tilde{E}$.

E^h is open. Since $\tilde{E}$ and each K_i are open, it suffices to show every $x \in \partial \tilde{E} \cap E^h$ is interior to E^h ; but in fact no point of $\partial \tilde{E}$ lies in E^h , because $E^h = \tilde{E} \cup \bigcup_i K_i$ is a union of open sets disjoint from $\partial \tilde{E}$ (as $\partial \tilde{E} \cap \tilde{E} = \emptyset$ and $\partial \tilde{E} \cap K_i = \emptyset$, the latter since K_i is open and $\subset \mathbb{R}^2 \setminus \tilde{E}$). Thus E^h is a union of open sets, hence open. It is bounded since $\tilde{E}$ and all K_i lie in any ball B with $\mathbb{R}^2 \setminus B \subset U_\infty$ (such B exists as U_∞ contains the complement of a large ball).

E^h is connected. $\tilde{E}$ is connected (Definition 4.9, Lemma 4.3). For each i , $\partial K_i \subset \partial \tilde{E}$ is nonempty (a bounded open $K_i \subsetneq \mathbb{R}^2$ has nonempty boundary), so pick $x \in \partial K_i \subset \partial \tilde{E}$; then $x \in \overline{K_i} \cap \tilde{E}$, so $\tilde{E} \cup K_i$ is connected (two connected sets with intersecting closures, both inside the connected $\overline{\tilde{E} \cup K_i}$; concretely $\overline{K_i} \cap \tilde{E} \ni x$ joins them). All the $\tilde{E} \cup K_i$ share the connected $\tilde{E}$, so their union E^h is connected.

Exterior connectedness. By the displayed identity $\overline{E^h} = \mathbb{R}^2 \setminus U_\infty$, so $\mathbb{R}^2 \setminus \overline{E^h} = U_\infty$, which is connected and unbounded by construction. Moreover $\partial E^h = \overline{E^h} \setminus E^h = (\mathbb{R}^2 \setminus U_\infty) \setminus E^h \subset \partial \tilde{E}$ (the points of $\mathbb{R}^2 \setminus U_\infty$ not in $E^h = \tilde{E} \cup \bigcup_i K_i$ lie in $\partial \tilde{E}$, as $\mathbb{R}^2 = \tilde{E} \sqcup \partial \tilde{E} \sqcup (\mathbb{R}^2 \setminus \tilde{E})$ and $\mathbb{R}^2 \setminus \tilde{E} = U_\infty \sqcup \bigcup_i K_i$). Conversely $\partial U_\infty = \overline{U_\infty} \setminus U_\infty \subset \partial \tilde{E}$ lies in ∂E^h since these points are limits of $U_\infty = \mathbb{R}^2 \setminus \overline{E^h}$ and of E^h (being in $\partial \tilde{E}$, hence limits of $\tilde{E} \subset E^h$). Thus $\partial(\mathbb{R}^2 \setminus \overline{E^h}) = \partial U_\infty = \partial E^h$, and the trichotomy $\mathbb{R}^2 = E^h \sqcup \partial E^h \sqcup (\mathbb{R}^2 \setminus \overline{E^h})$ holds. This is exactly the exterior-connectedness convention of Lemma 5.2, established with no appeal to the minimum principle, hence with no hypothesis on $\partial \Omega$.

(b). Clearly $\tilde{E} \subset E^h$ and the K_i are pairwise disjoint and disjoint from $\tilde{E}$, so $|E^h| = \mu_E + \sum_i |K_i| = \mu_E + H$ with $H = \sum_i |K_i| \geq 0$; each K_i is bounded with finite perimeter $P(K_i) < \infty$, since $P(\tilde{E}) < \infty$ and $\partial^* K_i \subset \partial^* \tilde{E}$ $\mathcal{H}^1$ -a.e. (see below).

For the perimeter, we identify the essential boundaries. The plane is partitioned $\mathcal{H}^2$ -a.e. into the measure-theoretic interiors of the disjoint open sets $\tilde{E}, K_1, K_2, \dots, U_\infty$, and the perimeters add up along their shared interfaces. Concretely, write $F_\infty := U_\infty$, $F_0 := \tilde{E}$, $F_i := K_i$. Each pair of these sets is separated by the level set $\{u = \hat{v}\} \cap \Omega$ or sits across $\partial \tilde{E}$, and by the locality and structure of the reduced boundary [9, Thm. 16.3, §16.2] the essential boundary of $\tilde{E}$ decomposes $\mathcal{H}^1$ -a.e. as

$$\partial^* \tilde{E} = (\partial^* \tilde{E} \cap \partial^* U_\infty) \sqcup \bigsqcup_i (\partial^* \tilde{E} \cap \partial^* K_i) \quad (\mathcal{H}^1\text{-a.e.}), \quad (46)$$

since every point of $\partial^* \tilde{E}$ has, on its outer side, density-one membership in exactly one of the complementary components $U_\infty, K_1, K_2, \dots$. Filling the bounded components K_i removes from the boundary exactly the interfaces $\partial^* \tilde{E} \cap \partial^* K_i$ (now interior to E^h) and keeps the outer interface $\partial^* \tilde{E} \cap \partial^* U_\infty = \partial^* E^h$; moreover, since each K_i has *all* of its essential boundary shared with $\tilde{E}$ or with other filled components (its complement in $\mathbb{R}^2$ is $\tilde{E} \cup U_\infty \cup \bigcup_{j \neq i} K_j$, whose density-one points exhaust $\partial^* K_i$), filling adds no new reduced boundary. Hence $\partial^* E^h = \partial^* \tilde{E} \cap \partial^* U_\infty = \partial^* \tilde{E} \setminus \bigsqcup_i \partial^* K_i$ ($\mathcal{H}^1$ -a.e.), and by additivity of $\mathcal{H}^1$ over (46),

$$P(E^h) = \mathcal{H}^1(\partial^* E^h) = \mathcal{H}^1(\partial^* \tilde{E}) - \sum_i \mathcal{H}^1(\partial^* \tilde{E} \cap \partial^* K_i) = P(\tilde{E}) - \sum_i P(K_i), \quad (47)$$

the last equality because $\partial^* K_i \subset \partial^* \tilde{E}$ $\mathcal{H}^1$ -a.e. (every essential-boundary point of K_i adjacent to $\tilde{E}$ is one of $\tilde{E}$ adjacent to K_i ; the components K_i do not touch each other or U_∞ across a set of positive $\mathcal{H}^1$ -measure, being separated by $\tilde{E}$). In particular $P(E^h) \leq P(\tilde{E})$, which is (40).

(c). Let the filled holes K_i have areas $A_i^h := |K_i|$ and perimeters $p_i := P(K_i)$, so $H = \sum_i A_i^h$. By (b), filling removes exactly the boundaries of the holes, $P(E^h) = P(\tilde{E}) - \sum_i p_i$ (eq. (47)). By the planar isoperimetric inequality (Theorem 2.9) applied to each hole, $p_i \geq 2\sqrt{\pi A_i^h}$; and by superadditivity of $t \mapsto \sqrt{t}$ on non-negative terms,

$$\sum_i p_i \geq 2\sqrt{\pi} \sum_i \sqrt{A_i^h} \geq 2\sqrt{\pi} \sqrt{\sum_i A_i^h} = 2\sqrt{\pi H}. \quad (48)$$

On the other hand the isoperimetric inequality for E^h itself, of area $\mu_E + H$, gives $P(E^h) \geq 2\sqrt{\pi(\mu_E + H)}$. Substituting both into (47),

$$2\sqrt{\pi(\mu_E + H)} \leq P(E^h) = P(\tilde{E}) - \sum_i p_i \leq P(\tilde{E}) - 2\sqrt{\pi H},$$

so that

$$P(\tilde{E}) \geq 2\sqrt{\pi} \left(\sqrt{\mu_E + H} + \sqrt{H} \right).$$

Squaring and using $\sqrt{(\mu_E + H)H} \geq \sqrt{\mu_E H}$,

$$P(\tilde{E})^2 \geq 4\pi \left(\mu_E + 2H + 2\sqrt{H(\mu_E + H)} \right) \geq 4\pi\mu_E + 8\pi\sqrt{H\mu_E},$$

whence $D_I(\tilde{E}) = P(\tilde{E})^2 - 4\pi\mu_E \geq 8\pi\sqrt{H\mu_E}$, i.e.

$$H \leq \frac{D_I(\tilde{E})^2}{64\pi^2\mu_E},$$

which is (41); the second inequality there uses $\mu_E \geq \pi/4$ (Lemma 4.8), giving the absolute constant $C = 1/(16\pi^3)$.

(d). By (33) and its proof, $D_I(\tilde{E}) \leq D_I(\hat{v}) + 4\pi S \leq 2D_I(\hat{v}) \leq 2A_0 \delta_T/\tau$, so (41) gives $H \leq C(\delta_T/\tau)^2$, and at $\tau = \sqrt{\delta_T}$, $H \leq C\delta_T$, of the same higher order as the spillover S of Lemma 4.8. The consequences follow:

Closeness and radius. Since $\tilde{E} \subset E^h$ and $\tilde{E} \subset \Omega$ (so $|\tilde{E} \setminus \Omega| = 0$), we split the symmetric difference: $|\Omega \setminus E^h| \leq |\Omega \setminus \tilde{E}|$ (as $\tilde{E} \subset E^h$ shrinks the part of Ω outside), while $|E^h \setminus \Omega| \leq |E^h \setminus \tilde{E}| + |\tilde{E} \setminus \Omega| = H + 0 = H$. Hence

$$|\Omega \Delta E^h| = |\Omega \setminus E^h| + |E^h \setminus \Omega| \leq |\Omega \setminus \tilde{E}| + H = |\Omega \Delta \tilde{E}| + H \leq C\sqrt{\delta_T} + C\delta_T \leq C\sqrt{\delta_T},$$

which is (42); no assertion that $E^h \subset \Omega$ is needed. The radius obeys $\hat{r}^h = \sqrt{(\mu_E + H)/\pi} = \hat{r}\sqrt{1 + H/\mu_E} = \hat{r} + O(H/\hat{r}) = \hat{r} + O(\delta_T)$ by $H \leq C\delta_T$ and $\hat{r} \geq 1/2$.

Isoperimetric deficit. Since $P(E^h) \leq P(\tilde{E})$ (40) and $|E^h| = \mu_E + H \geq \mu_E$,

$$\delta_{\text{iso}}(E^h) = \frac{P(E^h)}{2\sqrt{\pi|E^h|}} - 1 \leq \frac{P(\tilde{E})}{2\sqrt{\pi\mu_E}} - 1 = \delta_{\text{iso}}(\tilde{E}) \leq C \frac{\delta_T}{\tau},$$

which is the first bound in (43); and $\text{Exc}(E^h) = P(E^h) - 2\pi\hat{r}^h \leq P(\tilde{E}) - 2\pi\hat{r} + 2\pi(\hat{r} - \hat{r}^h) \leq \text{Exc}(\tilde{E}) + CH \leq C\delta_T/\tau$, using $|\hat{r} - \hat{r}^h| \leq CH$.

Ball distance. For any common centre, $|E^h \Delta B_{\hat{r}^h}(z)| \leq |\tilde{E} \Delta B_{\hat{r}}(z)| + |E^h \setminus \tilde{E}| + |B_{\hat{r}^h}(z) \Delta B_{\hat{r}}(z)| \leq a(\tilde{E}) + H + C|\hat{r}^h - \hat{r}| \leq a(\tilde{E}) + CH$; minimising over z gives $a(E^h) \leq a(\tilde{E}) + CH \leq a(\tilde{E}) + C\delta_T$, which is (44), and the parenthetical rate follows from $a(\tilde{E}) \leq C(\delta_T/\tau)^{1/2}$ and $\delta_T \leq (\delta_T/\tau)^{1/2}$ (as $\delta_T/\tau \leq 1$).

Scale-invariant deficit. Torsional rigidity is monotone under set inclusion: from the variational characterisation $T(F) = \max \{2 \int_F v - \int_F |\nabla v|^2 : v \in H_0^1(F)\}$ (attained at the torsion function of F) and the inclusion $H_0^1(\tilde{E}) \subset H_0^1(E^h)$ (extension by zero, valid as $\tilde{E} \subset E^h$ are open), one has $T(E^h) \geq T(\tilde{E})$ for $\tilde{E} \subset E^h$. Hence, with $|E^h| = \mu_E + H$,

$$1 - \delta_*(E^h) = \frac{8\pi T(E^h)}{|E^h|^2} \geq \frac{8\pi T(\tilde{E})}{(\mu_E + H)^2} = \frac{1 - \delta_*(\tilde{E})}{(1 + H/\mu_E)^2},$$

so

$$\delta_*(E^h) \leq 1 - \frac{1 - \delta_*(\tilde{E})}{(1 + H/\mu_E)^2} \leq \delta_*(\tilde{E}) + \left(1 - \frac{1}{(1 + H/\mu_E)^2}\right) \leq \delta_*(\tilde{E}) + \frac{2H}{\mu_E},$$

using $1 - (1+t)^{-2} \leq 2t$ for $t \geq 0$. With $\mu_E \geq \pi/4$ and $H \leq C\delta_T$ this is $\delta_*(E^h) \leq \delta_*(\tilde{E}) + C\delta_T \leq C\delta_*(\Omega)$ by (34) and $\delta_T = \frac{\pi}{8}\delta_*(\Omega)$, which is (45). $\square$

Remark 4.13 (The hole-filled inset is the object of Sections 5 and 6). Lemma 4.12 resolves, unconditionally, the global exterior-connectedness property required by the trapping annulus and the inscribed-ball lemma: the hole-filled inset E^h is connected and satisfies $\mathbb{R}^2 \setminus \overline{E^h}$ connected, with all the inset scales of Proposition 4.10 preserved up to an additive $O(\delta_T)$, of the same higher order as the spillover. From Section 5 onward the geometric reduction is therefore applied to E^h in place of $\tilde{E}$; the only change in any estimate is the replacement of μ_E by $\mu_E + H = \mu_E + O(\delta_T)$ and of $\hat{r}, \delta_{\text{iso}}, \text{Exc}, a, \delta_*$ by their E^h -counterparts, with identical rates. The mass H added by filling, like the spillover S subtracted by selecting the largest component, is quadratic in the isoperimetric deficit and hence negligible at the sharp scale.

Remark 4.14 (Self-contained deficit transfer; no forward dependence). The bound (34) is proved above without any appeal to the cutting principle Lemma 7.4 of Section 7: the passage from $E_{\hat{v}}$ to its largest component $\tilde{E}$ is handled by the exact additivity of torsion over components (36) and the Saint-Venant bound (37) on the spillover torsion, both internal to Sections 3 and 4. There is therefore *no* circular dependence on Section 7 in this section. (One could alternatively invoke Lemma 7.4; but its precise hypothesis — that the excised set lie in a thin concentric collar of $\tilde{E}$ — is *not* met by the spillover, whose components are in general not contained in the trapping annulus of $\tilde{E}$. The additivity argument avoids this issue entirely, which is why we prefer it.) For the record, Lemma 7.4 itself rests only on Green-function monotonicity and a boundary-layer barrier, so even the optional route would be acyclic.

Remark 4.15 (The scales, recorded). With $\tau = \sqrt{\delta_T}$ (Theorem 9.1) the inset $\tilde{E}$ satisfies $\delta_{\text{iso}} = \delta_{\text{iso}}(\tilde{E}) \leq C\sqrt{\delta_T}$ and $\text{Exc} = 2\pi\hat{r}\delta_{\text{iso}} \leq C\sqrt{\delta_T}$. Feeding δ_{iso} into the sharp quantitative isoperimetric inequality (Theorem 2.11) gives the ball-distance $a := |\tilde{E} \Delta B_{\hat{r}}(z)| \leq C_F \sqrt{\delta_{\text{iso}}} \mu_E \leq C\delta_T^{1/4}$ at the optimal centre z . The trapping width then satisfies $\eta \leq C(\text{Exc} + a)^{1/2} \leq C\delta_T^{1/8}$ (Theorem 5.10), an *absolute* small amplitude. These are the inputs to the annulus and fold sections; the deficit $\delta_*(\tilde{E}) \leq C\delta_*(\Omega)$ and the closeness $|\Omega \Delta \tilde{E}| \leq C\sqrt{\delta_T}$ are the inputs to the conclusion. We emphasise that closeness is carried by the *level* $\hat{v} \sim \sqrt{\delta_T}$ through Lemma 4.7, not by the amplitude η ; and that the deficit transfer Lemma 4.5 delivers $\delta_*(\tilde{E}) \lesssim \delta_*(\Omega)$ without recourse to the inflated single-level deficits.

5. THE TRAPPING ANNULUS

In this section we prove that the connected inset $\tilde{E}$ produced in Section 4 is trapped between two concentric circles whose radii differ by an *absolutely small* amount, controlled by the perimeter excess and the L^1 distance to a ball. The decisive point is that the width of the trapping annulus is bounded *without any second-order information* on $\partial\tilde{E}$: no interior or exterior ball condition, no C^2 regularity, no Magnanini–Poggesi oscillation estimate. The only inputs are the connectedness and the no-hole structure of the inset, the planar isoperimetric deficit, and the closeness to a ball furnished by Theorem 2.11.

Remark 5.1 (The inset is hole-filled from here on). From this section onward $\tilde{E}$ denotes the hole-filled inset E^h of Lemma 4.12: the saturation of the largest super-level component, with all bounded complementary components filled in. By Lemma 4.12(a) this set is bounded, open and connected, and globally hole-free — $\mathbb{R}^2 \setminus \tilde{E}$ is connected with $\partial(\mathbb{R}^2 \setminus \tilde{E}) = \partial\tilde{E}$ — *unconditionally*, with no hypothesis on $\partial\Omega$. By Lemma 4.12(b) its boundary is contained in that of the super-level component, $\partial\tilde{E} \subset \{u = \hat{v}\} \subset \Omega$, so it remains an $\mathcal{H}^1$ -rectifiable curve in the open set Ω , and by Lemma 4.12(d) it inherits *every* inset scale (μ_E , $\hat{r}$, δ_{iso} , Exc , a , δ_*) of Proposition 4.10 with the same rates, the filled mass $H \leq C\delta_T$ being of the same higher order as the spillover. All estimates below depend on $\tilde{E}$ only through these scales and through its connectedness and global hole-freeness, so they hold verbatim for the hole-filled inset. We keep the symbol $\tilde{E}$ to avoid notational clutter; nothing in the geometric reduction of Sections 5 to 7 is sensitive to the replacement.

The mechanism is purely geometric: the boundary of a connected, hole-free set must cross every circle of intermediate radius at least twice; summed against the radial coordinate through the

coarea formula, this forces a lower bound on the *angular* part of the perimeter, which a single Cauchy–Schwarz inequality converts into the deficit. We work throughout with the *measure-theoretic representative* of $\tilde{E}$ described in Theorem 2.13 and Lemmas 2.18, 4.1 and 4.12, in accordance with the standing convention of Section 4.1: $\hat{v}$ is a regular value of the (real-analytic, in Ω) torsion function u , and by Remark 5.1 $\tilde{E}$ is the hole-filled inset, a bounded open connected subset of Ω whose boundary lies in the level set $\{u = \hat{v}\}$ along which $\nabla u \neq 0$. Its topological boundary

$$\partial\tilde{E} \subset \{u = \hat{v}\} \subset \Omega$$

lies in the open set Ω and is an $\mathcal{H}^1$ -rectifiable curve along which $\nabla u \neq 0$; by Lemma 4.1(ii) the closure $\overline{\tilde{E}} \cap \Omega$ taken in Ω is a compact subset of Ω , and the only way $\overline{\tilde{E}}$ (closure in $\mathbb{R}^2$) could meet $\partial\Omega$ is on a set carrying no perimeter of $\tilde{E}$. We therefore use $\overline{\tilde{E}} \subset \subset \Omega$ as shorthand for this compactness in Ω of the representative, which is all the arguments below require; no regularity of $\partial\Omega$ is assumed, and for the punctured disc (Section 4.1) the apparent accumulation of $\overline{\tilde{E}}$ at the polar puncture is invisible to every quantity used. With this reading $\partial\tilde{E}$ coincides $\mathcal{H}^1$ -a.e. with the reduced and the essential boundary, and $\mathcal{H}^1(\partial\tilde{E}) = P(\tilde{E})$. The plane decomposes as the disjoint union

$$\mathbb{R}^2 = \tilde{E} \sqcup \partial\tilde{E} \sqcup (\mathbb{R}^2 \setminus \overline{\tilde{E}}), \quad (49)$$

the first and last sets being open. We record the precise topological consequence of the no-hole property that we shall use.

Lemma 5.2 (Exterior connectedness convention). *In the geometric annulus theorem below, the phrase “ E has no holes” means that $U := \mathbb{R}^2 \setminus \overline{E}$ is connected and $\partial U = \partial E$ for the chosen representative. Under this convention the set U used in Lemma 5.7 is connected by hypothesis.*

Proof. This is the convention used in the purely geometric part of the section. The identity $\partial U = \partial E$ is automatic for representatives satisfying the trichotomy (49) and $E = \text{int } \overline{E}$. $\square$

Remark 5.3 (Global exterior connectedness is supplied by hole-filling). Lemma 4.3 proves only that a super-level component of the torsion function has no complementary component compactly contained in Ω , which is weaker than the global exterior-connectedness convention above: a bounded hole might in principle touch $\partial\Omega$ and so evade the interior minimum-principle argument. This is exactly why we pass to the hole-filled inset of Remark 5.1: by Lemma 4.12(a) the saturation $\tilde{E} = E^h$ has $\mathbb{R}^2 \setminus \overline{\tilde{E}}$ connected with $\partial(\mathbb{R}^2 \setminus \overline{\tilde{E}}) = \partial\tilde{E}$ *unconditionally*, so it satisfies Lemma 5.2 with no hypothesis on $\partial\Omega$ and no appeal to the minimum principle. The annulus theorem Theorem 5.10 and the inscribed-ball lemma Lemma 5.13 below therefore apply to the hole-filled inset directly, and Lemma 4.12(d) guarantees that the deficit and closeness controls are preserved at the sharp scale (the filled mass being quadratic in the isoperimetric deficit). All statements of this section are accordingly *unconditional*.

Throughout, $z \in \mathbb{R}^2$ denotes the centre of an optimal ball realising the Fraenkel asymmetry of $\tilde{E}$ — the *exact* L^1 -optimal centre. This z is an interior point of $\tilde{E}$, $z \in \tilde{E}$, with a definite inscribed ball; this is *proved* for the inset, with the sharp rate and an absolute constant, in Lemma 5.13 below. (The proof is deferred to the end of the section because it reuses the trapping mechanism, but it removes any hypothesis on the location of z .) With $\hat{r} := \sqrt{|\tilde{E}|/\pi} \in [\frac{1}{2}, 1]$ (the bounds follow from

$\pi/4 \leq |\tilde{E}| \leq \pi$, Lemma 4.8)

$$a := |\tilde{E} \triangle B_{\hat{r}}(z)| = \min_{x_0 \in \mathbb{R}^2} |\tilde{E} \triangle B_{\hat{r}}(x_0)| = \mathcal{A}(\tilde{E}) |\tilde{E}|. \quad (50)$$

We write $e_r := (x - z)/|x - z|$ for the radial unit field (defined on $\mathbb{R}^2 \setminus \{z\}$), $e_\theta := e_r^\perp$ for the orthogonal angular field, ν for the outward unit normal on the rectifiable boundary, and

$$\begin{aligned} \nu_r &:= \nu \cdot e_r, & \nu_\theta &:= \nu \cdot e_\theta, & \nu_r^2 + \nu_\theta^2 &= 1 \quad (\mathcal{H}^1\text{-a.e. on } \partial\tilde{E}), \\ P_r &:= \int_{\partial\tilde{E}} |\nu_r| d\mathcal{H}^1, & P_\theta &:= \int_{\partial\tilde{E}} |\nu_\theta| d\mathcal{H}^1, & 0 &\leq P_r, P_\theta \leq P(\tilde{E}). \end{aligned}$$

Finally we record the standing scaling assumption of the construction, $\delta_T \leq \delta_\star$, together with the deficit scales carried in from Section 4 (here τ is the level-selection parameter of that section, kept symbolic until the optimisation in Section 9):

$$\text{Exc} := P(\tilde{E}) - 2\pi\hat{r} = 2\pi\hat{r} \delta_{\text{iso}}(\tilde{E}) \leq C \frac{\delta_T}{\tau}, \quad a \leq C_F \sqrt{\delta_{\text{iso}}(\tilde{E})} |\tilde{E}| \leq C \left(\frac{\delta_T}{\tau} \right)^{1/2}, \quad (51)$$

the first from Lemma 4.8, the second from the planar quantitative isoperimetric inequality Theorem 2.11. In particular $\text{Exc} + a \leq C(\delta_T/\tau)^{1/2}$ for $\delta_T/\tau \leq 1$, so the trapping width produced below will scale like $(\delta_T/\tau)^{1/4}$.

5.1. The inner and outer radii.

Definition 5.4 (Inner and outer radius). For the optimal centre $z \in \tilde{E}$ set

$$\rho_i := \sup \left\{ r > 0 : B_r(z) \subseteq \tilde{E} \right\}, \quad \rho_e := \inf \left\{ r > 0 : \tilde{E} \subseteq B_r(z) \right\}. \quad (52)$$

We will use Definition 5.4 under the hypothesis $z \in \tilde{E}$, which guarantees $\rho_i > 0$; that $z \in \tilde{E}$ holds in the application is proved by Lemma 5.13 (using the global hole-freeness of the hole-filled inset, Lemma 4.12(a)). The next lemma collects the elementary properties of these radii; they are the rigorous substitute for the (here unavailable) minimum and maximum of $|x - z|$ over a smooth boundary.

Lemma 5.5 (Basic properties of ρ_i, ρ_e). *Assume $z \in \tilde{E}$. Then $0 < \rho_i \leq \rho_e < \infty$ and*

$$B_{\rho_i}(z) \subseteq \tilde{E} \subseteq \overline{B_{\rho_e}(z)}. \quad (53)$$

Moreover $\rho_i = \min_{x \in \partial\tilde{E}} |x - z|$ and $\rho_e = \max_{x \in \partial\tilde{E}} |x - z| = \max_{x \in \tilde{E}} |x - z|$; both extrema are attained, and

$$\rho_i \leq \hat{r} \leq \rho_e. \quad (54)$$

Proof. Since $z \in \tilde{E}$ and $\tilde{E}$ is open, some $B_r(z) \subseteq \tilde{E}$, so $\rho_i > 0$; since $\tilde{E}$ is bounded, $\rho_e < \infty$, and $B_r(z) \subseteq \tilde{E} \subseteq B_{r'}(z)$ forces $r \leq r'$, whence $\rho_i \leq \rho_e$. For the left inclusion in (53), $B_{\rho_i}(z) = \bigcup_{r < \rho_i} B_r(z) \subseteq \tilde{E}$. For the right inclusion, $\tilde{E} \subseteq \overline{B_{\rho_e}(z)}$ because $\tilde{E} \subseteq B_r(z)$ for every $r > \rho_e$.

Set $\rho'_i := \min \left\{ |x - z| : x \in \partial\tilde{E} \right\}$ and $\rho'_e := \max \left\{ |x - z| : x \in \partial\tilde{E} \right\}$, both attained since $\partial\tilde{E}$ is compact and nonempty.

The outer radius. First, $\rho_e = \sup_{x \in \tilde{E}} |x - z|$: indeed $\tilde{E} \subseteq B_r(z)$ holds iff $r > \sup_{x \in \tilde{E}} |x - z|$, so the infimum defining ρ_e equals this supremum. We claim this supremum equals ρ'_e , the maximal boundary radius. On one hand $\rho'_e \leq \sup_{\tilde{E}} |x - z|$: a boundary point w of radius ρ'_e is a limit of

points of $\tilde{E}$, whose radii approach ρ'_e . On the other hand $\sup_{\tilde{E}} |x - z| \leq \rho'_e$: suppose some $x \in \tilde{E}$ had $|x - z| > \rho'_e$. Extend the ray from z through x to a far point y with $y \notin \tilde{E}$ (possible, $\tilde{E}$ being bounded). The set $\{t \in [0, 1] : (1 - t)x + ty \in \tilde{E}\}$ is closed, contains 0 but not 1; its supremum t^* gives a point $p = (1 - t^*)x + t^*y \in \tilde{E}$ with all nearby points further out on the ray outside $\tilde{E}$, so $p \in \partial\tilde{E}$. As p lies on the outward ray beyond x , $|p - z| \geq |x - z| > \rho'_e$, contradicting the maximality of ρ'_e . Hence $\rho_e = \sup_{\tilde{E}} |x - z| = \rho'_e$ is attained on $\partial\tilde{E}$, and (53) gives $\tilde{E} \subseteq \overline{B_{\rho_e}(z)}$.

The inner radius. No boundary point has radius $< \rho_i$, because the open ball $B_{\rho_i}(z) \subseteq \tilde{E}$ contains no point of $\partial\tilde{E}$; thus $\rho'_i \geq \rho_i$. Conversely $\rho'_i \leq \rho_i$: the open ball $B_{\rho'_i}(z)$ contains no boundary point (all of $\partial\tilde{E}$ has radius $\geq \rho'_i$), so by (49) it is partitioned by the two disjoint open sets $\tilde{E}$ and $\mathbb{R}^2 \setminus \tilde{E}$; being connected and containing $z \in \tilde{E}$, it lies entirely in $\tilde{E}$, whence $\rho_i \geq \rho'_i$ by definition of ρ_i . Hence $\rho_i = \rho'_i$, attained at some $w_i \in \partial\tilde{E}$ with $|w_i - z| = \rho_i$.

Position of $\hat{r}$. Finally $B_{\rho_i}(z) \subseteq \tilde{E}$ gives $\pi\rho_i^2 \leq |\tilde{E}| = \pi\hat{r}^2$, and $\tilde{E} \subseteq \overline{B_{\rho_e}(z)}$ gives $\pi\hat{r}^2 = |\tilde{E}| \leq \pi\rho_e^2$, which is (54). $\square$

5.2. The angular coarea identity. For $\rho > 0$ let

$$N(\rho) := \mathcal{H}^0(\partial\tilde{E} \cap \partial B_\rho(z)) \in \{0, 1, 2, \dots, \infty\} \quad (55)$$

be the number of boundary points on the circle of radius ρ about z .

Lemma 5.6 (Angular coarea identity). *With P_θ and N as above,*

$$P_\theta = \int_{\partial\tilde{E}} |\nu_\theta| d\mathcal{H}^1 = \int_0^\infty N(\rho) d\rho. \quad (56)$$

Proof. The radial function $f(x) := |x - z|$ is 1-Lipschitz on $\mathbb{R}^2$, with $\nabla f = e_r$ on $\mathbb{R}^2 \setminus \{z\}$. Let $\nabla^{\partial\tilde{E}} f$ denote its tangential gradient along the rectifiable curve $\partial\tilde{E}$, i.e. the orthogonal projection of ∇f onto the (approximate) tangent line, which $\mathcal{H}^1$ -a.e. is the line $\mathbb{R}\nu^\perp$. Writing $\nu = \nu_r e_r + \nu_\theta e_\theta$ and decomposing $e_r = (e_r \cdot \nu)\nu + (e_r \cdot \nu^\perp)\nu^\perp$,

$$|\nabla^{\partial\tilde{E}} f| = |e_r \cdot \nu^\perp| = \sqrt{1 - (e_r \cdot \nu)^2} = \sqrt{1 - \nu_r^2} = |\nu_\theta| \quad (\mathcal{H}^1\text{-a.e.}).$$

The coarea formula for Lipschitz functions on $\mathcal{H}^1$ -rectifiable sets (Theorem 2.6; see also [1, Theorem 2.93] and [5, §3.4.2]) gives

$$\int_{\partial\tilde{E}} |\nabla^{\partial\tilde{E}} f| d\mathcal{H}^1 = \int_0^\infty \mathcal{H}^0(\partial\tilde{E} \cap f^{-1}(\rho)) d\rho = \int_0^\infty N(\rho) d\rho,$$

which is (56). $\square$

We record the companion *weighted* identity, used once below. Let $\Theta : \mathbb{R}^2 \setminus \{z\} \rightarrow S^1$, $\Theta(x) = (x - z)/|x - z|$, be the angular projection (well defined on $\partial\tilde{E}$ since $z \in \tilde{E}$ is interior). A computation identical to the one above, now for Θ , gives $|\nabla^{\partial\tilde{E}} \Theta| = |\nu_r|/|x - z|$, so the coarea formula with the weight $g(x) = |x - z|$ yields

$$P_r = \int_{\partial\tilde{E}} |\nu_r| d\mathcal{H}^1 = \int_{\partial\tilde{E}} |x - z| |\nabla^{\partial\tilde{E}} \Theta| d\mathcal{H}^1 = \int_{S^1} \left(\sum_{x \in \partial\tilde{E} \cap \text{ray}(\theta)} |x - z| \right) d\theta, \quad (57)$$

where $\text{ray}(\theta) := \{z + \rho e^{i\theta} : \rho > 0\}$. (We only use the elementary consequence (60) below.)

5.3. Two crossings at every intermediate radius.

Lemma 5.7 ($N \geq 2$ on the trapping interval). *Assume $z \in \tilde{E}$, and recall $U := \mathbb{R}^2 \setminus \tilde{\tilde{E}}$ is connected with $\partial U = \partial \tilde{E}$ (Lemma 5.2, from the global hole-freeness of the hole-filled inset Lemma 4.12(a)). Then*

$$N(\rho) \geq 2 \quad \text{for every } \rho \in (\rho_i, \rho_e). \quad (58)$$

Proof. Fix $\rho \in (\rho_i, \rho_e)$ and let $C_\rho := \partial B_\rho(z)$, a circle (homeomorphic to S^1 , hence connected). By (49) every point of C_ρ lies in exactly one of the open sets $\tilde{E}$, $U := \mathbb{R}^2 \setminus \tilde{\tilde{E}}$, or in $\partial \tilde{E}$. We show that C_ρ meets both $\tilde{E}$ and U ; the conclusion then follows from a connectedness count.

Step 1: C_ρ meets $\tilde{E}$. The function $f(x) = |x - z|$ is continuous on the connected open set $\tilde{E}$ (open connected subsets of $\mathbb{R}^2$ are path-connected), so $f(\tilde{E})$ is an interval. It contains 0 (as $z \in \tilde{E}$, take $f(z) = 0$) and, by Lemma 5.5, values arbitrarily close to ρ_e from below (points of $\tilde{E}$ of radius $\rightarrow \rho_e$). Hence $f(\tilde{E}) \supseteq [0, \rho_e) \ni \rho$, so some point of $\tilde{E}$ has radius ρ , i.e. $C_\rho \cap \tilde{E} \neq \emptyset$.

Step 2: C_ρ meets U . By Lemma 5.5 there is a boundary point $w_i \in \partial \tilde{E}$ with $|w_i - z| = \rho_i$. For the good representative the topological boundaries coincide, $\partial U = \partial \tilde{\tilde{E}} = \partial \tilde{E}$, so $w_i \in \bar{U}$, and every neighbourhood of w_i meets the open set U ; thus U contains points of radius arbitrarily close to ρ_i , all of radius $\geq \rho_i$ since $B_{\rho_i}(z) \subseteq \tilde{E}$. Hence $\inf_U f = \rho_i$ (not necessarily attained). On the other hand U is unbounded (it is the complement of the bounded set $\tilde{\tilde{E}}$), so $\sup_U f = \infty$. Since U is connected and open, hence path-connected, $f(U)$ is an interval with infimum ρ_i and supremum ∞ , so $f(U) \supseteq (\rho_i, \infty) \ni \rho$. Therefore some point of U has radius ρ , i.e. $C_\rho \cap U \neq \emptyset$.

Step 3: at least two boundary crossings. Suppose, for contradiction, that $\mathcal{H}^0(C_\rho \cap \partial \tilde{E}) \leq 1$. Then $C_\rho \setminus \partial \tilde{E}$ is C_ρ with at most one point removed, hence connected. By (49), $C_\rho \setminus \partial \tilde{E} = (C_\rho \cap \tilde{E}) \sqcup (C_\rho \cap U)$ is a partition into two disjoint sets that are open in C_ρ and, by Steps 1–2, both nonempty. A connected topological space admits no such partition—a contradiction. Hence $N(\rho) = \mathcal{H}^0(C_\rho \cap \partial \tilde{E}) \geq 2$. $\square$

Remark 5.8 (Where connectedness and no-holes enter). Connectedness of $\tilde{E}$ is used in Step 1 to drive the radius continuously from 0 to near ρ_e ; the no-hole hypothesis (connectedness of $U = \mathbb{R}^2 \setminus \tilde{\tilde{E}}$) is used in Step 2 to drive the *exterior* radius continuously from near ρ_i out to ∞ . If $\tilde{E}$ had a hole, the bounded complementary component could sit inside a circle C_ρ without forcing a second crossing, and the argument would fail. The bound (58) holds for *every* $\rho \in (\rho_i, \rho_e)$, not merely almost every ρ ; the a.e. caveat enters only through the coarea formula (56), where $N(\rho)$ is finite for a.e. ρ because $P_\theta \leq P(\tilde{E}) < \infty$.

5.4. The radial-profile bound. We now bound the angular defect $P - P_r$ of the perimeter by the geometric quantities Exc and a . Define the *outer radial profile*

$$R(\theta) := \sup \left\{ \rho > 0 : z + \rho e^{i\theta} \in \tilde{E} \right\} \in [0, \rho_e] \quad (\theta \in S^1). \quad (59)$$

The function R is measurable: for each $t > 0$,

$$\{\theta : R(\theta) > t\} = \bigcup_{\substack{q \in \mathbb{Q} \\ q > t}} \{\theta : z + q e^{i\theta} \in \tilde{E}\},$$

and the sets in the union are open because $\tilde{E}$ is open and $\theta \mapsto z + q e^{i\theta}$ is continuous. Because $z \in \tilde{E}$ and $\tilde{E}$ is bounded and open, the ray $\text{ray}(\theta)$ starts inside $\tilde{E}$ and leaves it, so it meets $\partial \tilde{E}$; for a.e. θ

the meeting set is finite with outermost radius equal to $R(\theta)$, and

$$P_r = \int_{S^1} \left(\sum_{x \in \partial \tilde{E} \cap \text{ray}(\theta)} |x - z| \right) d\theta \geq \int_{S^1} R(\theta) d\theta, \quad (60)$$

the equality being (57) and the inequality holding because the inner sum is at least its largest term $R(\theta)$.

Lemma 5.9 (Radial-profile bound). *Assume $z \in \tilde{E}$. Then*

$$P(\tilde{E}) - P_r \leq \text{Exc} + \frac{2a}{\hat{r}} \leq \text{Exc} + 4a. \quad (61)$$

Proof. For each θ with $R(\theta) < \hat{r}$, every point $z + \rho e^{i\theta}$ with $R(\theta) < \rho < \hat{r}$ has radius exceeding $R(\theta)$, hence lies *outside* $\tilde{E}$ (by definition of R), while having radius $< \hat{r}$, hence inside $B_{\hat{r}}(z)$; thus the radial segment $\{R(\theta) < \rho < \hat{r}\}$ along direction θ lies in $B_{\hat{r}}(z) \setminus \tilde{E}$. Integrating in polar coordinates and using $\int_R^{\hat{r}} \rho d\rho = \frac{1}{2}(\hat{r}^2 - R^2) = \frac{1}{2}(\hat{r} - R)(\hat{r} + R) \geq \frac{\hat{r}}{2}(\hat{r} - R)$ (valid as $R \geq 0$),

$$a \geq |B_{\hat{r}}(z) \setminus \tilde{E}| \geq \int_{\{R < \hat{r}\}} \int_{R(\theta)}^{\hat{r}} \rho d\rho d\theta \geq \frac{\hat{r}}{2} \int_{S^1} (\hat{r} - R(\theta))_+ d\theta,$$

where $|B_{\hat{r}}(z) \setminus \tilde{E}| \leq |\tilde{E} \triangle B_{\hat{r}}(z)| = a$ by (50). Hence $\int_{S^1} (\hat{r} - R)_+ d\theta \leq 2a/\hat{r}$, and since $R \geq \hat{r} - (\hat{r} - R)_+$,

$$\int_{S^1} R(\theta) d\theta \geq 2\pi\hat{r} - \int_{S^1} (\hat{r} - R)_+ d\theta \geq 2\pi\hat{r} - \frac{2a}{\hat{r}}.$$

Combining with (60) and $P(\tilde{E}) = 2\pi\hat{r} + \text{Exc}$,

$$P(\tilde{E}) - P_r \leq P(\tilde{E}) - \int_{S^1} R d\theta \leq (2\pi\hat{r} + \text{Exc}) - \left(2\pi\hat{r} - \frac{2a}{\hat{r}}\right) = \text{Exc} + \frac{2a}{\hat{r}}.$$

The final inequality in (61) uses $\hat{r} \geq \frac{1}{2}$. □

5.5. The trapping theorem.

Theorem 5.10 (Trapping annulus). *Let $E \subseteq \mathbb{R}^2$ be a bounded, open, connected set, equal to the good representative described above (so ∂E is the topological boundary, $\mathcal{H}^1$ -rectifiable with $\mathcal{H}^1(\partial E) = P(E)$, and the trichotomy (49) holds), with no holes ($\mathbb{R}^2 \setminus \bar{E}$ connected), area $|E| = \pi\hat{r}^2$, isoperimetric deficit $\delta_{\text{iso}}(E)$, perimeter excess $\text{Exc} = P(E) - 2\pi\hat{r} = 2\pi\hat{r}\delta_{\text{iso}}(E)$, and ball distance $a = |E \triangle B_{\hat{r}}(z)|$ at the optimal centre z . Assume $z \in E$, and let ρ_i, ρ_e be as in (52). Then*

$$B_{\rho_i}(z) \subseteq E \subseteq \overline{B_{\rho_e}(z)}, \quad \rho_i \leq \hat{r} \leq \rho_e, \quad 4\eta^2 \leq 2(2\pi\hat{r} + \text{Exc}) \left(\text{Exc} + \frac{2a}{\hat{r}} \right), \quad (62)$$

where $\eta := \rho_e - \rho_i$. In particular, if $\hat{r} \in [\frac{1}{2}, 1]$ and $\text{Exc} \leq C_0$, then $\eta \leq C(C_0)(\text{Exc} + a)^{1/2}$. For the hole-filled inset $\tilde{E} = E^h$ of Section 4 (Remark 5.1; $\delta_T \leq \delta_*$, $\hat{r} \in [\frac{1}{2}, 1]$, and the scales (51)), which is connected and globally hole-free by Lemma 4.12(a) and where the hypothesis $z \in \tilde{E}$ at the exact L^1 -optimal centre z holds by Lemma 5.13 below,

$$\eta = \rho_e - \rho_i \leq C(\text{Exc} + a)^{1/2} \leq C \left(\frac{\delta_T}{\tau} \right)^{1/4}, \quad (63)$$

which at the optimal choice $\tau = \sqrt{\delta_T}$ of Section 9 reads $\eta \leq C\delta_T^{1/8}$. This rate uses the ball distance a at the exact optimal centre, with no recentring penalty, the exterior-connectedness hypothesis being available unconditionally (Lemma 4.12, Lemma 5.13).

Proof. The inclusions and $\rho_i \leq \hat{r} \leq \rho_e$ are Lemma 5.5; it remains to prove the width bound. We give the estimate in three steps and assemble.

Step 1 (geometry $\Rightarrow$ angular perimeter). By Lemma 5.7, $N(\rho) \geq 2$ for every $\rho \in (\rho_i, \rho_e)$, so Lemma 5.6 gives

$$P_\theta = \int_0^\infty N(\rho) d\rho \geq \int_{\rho_i}^{\rho_e} 2 d\rho = 2(\rho_e - \rho_i) = 2\eta. \quad (64)$$

Step 2 (Cauchy-Schwarz). Write $P := P(E)$. By the Cauchy-Schwarz inequality on ∂E (with the constant function 1),

$$P_\theta^2 = \left(\int_{\partial E} |\nu_\theta| d\mathcal{H}^1 \right)^2 \leq P \int_{\partial E} \nu_\theta^2 d\mathcal{H}^1 = P \int_{\partial E} (1 - \nu_r^2) d\mathcal{H}^1 = P \left(P - \int_{\partial E} \nu_r^2 d\mathcal{H}^1 \right),$$

using $\nu_r^2 + \nu_\theta^2 = 1$. A second application of Cauchy-Schwarz gives $\int_{\partial E} \nu_r^2 d\mathcal{H}^1 \geq (\int_{\partial E} |\nu_r| d\mathcal{H}^1)^2 / P = P_r^2 / P$, hence

$$P_\theta^2 \leq P \left(P - \frac{P_r^2}{P} \right) = P^2 - P_r^2 = (P - P_r)(P + P_r) \leq 2P(P - P_r), \quad (65)$$

the last step using $0 \leq P_r \leq P$.

Step 3 (radial-profile bound). By Lemma 5.9, $P - P_r \leq \text{Exc} + 2a/\hat{r}$.

Assembly. Combining (64), (65), Step 3 and $P = 2\pi\hat{r} + \text{Exc}$,

$$4\eta^2 \leq P_\theta^2 \leq 2P(P - P_r) \leq 2(2\pi\hat{r} + \text{Exc}) \left(\text{Exc} + \frac{2a}{\hat{r}} \right).$$

This is the scale-correct width bound in (62). If $\hat{r} \in [\frac{1}{2}, 1]$ and $\text{Exc} \leq C_0$, then $2\pi\hat{r} + \text{Exc} \leq 2\pi + C_0$ and $2a/\hat{r} \leq 4a$, so $\eta \leq C(C_0)(\text{Exc} + a)^{1/2}$. Finally, for the inset, inserting (51), $\text{Exc} + a \leq C(\delta_T/\tau) + C(\delta_T/\tau)^{1/2} \leq C(\delta_T/\tau)^{1/2}$ for $\delta_T/\tau \leq 1$, so $\eta \leq C(\delta_T/\tau)^{1/4}$, which is (63); at $\tau = \sqrt{\delta_T}$ this is $C\delta_T^{1/8}$. $\square$

Remark 5.11 (A posteriori: the annulus is centred near the unit circle). The bound (63) together with $\rho_i \leq \hat{r} \leq \rho_e$ yields

$$\hat{r} - \rho_i \leq \eta \leq C(\delta_T/\tau)^{1/4}, \quad \rho_e - \hat{r} \leq \eta \leq C(\delta_T/\tau)^{1/4},$$

so $\rho_i \geq \hat{r} - C(\delta_T/\tau)^{1/4} \geq \frac{1}{2}$ and $\rho_e \leq \hat{r} + C(\delta_T/\tau)^{1/4} \leq 2$ once $\delta_T \leq \delta_\star$ is small enough (recall $\hat{r} \in [\frac{1}{2}, 1]$ and $\delta_T/\tau \rightarrow 0$). These bounds (in particular $\rho_i \geq \frac{1}{2}$) are exactly the inputs used in Sections 6 and 7; they are *outputs* of the trapping, not assumptions of it.

5.6. The exact optimal centre is interior: the inscribed-ball lemma. It remains to discharge the sole structural hypothesis of Theorem 5.10: that the *exact* L^1 -optimal centre z (the one realising $a = |\tilde{E} \triangle B_{\hat{r}}(z)| = \mathcal{A}(\tilde{E})|\tilde{E}|$) is an interior point of $\tilde{E}$, with a definite inscribed ball. Using the global exterior-connectedness of the hole-filled inset (Lemma 4.12(a), Remark 5.1), we prove this at the sharp scale, so that the rate (63) holds with the ball distance a measured at the exact optimum — no recentring, hence no square-root loss. The mechanism is the trapping mechanism itself, run about z before knowing $z \in \tilde{E}$: the angular-coarea bound forces the angular perimeter P_θ to be *large* (a definite constant) whenever z fails to be interior, contradicting the smallness $P_\theta \leq C(\text{Exc} + a)^{1/2}$ that the very same Cauchy-Schwarz/radial-profile chain delivers.

We first isolate the two ingredients of the trapping chain that are valid for *any* centre $w \notin \partial\tilde{E}$, not only for $w \in \tilde{E}$. Recall $\partial\tilde{E}$ is a compact $\mathcal{H}^1$ -rectifiable curve in the open set Ω with $\mathcal{H}^1(\partial\tilde{E}) = P := P(\tilde{E})$.

Lemma 5.12 (Centre-free Cauchy–Schwarz and radial-profile bounds). *Fix any $w \in \mathbb{R}^2 \setminus \partial\tilde{E}$ and define, with $e_r^w := (x - w)/|x - w|$, $\nu_r^w := \nu \cdot e_r^w$, $\nu_\theta^w := \nu \cdot (e_r^w)^\perp$,*

$$P_r^w := \int_{\partial\tilde{E}} |\nu_r^w| d\mathcal{H}^1, \quad P_\theta^w := \int_{\partial\tilde{E}} |\nu_\theta^w| d\mathcal{H}^1, \quad N^w(\rho) := \mathcal{H}^0(\partial\tilde{E} \cap \partial B_\rho(w)).$$

Then $P_\theta^w = \int_0^\infty N^w(\rho) d\rho$, and

$$(P_\theta^w)^2 \leq 2P(P - P_r^w). \quad (66)$$

If moreover $a_w := |\tilde{E} \triangle B_{\hat{r}}(w)|$ denotes the ball distance at w , then

$$P - P_r^w \leq \text{Exc} + \frac{2a_w}{\hat{r}}. \quad (67)$$

Proof. The coarea identity $P_\theta^w = \int_0^\infty N^w d\rho$ is Lemma 5.6 verbatim with z replaced by w : its proof uses only that $f_w(x) := |x - w|$ is 1-Lipschitz with $|\nabla^{\partial\tilde{E}} f_w| = |\nu_\theta^w|$ $\mathcal{H}^1$ -a.e. on $\partial\tilde{E}$ (true since $w \notin \partial\tilde{E}$, so e_r^w is defined and smooth along $\partial\tilde{E}$), and the coarea formula on the rectifiable curve. The chain (66) is Step 2 of the proof of Theorem 5.10 (eq. (65)) verbatim: it is two Cauchy–Schwarz inequalities on $\partial\tilde{E}$ using $(\nu_r^w)^2 + (\nu_\theta^w)^2 = 1$ and $0 \leq P_r^w \leq P$, with no reference to the position of w . For (67), the proof of Lemma 5.9 applies with w in place of z : define the outer radial profile $R_w(\theta) := \sup \{ \rho > 0 : w + \rho e^{i\theta} \in \tilde{E} \} \in [0, \rho_e^w]$, where $\rho_e^w := \sup_{x \in \tilde{E}} |x - w| < \infty$; for each θ with $R_w(\theta) < \hat{r}$ the segment $\{R_w(\theta) < \rho < \hat{r}\}$ on the ray from w lies in $B_{\hat{r}}(w) \setminus \tilde{E}$ (its points have radius $> R_w(\theta)$, hence are outside $\tilde{E}$ by definition of the supremum, and radius $< \hat{r}$), so by polar integration and $\int_R^{\hat{r}} \rho d\rho \geq \frac{\hat{r}}{2}(\hat{r} - R)$ (valid as $R \geq 0$),

$$a_w \geq |B_{\hat{r}}(w) \setminus \tilde{E}| \geq \frac{\hat{r}}{2} \int_{S^1} (\hat{r} - R_w)_+ d\theta,$$

whence $\int_{S^1} R_w d\theta \geq 2\pi\hat{r} - 2a_w/\hat{r}$. Finally $P_r^w \geq \int_{S^1} R_w d\theta$ by the weighted coarea identity (57) with w in place of z (again valid because $w \notin \partial\tilde{E}$, so $\Theta_w(x) = (x - w)/|x - w|$ is defined on $\partial\tilde{E}$), the inner sum there being at least its outermost term $R_w(\theta)$. Combining with $P = 2\pi\hat{r} + \text{Exc}$ gives (67). $\square$

Lemma 5.13 (The exact optimal centre is interior; inscribed ball). *Let $\tilde{E} = E^h$ be the hole-filled inset (Remark 5.1; $\delta_T \leq \delta_\star$, connected, globally hole-free by Lemma 4.12(a), $\hat{r} \in [\frac{1}{2}, 1]$, scales (51)), and let z be an exact L^1 -optimal centre, $a = |\tilde{E} \triangle B_{\hat{r}}(z)| = \mathcal{A}(\tilde{E})|\tilde{E}|$. There is an absolute constant C_0 such that, after possibly shrinking the absolute threshold $\delta_\star$,*

$$z \in \tilde{E}, \quad \rho_i = \text{dist}(z, \partial\tilde{E}) \geq \hat{r} - C_0(\text{Exc} + a)^{1/2} \geq \frac{1}{2}. \quad (68)$$

In particular $B_{\rho_i}(z) \subseteq \tilde{E}$ with $\rho_i \geq \frac{1}{2} > 0$, so the hypothesis $z \in \tilde{E}$ of Theorem 5.10 holds at the exact optimum, unconditionally.

Proof. Set $\beta := 2P(\text{Exc} + 2a/\hat{r})$, an absolute small quantity: by (51) and $\hat{r} \in [\frac{1}{2}, 1]$ one has $P = 2\pi\hat{r} + \text{Exc} \leq 2\pi + C$ and $\text{Exc} + 2a/\hat{r} \leq \text{Exc} + 4a \leq C(\text{Exc} + a) \leq C(\delta_T/\tau)^{1/2}$, so

$$\beta \leq C_1(\text{Exc} + a) \leq C_1 C (\delta_T/\tau)^{1/2} \xrightarrow{\delta_T/\tau \rightarrow 0} 0, \quad (69)$$

with C_1 absolute. We show that if z is *not* an interior point with $\rho_i \geq \frac{1}{2}$, then $\beta \geq \frac{1}{4}$, which is impossible once $\delta_\star$ is small enough that the right side of (69) is $< \frac{1}{4}$; this proves (68).

Step 0: trichotomy for z . By (49), exactly one of $z \in \tilde{E}$, $z \in \partial\tilde{E}$, $z \in U := \mathbb{R}^2 \setminus \tilde{E}$ holds. We treat the two non-interior cases and show each forces $\beta \geq \frac{1}{4}$; we also show that in the interior case the inscribed radius is large.

Step 1: $z \in U$ is impossible. Suppose $z \in U$. Put $d := \text{dist}(z, \tilde{E}) > 0$ (positive since U is open). First, the trivial area bound: $B_{\min(d, \hat{r})}(z) \cap \tilde{E} = \emptyset$, so $\pi \min(d, \hat{r})^2 \leq |B_{\hat{r}}(z) \setminus \tilde{E}| \leq a$; since $a \leq C(\delta_T/\tau)^{1/2} < \pi \hat{r}^2$ for δ_T/τ small (as $\hat{r} \geq \frac{1}{2}$), this forces $d < \hat{r}$ and

$$d \leq \sqrt{a/\pi} \leq C(\delta_T/\tau)^{1/4} \leq \frac{1}{4} \quad (70)$$

for δ_T/τ small. Now we lower-bound P_θ^z . Because $z \notin \partial\tilde{E}$, Lemma 5.12 applies with $w = z$, giving $P_\theta^z = \int_0^\infty N^z(\rho) d\rho$ and $(P_\theta^z)^2 \leq 2P(\text{Exc} + 2a/\hat{r}) = \beta$. We claim $N^z(\rho) \geq 2$ for every $\rho \in (d, \rho_e^z)$, where $\rho_e^z := \sup_{x \in \tilde{E}} |x - z| \geq \hat{r}$ (the inequality because $\tilde{E} \subset \overline{B_{\rho_e^z}(z)}$ gives $\pi \hat{r}^2 = |\tilde{E}| \leq \pi(\rho_e^z)^2$). Indeed, let $C_\rho := \partial B_\rho(z)$ with $\rho \in (d, \rho_e^z)$.

- C_ρ meets $\tilde{E}$: the radial function $f(x) = |x - z|$ is continuous on the connected open set $\tilde{E}$, so $f(\tilde{E})$ is an interval; it has infimum $d = \inf_{\tilde{E}} f$ (attained in $\tilde{E}$, approached within $\tilde{E}$) and supremum ρ_e^z , so $f(\tilde{E}) \supseteq (d, \rho_e^z) \ni \rho$.
- C_ρ meets U : by Lemma 5.2 the set U is connected, open and unbounded, and $z \in U$, so $f(U)$ is an interval with infimum $f(z) = 0$ and supremum $+\infty$; hence $f(U) \supseteq (0, \infty) \ni \rho$.

Thus $C_\rho \setminus \partial\tilde{E} = (C_\rho \cap \tilde{E}) \sqcup (C_\rho \cap U)$ is a disjoint union of two nonempty relatively open subsets of the connected curve C_ρ ; were $N^z(\rho) = \mathcal{H}^0(C_\rho \cap \partial\tilde{E}) \leq 1$, removing at most one point would leave C_ρ connected, contradicting the existence of such a partition. Hence $N^z(\rho) \geq 2$. Therefore, using (70) and $\rho_e^z \geq \hat{r} \geq \frac{1}{2}$,

$$\sqrt{\beta} \geq P_\theta^z = \int_0^\infty N^z d\rho \geq \int_d^{\rho_e^z} 2 d\rho = 2(\rho_e^z - d) \geq 2\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{1}{2},$$

so $\beta \geq \frac{1}{4}$, the desired contradiction. Hence $z \notin U$.

Step 2: $z \in \partial\tilde{E}$ is impossible. Suppose $z \in \partial\tilde{E}$. We cannot apply Lemma 5.12 directly at $w = z$, but we apply it at a sequence $w_k \rightarrow z$, $w_k \in U$, $w_k \notin \partial\tilde{E}$ (such w_k exist: $z \in \partial\tilde{E} = \partial U$ by Lemma 5.2, so every neighbourhood of z meets the open set U). Set $d_k := \text{dist}(w_k, \tilde{E}) \geq 0$, with $d_k \rightarrow 0$ as $w_k \rightarrow z \in \partial\tilde{E}$. Exactly as in Step 1 ($w_k \in U$) we obtain $N^{w_k}(\rho) \geq 2$ for $\rho \in (d_k, \rho_e^{w_k})$, where $\rho_e^{w_k} = \sup_{x \in \tilde{E}} |x - w_k| \geq \hat{r}$, and

$$P_\theta^{w_k} \geq 2(\rho_e^{w_k} - d_k) \geq 2(\hat{r} - d_k) \xrightarrow{k \rightarrow \infty} 2\hat{r} \geq 1.$$

On the other hand $a_{w_k} = |\tilde{E} \triangle B_{\hat{r}}(w_k)| \leq a + 4\pi\hat{r}|w_k - z| \rightarrow a$, since $B_{\hat{r}}(z) \triangle B_{\hat{r}}(w_k) \subseteq B_{\hat{r}+|w_k-z|}(z) \setminus B_{\hat{r}-|w_k-z|}(z)$ has measure $4\pi\hat{r}|w_k - z|$; so by (66)–(67), $(P_\theta^{w_k})^2 \leq 2P(\text{Exc} + 2a_{w_k}/\hat{r})$, whose right side tends to $2P(\text{Exc} + 2a/\hat{r}) = \beta$. Fix $\varepsilon > 0$. For k large, $d_k < \varepsilon$ and the right side is $< \beta + \varepsilon$, so $(2\hat{r} - 2\varepsilon)^2 \leq (P_\theta^{w_k})^2 < \beta + \varepsilon$; as $\hat{r} \geq \frac{1}{2}$ and $\varepsilon > 0$ is arbitrary this gives $1 \leq 4\hat{r}^2 \leq \beta$, so again $\beta \geq 1 > \frac{1}{4}$, a contradiction. Hence $z \notin \partial\tilde{E}$.

Step 3: interior case and inscribed radius. By Steps 1–2, $z \in \tilde{E}$. Then Definition 5.4 gives $\rho_i = \text{dist}(z, \partial\tilde{E}) > 0$ and $B_{\rho_i}(z) \subseteq \tilde{E}$ (Lemma 5.5), and Theorem 5.10 applies at this very z with

the exact ball distance a : it yields $\eta = \rho_e - \rho_i \leq C(\text{Exc} + a)^{1/2}$. Since $\rho_i \leq \hat{r} \leq \rho_e$ (Lemma 5.5), $\hat{r} - \rho_i \leq \rho_e - \rho_i = \eta$, whence

$$\rho_i \geq \hat{r} - \eta \geq \hat{r} - C(\text{Exc} + a)^{1/2} =: \hat{r} - C_0(\text{Exc} + a)^{1/2}.$$

By (51), $C_0(\text{Exc} + a)^{1/2} \leq C_0 C(\delta_T/\tau)^{1/4} \leq \hat{r} - \frac{1}{2}$ once $\delta_\star$ is small (recall $\hat{r} \geq \frac{1}{2}$ and, in the application $\tau = \sqrt{\delta_T}$, $\hat{r} \rightarrow 1$ so $\hat{r} \geq \frac{3}{4}$ and $\hat{r} - \frac{1}{2} \geq \frac{1}{4} \geq C_0 C \delta_T^{1/8}$ for δ_T small). Hence $\rho_i \geq \frac{1}{2}$, which is (68). $\square$

Remark 5.14 (Why this is the sharp, not the recentred, rate). The naive route recenters from the exact optimum z to a nearby interior point $z_0 \in \tilde{E}$ with $|z - z_0|$ possibly as large as $\text{dist}(z, \tilde{E}) \sim a^{1/2}$, and pays the *linear* symmetric-difference penalty $a_0 \leq a + 4\pi\hat{r}|z - z_0| \sim a^{1/2}$, one square root worse than a ; fed into (62) this gives only $\eta \leq C(\text{Exc} + a_0)^{1/2} \sim \delta_T^{1/16}$ at $\tau = \sqrt{\delta_T}$. Lemma 5.13 avoids recentring altogether: it proves the exact optimum is *already* interior with $\rho_i \geq \frac{1}{2}$, so the trapping is run at z itself and the ball distance entering (62) is the optimal $a \sim \delta_T^{1/4}$, giving the sharp $\eta \leq C(\text{Exc} + a)^{1/2} \sim \delta_T^{1/8}$. The decisive point is that the obstruction to interiority is charged not to the *area* a (which only bounds $\text{dist}(z, \tilde{E}) \lesssim a^{1/2}$) but to the *angular perimeter*: a non-interior centre would force $P_\theta \geq \frac{1}{2}$ a definite constant, whereas the deficits force $P_\theta \leq C(\text{Exc} + a)^{1/2} \rightarrow 0$. This is the rigorous form of route (i) of the program: the cost of the exterior pocket reaching z is the two crossings it forces at *every* intermediate radius, summed by the coarea identity, i.e. a perimeter cost, not an area cost.

Remark 5.15 (The a-posteriori check is now a theorem). The self-consistency observation of Remark 5.11 — that the trapping output $\rho_i \geq \frac{1}{2}$ is compatible with z being interior — is *upgraded to a proof* by Lemma 5.13: interiority is established *a priori*, by the same angular-coarea mechanism, with no appeal to the conclusion. Consequently the graph-defect bound Corollary 6.5, the closeness $|\Omega \Delta G|$ of Sections 6 and 7, and the final exponent in Theorem 9.2 all rest on the *sharp* rate $\eta \sim \delta_T^{1/8}$ unconditionally, the global hole-freeness being supplied by the hole-filled inset (Lemma 4.12); the degenerate boundary configuration of the optimal centre does *not* occur. The inputs of this section remain exactly connectedness, the global hole property, Exc and a (the Serrin defect D_H is not used).

Remark 5.16 (What is, and is not, used). The trapping width $\eta \leq C(\text{Exc} + a)^{1/2}$ is proved with an *absolute* constant and *no* second-order information on $\partial\tilde{E}$: no interior or exterior ball condition, no curvature bound, no harmonic oscillation estimate. The ingredients are exactly: the connectedness and global hole-freeness of $\tilde{E}$ (Lemmas 4.3 and 4.12); the coarea formula on a rectifiable curve (Theorem 2.6); the isoperimetric excess Exc and the FMP closeness a (Theorem 2.11, via (51)). Notably, the Cauchy–Schwarz/Serrin defect D_H is *not* used in this section. The single arithmetic of the proof is the chain $4\eta^2 \leq P_\theta^2 \leq 2P(P - P_r) \leq 2P(\text{Exc} + 2a/\hat{r})$, in which the geometric lower bound (two crossings per circle) is converted into the deficit by one Cauchy–Schwarz and one elementary radial-profile estimate. The hypothesis $z \in \tilde{E}$ — the interiority of the exact L^1 -optimal centre, used to start the radius at 0 and to define the angular projection — is itself *proved* from the same ingredients in Lemma 5.13 (the very same chain, run before interiority is known, forces a definite P_θ at a non-interior centre, contradicting the deficit smallness). Consequently the *sharp rate* $\eta \leq C(\text{Exc} + a)^{1/2} \sim \delta_T^{1/8}$ for the inset’s exact FMP centre follows unconditionally, the exterior-connectedness being supplied by the hole-filled inset (Lemma 4.12, Remark 5.1).

6. FOLD CONTENT AND THE RADIAL GRAPH

The trapping annulus of Theorem 5.10 confines $\partial\tilde{E}$ to a thin shell $\{\rho_i \leq |x - z| \leq \rho_e\}$ of width $\eta = \rho_e - \rho_i$, so the inset has *small amplitude*. To convert this into a radial *graph* about the centre z we must rule out radial folds: rays from z meeting $\partial\tilde{E}$ more than once. This section bounds the total *fold content* — the obstruction to being a graph — by the geometric deficit, with *no* square-root loss from the quantitative isoperimetric inequality, and then realises $\tilde{E}$ as a measurable radial graph up to a controlled symmetric difference.

Throughout, $\tilde{E}$ is the hole-filled inset of Definition 4.9 and Lemma 4.12 (Remark 5.1), connected and globally hole-free, so the global exterior-connectedness needed to invoke the trapping annulus of Theorem 5.10 holds unconditionally (Lemma 4.12(a)). We write with $\mu_E := |\tilde{E}|$, $\hat{r} := \sqrt{\mu_E/\pi}$, isoperimetric deficit $\delta_{\text{iso}} := \delta_{\text{iso}}(\tilde{E})$, perimeter excess $\text{Exc} := P(\tilde{E}) - 2\pi\hat{r} = 2\pi\hat{r}\delta_{\text{iso}}$, and (Theorem 5.10) trapping

$$B_{\rho_i}(z) \subset \tilde{E} \subset B_{\rho_e}(z), \quad \rho_i \leq \hat{r} \leq \rho_e, \quad \eta := \rho_e - \rho_i \leq C(\text{Exc} + a)^{1/2}, \quad (71)$$

where z is the optimal centre and $a := |\tilde{E} \Delta B_{\hat{r}}(z)|$. In particular z lies in the interior of $\tilde{E}$ (indeed $B_{\rho_i}(z) \subset \tilde{E}$), and $\partial\tilde{E}$ avoids the singular point:

$$\partial\tilde{E} \subset \{x : \rho_i \leq |x - z| \leq \rho_e\}. \quad (72)$$

We write $e_r := (x - z)/|x - z|$, $e_\theta := e_r^\perp$ for the unit radial and angular vectors at $x \neq z$, ν for the outward unit normal on the reduced boundary $\partial^*\tilde{E}$, and

$$\nu_r := \nu \cdot e_r$$

for the radial part of ν . (As is standard for sets of finite perimeter, all boundary integrals are over the reduced boundary $\partial^*\tilde{E}$, on which ν is defined $\mathcal{H}^1$ -a.e.; we write $\partial\tilde{E}$ for it throughout, noting that for the hole-free finite-perimeter set $\tilde{E}$ the reduced and essential boundaries coincide $\mathcal{H}^1$ -a.e. by the De Giorgi–Federer structure theorem.) For $\theta \in S^1$ let

$$M(\theta) := \mathcal{H}^0(\partial\tilde{E} \cap R_\theta), \quad R_\theta := \{z + \rho e^{i\theta} : \rho > 0\} \quad (73)$$

be the number of crossings of $\partial\tilde{E}$ by the open ray in direction θ . By the one-dimensional structure of slices of finite-perimeter sets [1, §2.4, §3.11], for a.e. θ the radial slice

$$\tilde{E}_\theta := \{\rho > 0 : z + \rho e^{i\theta} \in \tilde{E}\}$$

is, up to an $\mathcal{H}^1$ -null set, a finite union of intervals; since $B_{\rho_i}(z) \subset \tilde{E} \subset B_{\rho_e}(z)$, the innermost interval starts at 0 and all endpoints lie in $[\rho_i, \rho_e]$, so

$$\tilde{E}_\theta = (0, r_1) \cup (r_2, r_3) \cup \cdots \cup (r_{2m}, r_{2m+1}), \quad \rho_i \leq r_1 \leq \cdots \leq r_{2m+1} \leq \rho_e, \quad M(\theta) = 2m(\theta) + 1. \quad (74)$$

The intervals (r_{2i}, r_{2i+1}) , $i \geq 1$, are the *caps*; $m(\theta)$ is the number of caps on the ray, and $\tilde{E}$ is a radial graph about z exactly when $M \equiv 1$ (no caps).

6.1. Point-source flux identities.

Lemma 6.1 (Two-flux identities). *Let $E \subset \mathbb{R}^2$ be a set of finite perimeter and suppose that, for some $r_0 > 0$,*

$$B_{r_0}(z) \subset E, \quad \partial^* E \subset \{x : |x - z| \geq r_0\}.$$

With $e_r = (x - z)/|x - z|$, $\nu_r = \nu \cdot e_r$, and $M(\theta)$ as in (73), the following hold:

$$(i) \quad \int_{\partial E} \frac{|\nu_r|}{|x - z|} d\mathcal{H}^1 = \int_{S^1} M(\theta) d\theta, \quad (75a)$$

$$(ii) \quad \int_{\partial E} \frac{\nu_r}{|x - z|} d\mathcal{H}^1 = 2\pi, \quad (75b)$$

$$(iii) \quad \int_{\partial E} \nu_r d\mathcal{H}^1 = \int_E \frac{dx}{|x - z|} =: I_E. \quad (75c)$$

We refer to the three identities collectively as (75).

Proof. We may translate so that $z = 0$; then $e_r = x/|x|$ and $|x - z| = |x|$. Set $A := \{x : r_0 \leq |x|\}$, a closed set containing $\partial^* E$ on which all three integrands are smooth and bounded. Since $B_{r_0}(0) \subset E$, for every $0 < \varepsilon < r_0$ the circle $\partial B_\varepsilon(0)$ lies in the measure-theoretic interior $E^{(1)}$.

(i) *The angular coarea identity.* Let $\Theta(x) := x/|x| \in S^1$ be the angular projection. On the annulus A this map is Lipschitz. The reduced boundary $\partial^* E$ is a countably 1-rectifiable set contained in A ; let τ denote a unit tangent to $\partial^* E$ (defined $\mathcal{H}^1$ -a.e., orthogonal to ν). The tangential Jacobian of Θ along $\partial^* E$ is

$$J_{\partial^* E} \Theta = \frac{|e_\theta \cdot \tau|}{|x|}.$$

Since $\{e_r, e_\theta\}$ and $\{\nu, \tau\}$ are each orthonormal bases of $\mathbb{R}^2$ with $e_r \perp e_\theta$ and $\nu \perp \tau$, the angle between e_θ and τ equals the angle between e_r and ν ; hence $|e_\theta \cdot \tau| = |e_r \cdot \nu| = |\nu_r|$, so

$$J_{\partial^* E} \Theta = \frac{|\nu_r|}{|x|}.$$

The coarea formula for the Lipschitz map $\Theta : \partial^* E \rightarrow S^1$ on the 1-rectifiable set $\partial^* E$ [1, Theorem 2.93] gives

$$\int_{\partial^* E} \frac{|\nu_r|}{|x|} d\mathcal{H}^1 = \int_{\partial^* E} J_{\partial^* E} \Theta d\mathcal{H}^1 = \int_{S^1} \mathcal{H}^0(\partial^* E \cap \Theta^{-1}(\theta)) d\theta = \int_{S^1} M(\theta) d\theta,$$

because $\Theta^{-1}(\theta) \cap A = R_\theta \cap A \supset \partial^* E \cap R_\theta$ and $\partial^* E \subset A$. This is (75a).

(ii) *The point-source field, singularity excised.* Consider the vector field $X(x) := x/|x|^2$, defined and smooth on $\mathbb{R}^2 \setminus \{0\}$, with

$$\operatorname{div} X = \frac{2}{|x|^2} - \frac{2}{|x|^2} = 0 \quad (x \neq 0)$$

(directly: $\partial_i(x_i |x|^{-2}) = |x|^{-2} + x_i(-2x_i |x|^{-4})$, summing $i = 1, 2$ gives $2|x|^{-2} - 2|x|^{-2} = 0$). On $\partial^* E$ we have $X \cdot \nu = (x \cdot \nu)/|x|^2 = \nu_r/|x|$. Fix $0 < \varepsilon < r_0$ and apply the Gauss–Green theorem for sets of finite perimeter [5, Theorem 5.16] to the field X , which is C^1 and bounded on the closed set $\{|x| \geq \varepsilon/2\}$, over the finite-perimeter set $E_\varepsilon := E \setminus \overline{B_\varepsilon(0)}$. Since $\partial B_\varepsilon \subset E^{(1)}$ (shown above), the reduced boundary $\partial^* E_\varepsilon$ splits ($\mathcal{H}^1$ -a.e.) into $\partial^* E$ (with outward normal ν , and on which X is

integrable since $\partial^* E \subset A$) and the inner circle ∂B_ε (with outward-for- E_ε normal $-x/\varepsilon$):

$$0 = \int_{E_\varepsilon} \operatorname{div} X \, dx = \int_{\partial^* E} X \cdot \nu \, d\mathcal{H}^1 + \int_{\partial B_\varepsilon} X \cdot \left(-\frac{x}{\varepsilon}\right) d\mathcal{H}^1.$$

On ∂B_ε , $X = x/\varepsilon^2$ and $X \cdot (-x/\varepsilon) = -|x|^2/\varepsilon^3 = -1/\varepsilon$, so the second term equals $-\varepsilon^{-1} \mathcal{H}^1(\partial B_\varepsilon) = -\varepsilon^{-1} \cdot 2\pi\varepsilon = -2\pi$, independent of ε . Hence

$$\int_{\partial^* E} \frac{\nu_r}{|x|} d\mathcal{H}^1 = \int_{\partial^* E} X \cdot \nu \, d\mathcal{H}^1 = 2\pi,$$

which is (75b). (No limit in ε is even needed: the left side does not depend on ε , and the inner-circle flux is exactly 2π for every admissible ε .)

(iii) *The radial field, singularity integrable.* Consider $Y(x) := e_r = x/|x|$, smooth on $\mathbb{R}^2 \setminus \{0\}$, with

$$\operatorname{div} Y = \sum_i \partial_i \frac{x_i}{|x|} = \sum_i \left(\frac{1}{|x|} - \frac{x_i^2}{|x|^3} \right) = \frac{2}{|x|} - \frac{|x|^2}{|x|^3} = \frac{1}{|x|} \quad (x \neq 0).$$

Again fix $0 < \varepsilon < r_0$ and apply Gauss–Green to the C^1 bounded field Y over $E_\varepsilon = E \setminus \overline{B_\varepsilon(0)}$:

$$\int_{E_\varepsilon} \frac{dx}{|x|} = \int_{E_\varepsilon} \operatorname{div} Y \, dx = \int_{\partial^* E} Y \cdot \nu \, d\mathcal{H}^1 + \int_{\partial B_\varepsilon} Y \cdot \left(-\frac{x}{\varepsilon}\right) d\mathcal{H}^1.$$

On ∂B_ε , $Y = x/\varepsilon$ and $Y \cdot (-x/\varepsilon) = -|x|^2/\varepsilon^2 = -1$, so the inner term is $-\mathcal{H}^1(\partial B_\varepsilon) = -2\pi\varepsilon$. Thus

$$\int_{\partial^* E} \nu_r d\mathcal{H}^1 = \int_{E_\varepsilon} \frac{dx}{|x|} + 2\pi\varepsilon.$$

Now let $\varepsilon \downarrow 0$. The left side is fixed. On the right, $|x|^{-1} \in L^1_{\text{loc}}(\mathbb{R}^2)$ (in polar coordinates $\int_{B_R} |x|^{-1} dx = 2\pi R < \infty$) and E has finite measure, so $|x|^{-1} \mathbf{1}_E \in L^1(\mathbb{R}^2)$; by dominated convergence $\int_{E_\varepsilon} |x|^{-1} dx \rightarrow \int_E |x|^{-1} dx = I_E$, while $2\pi\varepsilon \rightarrow 0$. Therefore $\int_{\partial^* E} \nu_r d\mathcal{H}^1 = I_E$, which is (75c). The constant I_E is finite because $|x|^{-1} \mathbf{1}_E \in L^1$. $\square$

Remark 6.2 (On the singularity). The two divergence-theorem identities are genuinely different at the source. In (75b) the field $X = x/|x|^2$ has a non-integrable $|x|^{-1}$ growth at 0 and a non-zero residual flux 2π through any small circle; excising B_ε is mandatory, and the 2π is precisely that residue (the planar Green-function source). In (75c) the field $Y = e_r$ is bounded, $\operatorname{div} Y = |x|^{-1}$ is integrable, and the inner-circle flux $2\pi\varepsilon \rightarrow 0$ vanishes; the excision is a convenience and the resulting I_E is a finite, well-defined number. We invoke Lemma 6.1 for $E = \tilde{E}$, whose reduced boundary satisfies the hypothesis $\partial^* \tilde{E} \subset \{|x - z| \geq \rho_i\}$ with $r_0 = \rho_i$ by (72), and whose interior contains z since $B_{\rho_i}(z) \subset \tilde{E}$.

6.2. Sharp fold content. The fold content is the deviation of M from the graph value 1. We bound its total mass with no quantitative-isoperimetric square-root loss; this is the crux of the sharp reduction.

Proposition 6.3 (Sharp fold content). *With the inset $\tilde{E}$ and the trapping data (71),*

$$\int_{S^1} (M(\theta) - 1) d\theta = 2 \int_{\partial \tilde{E} \cap \{\nu_r < 0\}} \frac{|\nu_r|}{|x - z|} d\mathcal{H}^1 \leq \frac{1}{\rho_i} \left(\operatorname{Exc} + \frac{a\eta}{\rho_i \hat{r}} \right) \leq C(\operatorname{Exc} + a\eta), \quad (76)$$

with C absolute. In particular the integrand is non-negative, so $M \geq 1$ a.e., and

$$|\{\theta \in S^1 : M(\theta) \geq 3\}| \leq \frac{1}{2} \int_{S^1} (M-1) d\theta \leq C(\text{Exc} + a\eta). \quad (77)$$

Proof. The fold identity. Subtract (75b) from (75a) of Lemma 6.1 (legitimate by Remark 6.2); since $\int_{S^1} 1 d\theta = 2\pi$ equals the right-hand side of (75b),

$$\int_{S^1} (M-1) d\theta = \int_{\partial \tilde{E}} \frac{|\nu_r| - \nu_r}{|x-z|} d\mathcal{H}^1 = 2 \int_{\partial \tilde{E} \cap \{\nu_r < 0\}} \frac{|\nu_r|}{|x-z|} d\mathcal{H}^1 \geq 0,$$

using $|\nu_r| - \nu_r = 0$ where $\nu_r \geq 0$ and $= 2|\nu_r|$ where $\nu_r < 0$. This is the first equality of (76). (Pointwise $M(\theta) = 2m(\theta) + 1 \geq 1$ a.e. by the slice structure (74), since every slice contains $(0, r_1)$; the content of the identity is the *global* bound on the excess $\int (M-1)$, which is exactly twice the inward-pointing radial flux.)

From the radial fold flux to $P_r - I_{\tilde{E}}$. On the folded part $\partial \tilde{E} \cap \{\nu_r < 0\}$ we bound $|x-z| \geq \rho_i$ (by (72)), so

$$\int_{S^1} (M-1) d\theta \leq \frac{2}{\rho_i} \int_{\partial \tilde{E} \cap \{\nu_r < 0\}} |\nu_r| d\mathcal{H}^1 = \frac{2F_r}{\rho_i}, \quad F_r := \int_{\partial \tilde{E} \cap \{\nu_r < 0\}} |\nu_r| d\mathcal{H}^1. \quad (78)$$

Write $P_r := \int_{\partial \tilde{E}} |\nu_r| d\mathcal{H}^1$ for the total radial perimeter. Splitting according to the sign of ν_r ,

$$\int_{\partial \tilde{E}} \nu_r d\mathcal{H}^1 = \int_{\{\nu_r \geq 0\}} |\nu_r| d\mathcal{H}^1 - \int_{\{\nu_r < 0\}} |\nu_r| d\mathcal{H}^1 = (P_r - F_r) - F_r = P_r - 2F_r.$$

By (75c), the left side is $I_{\tilde{E}}$, hence

$$2F_r = P_r - I_{\tilde{E}}. \quad (79)$$

Upper bound on P_r . Since $|\nu_r| = |\nu \cdot e_r| \leq |\nu| = 1$ pointwise, the radial perimeter is dominated by the perimeter:

$$P_r = \int_{\partial \tilde{E}} |\nu_r| d\mathcal{H}^1 \leq \int_{\partial \tilde{E}} 1 d\mathcal{H}^1 = P(\tilde{E}) = 2\pi\hat{r} + \text{Exc}.$$

The cancellation: lower bound on $I_{\tilde{E}}$. Here is the sharp step. Because $|\tilde{E}| = \mu_E = \pi\hat{r}^2 = |B_{\hat{r}}(z)|$, the indicators have equal mass, so $\int (\mathbf{1}_{\tilde{E}} - \mathbf{1}_{B_{\hat{r}}(z)}) dx = 0$ and we may subtract *any* constant from the kernel without changing the integral. Subtract the constant $1/\hat{r}$:

$$I_{\tilde{E}} - \int_{B_{\hat{r}}(z)} \frac{dx}{|x-z|} = \int (\mathbf{1}_{\tilde{E}} - \mathbf{1}_{B_{\hat{r}}(z)}) \frac{dx}{|x-z|} = \int (\mathbf{1}_{\tilde{E}} - \mathbf{1}_{B_{\hat{r}}(z)}) \left(\frac{1}{|x-z|} - \frac{1}{\hat{r}} \right) dx,$$

the last equality because $\int (\mathbf{1}_{\tilde{E}} - \mathbf{1}_{B_{\hat{r}}(z)}) \cdot \hat{r}^{-1} dx = \hat{r}^{-1} (|\tilde{E}| - |B_{\hat{r}}(z)|) = 0$. In polar coordinates, $\int_{B_{\hat{r}}(z)} |x-z|^{-1} dx = \int_0^{\hat{r}} \int_{S^1} s^{-1} s d\theta ds = 2\pi\hat{r}$, so

$$I_{\tilde{E}} - 2\pi\hat{r} = \int_{\tilde{E} \Delta B_{\hat{r}}(z)} \pm \left(\frac{1}{|x-z|} - \frac{1}{\hat{r}} \right) dx, \quad (80)$$

the integrand being supported on the symmetric difference $\tilde{E} \Delta B_{\hat{r}}(z)$ (with sign $+$ on $\tilde{E} \setminus B_{\hat{r}}(z)$ and $-$ on $B_{\hat{r}}(z) \setminus \tilde{E}$). Now $\tilde{E} \Delta B_{\hat{r}}(z)$ lies in the closed annulus $\{\rho_i \leq |x-z| \leq \rho_e\}$: indeed $B_{\rho_i}(z) \subset \tilde{E} \cap B_{\hat{r}}(z)$ (as $\rho_i \leq \hat{r}$), so neither $\tilde{E}$ nor $B_{\hat{r}}(z)$ differs from the other inside $B_{\rho_i}(z)$; and $\tilde{E} \cup B_{\hat{r}}(z) \subset B_{\rho_e}(z)$ (as $\hat{r} \leq \rho_e$ and $\tilde{E} \subset B_{\rho_e}(z)$), so neither differs outside $B_{\rho_e}(z)$. On that annulus,

with both $|x - z|$ and $\hat{r}$ in $[\rho_i, \rho_e]$,

$$\left| \frac{1}{|x - z|} - \frac{1}{\hat{r}} \right| = \frac{|\hat{r} - |x - z||}{\hat{r} |x - z|} \leq \frac{\rho_e - \rho_i}{\hat{r} \rho_i} = \frac{\eta}{\rho_i \hat{r}}. \quad (81)$$

Therefore, taking absolute values in (80) and using (81),

$$|I_{\tilde{E}} - 2\pi\hat{r}| \leq \frac{\eta}{\rho_i \hat{r}} |\tilde{E} \Delta B_{\hat{r}}(z)| = \frac{a\eta}{\rho_i \hat{r}}. \quad (82)$$

Conclusion. Combining (79) with the bound on P_r and the lower bound $I_{\tilde{E}} \geq 2\pi\hat{r} - a\eta/(\rho_i \hat{r})$ from (82),

$$2F_r = P_r - I_{\tilde{E}} \leq (2\pi\hat{r} + \text{Exc}) - \left(2\pi\hat{r} - \frac{a\eta}{\rho_i \hat{r}}\right) = \text{Exc} + \frac{a\eta}{\rho_i \hat{r}}.$$

Inserting into (78) yields the middle inequality of (76). Finally, in the optimised inset regime of Theorem 9.1, (35) gives $\mu_E = \pi - O(\sqrt{\delta_T})$, hence $\hat{r} \geq \frac{3}{4}$ after shrinking $\delta_\star$, and Theorem 5.10 gives $\eta \leq C\delta_T^{1/8} \leq \frac{1}{4}$. Thus $\rho_i \geq \hat{r} - \eta \geq \frac{1}{2}$ (using $\rho_e = \rho_i + \eta$ and $\rho_i \leq \hat{r} \leq \rho_e$). Therefore $1/\rho_i \leq 2$ and $1/(\rho_i^2 \hat{r}) \leq 1/(\frac{1}{4} \cdot \frac{3}{4}) \leq 6$, giving the absolute bound $C(\text{Exc} + a\eta)$. The fold-angle estimate (77) follows because $M - 1 \geq 2$ on $\{M \geq 3\}$ (as $M = 2m + 1$ is odd, $M \geq 3$ means $m \geq 1$), so $2|\{M \geq 3\}| \leq \int_{\{M \geq 3\}} (M - 1) \leq \int_{S^1} (M - 1)$. $\square$

Remark 6.4 (Why the factor η matters: the improvement over the naive bound). The naive estimate pulls $|x - z|^{-1} \leq \rho_i^{-1}$ out of the integral defining $I_{\tilde{E}} - 2\pi\hat{r}$ before subtracting any constant, giving

$$|I_{\tilde{E}} - 2\pi\hat{r}| = \left| \int (\mathbf{1}_{\tilde{E}} - \mathbf{1}_{B_{\hat{r}}(z)}) \frac{1}{|x - z|} \right| \leq \frac{1}{\rho_i} |\tilde{E} \Delta B_{\hat{r}}(z)| = \frac{a}{\rho_i} \leq Ca,$$

hence $\int (M - 1) \leq C(\text{Exc} + a)$. The two estimates differ by the factor $\eta = \rho_e - \rho_i$ on the a -term. This is decisive. By the quantitative isoperimetric inequality (Theorem 2.11), $a \leq C_F \sqrt{\delta_{\text{iso}}} \mu_E$ carries an unavoidable *square-root* of the isoperimetric deficit; with $\delta_{\text{iso}} \sim \delta_T/\tau \sim \sqrt{\delta_T}$ at the boundary scale this gives only $a \sim \delta_T^{1/4}$, and the naive fold content $\text{Exc} + a \sim \delta_T^{1/4}$ would cap the whole argument at $\delta_T^{1/4}$ closeness. The cancellation (82) replaces a by $a\eta$; since $\eta \leq C(\text{Exc} + a)^{1/2} \leq Ca^{1/2} \sim \delta_T^{1/8}$, the product $a\eta \sim \delta_T^{3/8}$, and through Corollary 6.5 the graph defect becomes $\text{gd} \sim \eta a\eta \sim \delta_T^{1/8} \cdot \delta_T^{3/8} = \delta_T^{1/2}$, matching the closeness $O(\sqrt{\delta_T})$. The single factor η — the legitimacy of subtracting the constant $1/\hat{r}$, because $|\tilde{E}| = |B_{\hat{r}}(z)|$, together with the confinement of $\tilde{E} \Delta B_{\hat{r}}(z)$ to the thin η -annulus — is exactly what removes the FMP square-root from the fold content and makes the reduction sharp.

6.3. The graph defect. The *graph defect* of $\tilde{E}$ about z ,

$$\text{gd}(\tilde{E}) := \inf_{\rho: S^1 \rightarrow (0, \infty)} |\tilde{E} \Delta \{z + r\omega : 0 \leq r < \rho(\omega)\}|,$$

measures how far $\tilde{E}$ is, in L^1 , from being a radial graph about z . The fold content controls it directly.

Corollary 6.5 (Graph defect). *The graph defect of $\tilde{E}$ about the centre z satisfies*

$$\text{gd}(\tilde{E}) \leq \rho_e \eta \int_{S^1} m(\theta) d\theta = \frac{1}{2} \rho_e \eta \int_{S^1} (M - 1) d\theta \leq C \eta (\text{Exc} + a\eta), \quad (83)$$

with C absolute.

Proof. Use the *core profile* $\rho_0(\theta) := r_1(\theta) \in [\rho_i, \rho_e]$, the innermost crossing radius from (74); it is defined for a.e. θ . Set $G_0 := \{z + re^{i\theta} : 0 \leq r < \rho_0(\theta)\}$. By construction the slice of G_0 in direction

θ is $(0, \rho_0(\theta)) = (0, r_1(\theta)) \subset \tilde{E}_\theta$, so $G_0 \subset \tilde{E}$ (up to $\mathcal{H}^1$ -null directions), and the slice of $\tilde{E} \setminus G_0$ is exactly the union of caps $(r_2, r_3) \cup \dots \cup (r_{2m}, r_{2m+1})$. By the polar-coordinate area formula (Fubini on $\mathbb{R}^2 \setminus \{z\}$),

$$|\tilde{E} \setminus G_0| = \int_{S^1} \sum_{i=1}^{m(\theta)} \int_{r_{2i}}^{r_{2i+1}} r \, dr \, d\theta.$$

On each ray with $m(\theta) \geq 1$, all cap endpoints lie in $[\rho_i, \rho_e]$, so the caps have total radial length $\sum_i (r_{2i+1} - r_{2i}) \leq r_{2m+1} - r_2 \leq \rho_e - \rho_i = \eta$, and each cap sits at radius $r \leq \rho_e$; hence

$$\sum_{i=1}^{m(\theta)} \int_{r_{2i}}^{r_{2i+1}} r \, dr \leq \rho_e \sum_{i=1}^{m(\theta)} (r_{2i+1} - r_{2i}) \leq \rho_e \eta \mathbf{1}_{\{m(\theta) \geq 1\}}.$$

Therefore

$$\text{gd}(\tilde{E}) \leq |\tilde{E} \triangle G_0| = |\tilde{E} \setminus G_0| \leq \rho_e \eta |\{\theta : m(\theta) \geq 1\}| \leq \rho_e \eta \int_{S^1} m \, d\theta,$$

where the last step uses $\mathbf{1}_{\{m \geq 1\}} \leq m$ (as m is a non-negative integer). Since $M = 2m + 1$, $\int_{S^1} m \, d\theta = \frac{1}{2} \int_{S^1} (M - 1) \, d\theta$, which is the displayed equality; and Proposition 6.3 together with $\rho_e \leq C\hat{r} \leq C$ (for $\delta_T \leq \delta_\star$) gives the final bound $\leq C \eta (\text{Exc} + a\eta)$. $\square$

6.4. Realization as a measurable radial graph. The core profile $\rho_0 = r_1$ is, a priori, only *measurable* in θ (it is defined ray by ray from the one-dimensional slices, with no modulus of continuity and no bounded variation available — a cap on a ray inserts a jump that the slice structure does not control). We do not attempt to regularise it. Instead we record that the measurable core graph G_0 is *already* an admissible input to the nearly-spherical estimate of Section 8, whose hypotheses — a bounded open star-shaped set trapped between two concentric circles, with measurable radial profile and small L^∞ amplitude — ask for no slope, no continuity, and no bounded variation. This removes the regularity step entirely.

Proposition 6.6 (Measurable radial-graph realization). *Let $\rho_0 = r_1 : S^1 \rightarrow [\rho_i, \rho_e]$ be the innermost-crossing core profile of Corollary 6.5, and let*

$$G_0 := \{z + r\omega(\theta) : \theta \in S^1, 0 \leq r < \rho_0(\theta)\}$$

be the associated core graph, taken in its open star-shaped representative (equivalently, replace ρ_0 by its lower-semicontinuous envelope along rays, which alters G_0 only on an $\mathcal{H}^2$ -null set and hence changes none of the quantities below). Then:

(i) ρ_0 is measurable with values in $[\rho_i, \rho_e]$, and G_0 is a bounded open set, star-shaped about z , with

$$B_{\rho_i}(z) \subset G_0 \subset B_{\rho_e}(z); \quad (84)$$

(ii) writing $\hat{r}_{G_0} := \sqrt{|G_0|/\pi} \in [\rho_i, \rho_e]$ and $\varphi_0 := \rho_0/\hat{r}_{G_0} - 1$, the amplitude obeys

$$\|\varphi_0\|_{L^\infty(S^1)} \leq \frac{\rho_e - \rho_i}{\rho_i} = \frac{\eta}{\rho_i} \leq 2\eta; \quad (85)$$

(iii) $G_0 \subset \tilde{E}$ and

$$|\tilde{E} \triangle G_0| = |\tilde{E} \setminus G_0| \leq C \eta (\text{Exc} + a\eta). \quad (86)$$

Consequently G_0 satisfies the hypotheses of Theorem 8.1 on the amplitude, the trapping ball, and the volume normalisation; only the quadratic mode-one bound (112) remains to be supplied, which is

done at the optimal centre in Section 9. No mollification is performed and no continuity of ρ_0 is claimed or used.

Proof. (i). The profile $\rho_0 = r_1$ is measurable: it is the essential infimum, along the ray R_θ , of the radii at which the slice $\tilde{E}_\theta$ first exits $\tilde{E}$, and $\theta \mapsto r_1(\theta)$ is measurable because for each t the set $\{\theta : r_1(\theta) > t\} = \{\theta : (0, t) \subset \tilde{E}_\theta \text{ up to } \mathcal{H}^1\text{-null}\}$ is, up to a null set, the slice projection of the open set $\tilde{E} \cap (B_t(z) \setminus \{z\})$, hence measurable (as in the measurability of R in (59)). By (74) every slice contains the innermost interval $(0, r_1)$ with $r_1 \in [\rho_i, \rho_e]$, so $\rho_0 \in [\rho_i, \rho_e]$. Passing to the lower-semicontinuous envelope of ρ_0 along rays makes $G_0 = \{r < \rho_0\}$ open; this changes ρ_0 , hence ∂G_0 , only on an $\mathcal{H}^1$ -null set of directions, so $|G_0|$ and all symmetric differences are unchanged. Then $B_{\rho_i}(z) \subset G_0$ because every ray contains $(0, \rho_i) \subset (0, r_1)$, and $G_0 \subset B_{\rho_e}(z)$ because $\rho_0 \leq \rho_e$; this is (84), and star-shapedness about z is immediate from the radial definition.

(ii). From (84), $\pi\rho_i^2 \leq |G_0| \leq \pi\rho_e^2$, so $\hat{r}_{G_0} = \sqrt{|G_0|/\pi} \in [\rho_i, \rho_e]$. For every θ both $\rho_0(\theta)$ and $\hat{r}_{G_0}$ lie in $[\rho_i, \rho_e]$, whence $|\rho_0 - \hat{r}_{G_0}| \leq \rho_e - \rho_i = \eta$ and $\|\varphi_0\|_\infty = \|\rho_0 - \hat{r}_{G_0}\|_\infty / \hat{r}_{G_0} \leq \eta/\rho_i \leq 2\eta$, using $\rho_i \geq \frac{1}{2}$ (Remark 5.11).

(iii). By construction the slice of G_0 in direction θ is $(0, \rho_0(\theta)) = (0, r_1(\theta)) \subset \tilde{E}_\theta$, so $G_0 \subset \tilde{E}$ up to $\mathcal{H}^1$ -null directions, and $|\tilde{E} \Delta G_0| = |\tilde{E} \setminus G_0|$. The bound (86) is exactly Corollary 6.5: the slice of $\tilde{E} \setminus G_0$ is the union of caps $(r_2, r_3) \cup \dots \cup (r_{2m}, r_{2m+1})$, of total area $\leq \rho_e \eta \int_{S^1} m d\theta = \frac{1}{2} \rho_e \eta \int_{S^1} (M-1) d\theta \leq C\eta(\text{Exc} + a\eta)$. $\square$

Remark 6.7 (Scales, and the role in the reduction). Specialising to the inset at level $\hat{v} \sim \tau = \sqrt{\delta_T}$ (so $\delta_{\text{iso}} \leq C\delta_T/\tau \leq C\sqrt{\delta_T}$, $\text{Exc} \leq C\sqrt{\delta_T}$, and by Theorem 2.11 $a \leq C\sqrt{\delta_{\text{iso}}} \leq C\delta_T^{1/4}$), the annulus width is $\eta \leq C(\text{Exc} + a)^{1/2} \leq Ca^{1/2} \leq C\delta_T^{1/8}$, an *absolute* small amplitude. Then Corollary 6.5 gives

$$\text{gd}(\tilde{E}) \leq C\eta(\text{Exc} + a\eta) \leq C(\delta_T^{1/8}\delta_T^{1/2} + \delta_T^{1/4}\delta_T^{1/4}) = C(\delta_T^{5/8} + \delta_T^{1/2}) \leq C\sqrt{\delta_T},$$

and Proposition 6.6 exhibits the measurable core graph G_0 of amplitude $\|\varphi_0\|_\infty \leq 2\eta \leq C\delta_T^{1/8}$ with $|\tilde{E} \Delta G_0| \leq C\sqrt{\delta_T}$. These are precisely the scales fed into the cutting estimate (Lemma 7.4) and the assembly (Theorem 9.1): the graph defect balances the level-carried closeness $|\Omega \Delta \tilde{E}| \sim \sqrt{\delta_T}$ at the optimal exponent, and the measurable-profile form of Theorem 8.1 consumes G_0 directly, with no intervening regularisation.

7. CUTTING THE CAPS LOWERS THE DEFICIT

The construction of Sections 4 to 6 produces from Ω a connected, hole-free inset $\tilde{E}$ trapped in a thin annulus $B_{\rho_i}(z) \subset \tilde{E} \subset B_{\rho_e}(z)$, together with a radial core profile whose graph $G_0 = \{z + r\omega : r < \rho_0(\omega)\}$ satisfies $G_0 \subset \tilde{E}$ and $|\tilde{E} \setminus G_0| = \text{gd}(\tilde{E})$ (Corollary 6.5). Manufacturing the honest graph from $\tilde{E}$ therefore amounts to *excising* the radial caps $\tilde{E} \setminus G_0$, all of which lie in the collar $\{\rho_i < |x - z| < \rho_e\}$. The naive worry is that removing material can only destroy torsion and hence inflate the scale-invariant deficit δ_* . The opposite is true: caps in a thin collar carry very little torsion, and the equal-area comparison ball shrinks faster than the torsion is lost, so δ_* *decreases*. This section proves the two facts that make this precise. They are sign-critical, so we prove them in full and check the inequalities digit by digit.

Throughout, for a bounded open set $D \subset \mathbb{R}^2$ the *torsion function* $u_D \in H_0^1(D)$ is the weak solution of $-\Delta u_D = 1$ in D , $u_D = 0$ on ∂D , and $T(D) := \int_D u_D$ is its torsional rigidity; for a general set of finite measure D we set $T(D) := T(\text{int } D)$ on the open representative produced by our construction, which is all we use. Recall (Theorem 2.13) that u_D is real-analytic in D and, by the comparison principle for the Laplacian, $u_D \geq 0$. The variational characterisation we use repeatedly is that u_D minimises the Dirichlet energy functional $J_D(v) := \frac{1}{2} \int_D |\nabla v|^2 - \int_D v$ over $v \in H_0^1(D)$, with

$$\min_{v \in H_0^1(D)} J_D(v) = J_D(u_D) = -\frac{1}{2} T(D), \quad \int_D |\nabla u_D|^2 = \int_D u_D = T(D), \quad (87)$$

the second identity being the Euler–Lagrange relation tested against u_D itself.

7.1. Torsion loss under radial-cap cutting. The next lemma is the analytic heart of the section: it controls the torsion lost when one cuts radial caps off a set trapped in a thin collar. Unlike a naive measure-based bound — which is false in general, since excising a zero-area set of positive capacity (a slit) changes the torsion — the estimate is variational and exploits the specific nested radial-cap geometry produced in Section 6.

Lemma 7.1 (Torsion loss under radial-cap cutting). *Let $z \in \mathbb{R}^2$ and $0 < \rho_i \leq \rho_e$, and put*

$$\tilde{\eta} := \frac{1}{4}(\rho_e^2 - \rho_i^2). \quad (88)$$

Let $G \subset E \subset \mathbb{R}^2$ be bounded open sets with

$$B_{\rho_i}(z) \subset G \subset E \subset B_{\rho_e}(z), \quad E \setminus G \subset \{\rho_i < |x - z| < \rho_e\}, \quad (89)$$

and put $g := |E \setminus G|$. Then $0 \leq u_G \leq u_E$ on G , the difference $h := u_E - u_G$ (extended by u_E on $E \setminus G$, equivalently $h = u_E - \bar{u}_G$ where $\bar{u}_G \in H_0^1(E)$ is the extension of u_G by zero) lies in $H_0^1(E)$ and is harmonic in G , and

$$0 \leq T(E) - T(G) = \int_{E \setminus G} u_E + \int_G h \leq \tilde{\eta} g + \int_G h. \quad (90)$$

Moreover the boundary-layer term $\int_G h$ is non-negative, satisfies the energy identity

$$\int_G h \leq T(E) - T(G) = \int_E |\nabla h|^2, \quad (91)$$

and obeys $0 \leq h \leq \tilde{\eta}$ throughout G , whence the crude bound $\int_G h \leq \tilde{\eta} |G|$.

Proof. Step A: the exact decomposition. Since $G \subset E$, the restriction $u_E|_G$ solves $-\Delta u_E = 1 = -\Delta u_G$ in G with $u_E \geq 0 = u_G$ on ∂G ; by the comparison principle on G the difference $h := u_E - u_G$ is then ≥ 0 , i.e. $0 \leq u_G \leq u_E$ on G . Being a difference of two solutions of $-\Delta(\cdot) = 1$, h is harmonic in G , with boundary trace $h = u_E$ on ∂G (as $u_G = 0$ there). Splitting $E = G \cup (E \setminus G)$ and using $u_G = 0$ on $E \setminus G$,

$$T(E) - T(G) = \int_E u_E - \int_G u_G = \int_{E \setminus G} u_E + \int_G (u_E - u_G) = \int_{E \setminus G} u_E + \int_G h, \quad (92)$$

which is the equality in (90); both summands are ≥ 0 , so $T(E) - T(G) \geq 0$.

Step B: the cap term. Since $E \subset B_{\rho_e}(z)$, the comparison principle gives $u_E \leq u_{B_{\rho_e}(z)}$ on E : the difference $u_{B_{\rho_e}(z)} - u_E$ is harmonic in E (both solve $-\Delta(\cdot) = 1$) and equals $u_{B_{\rho_e}(z)} \geq 0$ on ∂E , hence

is ≥ 0 in E by the minimum principle. The disc torsion function is explicit,

$$u_{B_{\rho_e}(z)}(y) = \frac{\rho_e^2 - |y - z|^2}{4}. \quad (93)$$

On the caps $E \setminus G \subset \{\rho_i < |x - z| < \rho_e\}$ one has $|x - z| \geq \rho_i$, so

$$u_E \leq \frac{\rho_e^2 - |x - z|^2}{4} \leq \frac{\rho_e^2 - \rho_i^2}{4} = \tilde{\eta} \quad \text{on } E \setminus G, \quad (94)$$

and therefore $\int_{E \setminus G} u_E \leq \tilde{\eta} g$. This is the main term and is fully rigorous; combined with (92) it gives the inequality in (90).

Step C: the boundary-layer term, energy form and crude bound. Let $\bar{u}_G \in H_0^1(E)$ denote the extension of u_G by zero outside G ; then $h = u_E - \bar{u}_G \in H_0^1(E)$ agrees with the previously defined h on G and equals u_E on $E \setminus G$. Since u_E minimises J_E on $H_0^1(E)$ and $\bar{u}_G$ minimises J_E restricted to the subspace $H_0^1(G) \subset H_0^1(E)$ (its minimum there being $J_E(\bar{u}_G) = J_G(u_G) = -\frac{1}{2}T(G)$), the parallelogram identity for the quadratic functional J_E gives the exact energy balance

$$\frac{1}{2} \int_E |\nabla h|^2 = J_E(\bar{u}_G) - J_E(u_E) = \left(-\frac{1}{2}T(G)\right) - \left(-\frac{1}{2}T(E)\right) = \frac{1}{2}(T(E) - T(G)), \quad (95)$$

using (87) for both $D = E$ and $D = G$. Hence $T(E) - T(G) = \int_E |\nabla h|^2$, the equality in (91); combined with (90) and $\int_{E \setminus G} u_E \geq 0$ this gives $\int_G h \leq T(E) - T(G) = \int_E |\nabla h|^2$.

Finally, the bounds $0 \leq h \leq \tilde{\eta}$ in G : non-negativity was shown in Step A. For the upper bound, h is harmonic in G with boundary trace $h = u_E$ on ∂G , and on ∂G one has $u_E \leq \tilde{\eta}$ — indeed $\partial G \subset \overline{B_{\rho_e}(z)}$, so the barrier (93) gives $u_E \leq u_{B_{\rho_e}(z)} \leq \tilde{\eta}$ on the part of ∂G lying in $\{|x - z| \geq \rho_i\}$ (which contains $\partial G \cap E$, the interior cap-floor boundary, since $E \setminus G$ is confined to the collar by (89)), while $u_E = 0$ on $\partial G \cap \partial E$. By the maximum principle $h \leq \tilde{\eta}$ throughout G , whence $\int_G h \leq \tilde{\eta}|G|$. $\square$

The cap term $\tilde{\eta} g$ in (90) is exactly what the deficit algebra of Lemma 7.4 can afford. The boundary-layer term $\int_G h$ is genuinely smaller in the application but requires care; we isolate it next.

Remark 7.2 (The boundary-layer term and the residual analytic point). The cap term $\int_{E \setminus G} u_E \leq \tilde{\eta} g$ of Step B is rigorous and, as we show in Lemma 7.4, by itself drives the deficit down once the harmonic boundary-layer term $\int_G h$ is controlled at the same order. The sharp description of $\int_G h$ is, by Green's identity in G (legitimate when ∂G admits an outward normal $\mathcal{H}^1$ -a.e.; for the rough star-shaped G below it is the limiting form along a smooth exhaustion),

$$\int_G h = \int_G h(-\Delta u_G) = - \int_{\partial G} h \partial_\nu u_G d\mathcal{H}^1 = \int_{\partial G} u_E |\partial_\nu u_G| d\mathcal{H}^1 = \int_\Gamma u_E |\partial_\nu u_G| d\mathcal{H}^1, \quad (96)$$

where $\partial_\nu u_G \leq 0$ on ∂G (as $u_G \geq 0 = u_E$ on ∂G) and the integral is supported on the interior cap-floor boundary $\Gamma := \partial G \cap E$, because $u_E = 0$ on the remainder $\partial G \cap \partial E$. Since $u_E \leq \tilde{\eta}$ on Γ by (94), this yields the clean bound

$$\int_G h \leq \tilde{\eta} \Phi, \quad \Phi := \int_\Gamma |\partial_\nu u_G| d\mathcal{H}^1, \quad (97)$$

where Φ is the flux of u_G through the interior cap-floor. The total flux through the *whole* boundary is exactly $\int_{\partial G} |\partial_\nu u_G| = \int_G (-\Delta u_G) = |G|$, so $\Phi \leq |G|$ unconditionally; the substance is that Γ is only the small, fold-supported part of ∂G . For a boundary with a one-sided gradient bound $\|\nabla u_G\|_{L^\infty(\partial G)} \leq C$ (an $O(1)$ quantity for the disc-trapped torsion function, by the barrier $u_G \leq u_{B_{\rho_e}(z)}$ and interior

estimates) one has $\Phi \leq C\mathcal{H}^1(\Gamma)$, so it would suffice to bound the length $\mathcal{H}^1(\Gamma)$ of the interior cap-floor by the fold content of Proposition 6.3. This is the *single* residual analytic point: the merely measurable core profile $\rho_0 = r_1$ of Corollary 6.5 carries no slope control, so $\mathcal{H}^1(\Gamma)$ — which involves the radial derivative of ρ_0 — is not controlled by the fold content (a rough graph can have large perimeter over a small angular sector). We therefore record the boundary-layer control as the explicit scoped hypothesis

$$\int_G h \leq \tilde{\eta} \Phi \leq \tilde{\eta} g \quad (98)$$

(equivalently $\Phi \leq g$: the flux through the interior cap-floor is dominated by the excised area), which holds whenever ∂G has the regularity making (96) valid and $\mathcal{H}^1(\Gamma) \leq C(\text{Exc} + a\eta)$; under (98) the cutting estimate below is unconditional. We do not claim (98) for an arbitrary measurable star-shaped G , and we flag every downstream use of it explicitly.

Remark 7.3 (The barrier constant). The width of the collar enters only through $\tilde{\eta} = \frac{1}{4}(\rho_e^2 - \rho_i^2)$, the maximal value of the disc torsion function on the inner sphere. With $\rho_i = \hat{r} - O(\eta)$ and $\rho_e = \hat{r} + O(\eta)$ one has $\tilde{\eta} = \frac{1}{4}(\rho_e + \rho_i)(\rho_e - \rho_i) \leq \frac{1}{2}\rho_e \eta \leq C\eta$, so $\tilde{\eta}$ is of the same order as the annulus width $\eta = \rho_e - \rho_i$. In the unit-scale normalisation $\rho_e = 1 + \eta$, $\rho_i = 1 - \eta$ one gets exactly $\tilde{\eta} = \eta$.

7.2. Cutting boundary material decreases δ_* . Recall the scale-invariant relative deficit of a finite-measure set D ,

$$\delta_*(D) = \frac{T(B_D) - T(D)}{T(B_D)}, \quad T(B_D) = \frac{|D|^2}{8\pi}, \quad (99)$$

where B_D is any ball with $|B_D| = |D|$ (so $T(B_D) = |D|^2/(8\pi)$ by scaling from $T(B_1) = \pi/8$). Equivalently, $\delta_*(D) = 1 - 8\pi T(D)/|D|^2$. We now show that excising radial caps confined to a thin collar lowers δ_* , provided the torsion loss per unit excised area stays below the torsion density of the larger set. The hypothesis is stated directly as this *loss-slope* condition, which is exactly what the sign-critical algebra consumes and which the geometric input of Lemma 7.1 supplies.

Lemma 7.4 (Cutting boundary material decreases δ_*). *Let $z \in \mathbb{R}^2$, $0 < \rho_i \leq \rho_e$ and $\tilde{\eta} = \frac{1}{4}(\rho_e^2 - \rho_i^2)$. Let E, G be bounded open sets with*

$$B_{\rho_i}(z) \subset G \subset E \subset B_{\rho_e}(z), \quad E \setminus G \subset \{\rho_i < |x - z| < \rho_e\}, \quad (100)$$

put $V := |E|$, $g := |E \setminus G| = V - |G|$, and suppose the torsion-loss bound

$$T(E) - T(G) \leq \lambda g \quad \text{with} \quad \lambda \leq \frac{T(E)}{|E|} \quad (101)$$

holds. Then $\delta_(G) \leq \delta_*(E)$.*

Proof. Write $T := T(E)$; then $|G| = V - g$ and, since $\emptyset \neq B_{\rho_i}(z) \subset G \subset E$, we have $0 \leq g \leq V$ and $V > 0$.

Step 1: lower bound for $T(G)$. This is the hypothesis (101) rearranged:

$$T(G) \geq T - \lambda g, \quad \lambda \leq \frac{T}{V}. \quad (102)$$

Step 2: the deficit comparison reduces to a torsion inequality. By (99), with $|E| = V$ and $|G| = V - g$,

$$\delta_*(E) - \delta_*(G) = \left(1 - \frac{8\pi T}{V^2}\right) - \left(1 - \frac{8\pi T(G)}{(V-g)^2}\right) = 8\pi \left[\frac{T(G)}{(V-g)^2} - \frac{T}{V^2} \right]. \quad (103)$$

Thus $\delta_*(G) \leq \delta_*(E)$ holds if and only if $\frac{T(G)}{(V-g)^2} \geq \frac{T}{V^2}$, i.e. (multiplying by $(V-g)^2 \geq 0$) if and only if

$$T(G) \geq T \frac{(V-g)^2}{V^2} = T \left(1 - \frac{g}{V}\right)^2. \quad (104)$$

We now prove (104). (When $g = V$, $|G| = 0$ is impossible since $G \supset B_{\rho_i}(z)$, so in fact $g < V$; but the argument below covers $g \leq V$ verbatim.)

Step 3: the sign-critical chain. Expand the right-hand side of (104):

$$T \left(1 - \frac{g}{V}\right)^2 = T - \frac{2Tg}{V} + \frac{Tg^2}{V^2}. \quad (105)$$

Subtracting (105) from the lower bound (102) for $T(G)$,

$$\begin{aligned} T(G) - T \left(1 - \frac{g}{V}\right)^2 &\geq (T - \lambda g) - \left(T - \frac{2Tg}{V} + \frac{Tg^2}{V^2}\right) \\ &= \frac{2Tg}{V} - \frac{Tg^2}{V^2} - \lambda g = g \left(\frac{2T}{V} - \frac{Tg}{V^2} - \lambda \right). \end{aligned} \quad (106)$$

We bound the bracket from below. Since $0 \leq g \leq V$ we have $0 \leq g/V \leq 1$, hence $2 - g/V \geq 1$, and therefore

$$\frac{2T}{V} - \frac{Tg}{V^2} = \frac{T}{V} \left(2 - \frac{g}{V}\right) \geq \frac{T}{V} \geq \lambda, \quad (107)$$

the last inequality being exactly the loss-slope hypothesis (101) ($\lambda \leq T/V = T(E)/|E|$). Consequently the bracket in (106) is $\geq \frac{T}{V} - \lambda \geq 0$, and since $g \geq 0$ the whole expression (106) is ≥ 0 . This is (104), which by Step 2 yields $\delta_*(G) \leq \delta_*(E)$. $\square$

Remark 7.5 (Supplying the loss-slope hypothesis from Lemma 7.1). Lemma 7.1 furnishes (101) as follows. Its hypotheses coincide with (100), and (90) gives the exact torsion loss $T(E) - T(G) = \int_{E \setminus G} u_E + \int_G h \leq \tilde{\eta} g + \int_G h$. Under the boundary-layer control (98) of Remark 7.2 — $\int_G h \leq \tilde{\eta} g$ — this reads

$$T(E) - T(G) \leq 2\tilde{\eta} g, \quad (108)$$

so (101) holds with the explicit slope $\lambda = 2\tilde{\eta}$, provided $2\tilde{\eta} \leq T(E)/|E|$. Even with only the crude bound $\int_G h \leq \tilde{\eta}|G|$ of Lemma 7.1 one obtains the loss $\leq \tilde{\eta}(g + |G|) = \tilde{\eta}|E|$, but that slope is $\tilde{\eta}|E|/g$, which is *not* below T/V at the relevant scales; the gain from (98) is precisely the localisation of the boundary-layer term to the cap region, replacing the slope $\tilde{\eta}|E|/g$ by the absolute $2\tilde{\eta}$.

Remark 7.6 (The loss-slope hypothesis (101) holds with room to spare). In the application $E = \tilde{E}$ is the inset of Definition 4.9, of measure $\mu_E = |\tilde{E}| = \pi - O(\sqrt{\delta_T}) \geq \pi/2$ and deficit $\delta_*(\tilde{E}) \leq \frac{1}{2}$ for $\delta_T \leq \delta_*$. Then the torsion density is an *absolute* constant bounded below,

$$\frac{T(\tilde{E})}{|\tilde{E}|} = (1 - \delta_*(\tilde{E})) \frac{|\tilde{E}|}{8\pi} \geq \left(1 - \frac{1}{2}\right) \cdot \frac{\pi/2}{8\pi} = \frac{1}{32},$$

using $T(\tilde{E}) = (1 - \delta_*(\tilde{E}))T(B_{\tilde{E}}) = (1 - \delta_*(\tilde{E})) \left| \tilde{E} \right|^2 / (8\pi)$ from (99). On the other hand the slope $\lambda = 2\tilde{\eta}$ supplied by Remark 7.5 vanishes: $\tilde{\eta} \leq \frac{1}{2}\rho_e \eta \leq C\eta \leq C(\delta_T/\tau)^{1/4} \rightarrow 0$ as $\delta_T \rightarrow 0$ (Theorem 5.10, with $\tau = \sqrt{\delta_T}$ giving $\eta \leq C\delta_T^{1/8}$). Hence $\lambda = 2\tilde{\eta} \leq \frac{1}{32} \leq T(\tilde{E})/\left| \tilde{E} \right|$ for all $\delta_T \leq \delta_*$, and the loss-slope condition (101) is satisfied with a fixed margin. The crucial structural point is that the comparison is between a *vanishing* slope $\lambda = 2\tilde{\eta} \rightarrow 0$ and an *absolute* torsion density $\geq 1/32$, so the inequality $\lambda \leq T(\tilde{E})/\left| \tilde{E} \right|$ holds robustly; the threshold $1/32$ is not sharp and is recorded only to fix δ_* . This is exactly where the boundary-layer control (98) enters: it is what turns the exact loss $\int_{E \setminus G} u_E + \int_G h$ into a bound with absolute slope $2\tilde{\eta}$ rather than the much larger $\tilde{\eta} \left| \tilde{E} \right|/g$.

Remark 7.7 (The application: cutting the caps). Lemma 7.4 is used in the assembly (Theorem 9.1) with $E = \tilde{E}$ and $G = G_0 = \{z + r\omega : r < \rho_0(\omega)\}$ the measurable radial core graph of Corollary 6.5 and Proposition 6.6, taken in its open star-shaped representative. The inclusions $B_{\rho_i}(z) \subset G_0 \subset \tilde{E} \subset B_{\rho_e}(z)$ hold by Corollary 6.5 and Theorem 5.10, the difference $\tilde{E} \setminus G_0$ is the union of the radial caps, all contained in the collar $\{\rho_i < |x - z| < \rho_e\}$, and the exact torsion-loss decomposition $T(\tilde{E}) - T(G_0) = \int_{\tilde{E} \setminus G_0} u_{\tilde{E}} + \int_{G_0} h \leq \tilde{\eta}g + \int_{G_0} h$ of Lemma 7.1 holds unconditionally. Under the boundary-layer control (98) of Remark 7.2 the loss is $\leq 2\tilde{\eta}g$ (Remark 7.5); together with the absolute torsion-density lower bound of Remark 7.6, the loss-slope condition (101) holds with $\lambda = 2\tilde{\eta}$, and Lemma 7.4 gives that cutting the caps lowers the deficit,

$$\delta_*(G_0) \leq \delta_*(\tilde{E}).$$

This is the one place where the geometry of Lemma 7.4 is needed: the discarded material is precisely low-torsion boundary material in the η -collar about the *single* centre z . The deduction is unconditional except for the single residual analytic point (98) isolated in Remark 7.2 — the localisation of the harmonic boundary-layer term to the cap region; the cap term $\tilde{\eta}g$ and the entire deficit algebra are rigorous.

Remark 7.8 (The other deficit step is *not* an instance of Lemma 7.4). The earlier reduction step — passing from the full superlevel set $\{u > \hat{v}\}$ to its largest connected component $\tilde{E}$, i.e. discarding the spillover components of total measure $\left| \{u > \hat{v}\} \setminus \tilde{E} \right| \leq D_I(\hat{v})^2/(16\pi^3)$ (Lemma 4.8) — is *not* covered by Lemma 7.4, and we do not claim it is. Lemma 7.4 requires $G \subset E \subset B_{\rho_e}(z)$ with $E \setminus G$ inside one collar about z ; but the discarded spillover components are distinct level components, which need not lie in the trapping ball $B_{\rho_e}(z)$ of $\tilde{E}$ (they can sit elsewhere in Ω), so the hypothesis $E \subset B_{\rho_e}(z)$ may fail for $E = \{u > \hat{v}\}$. The correct tool for that step is the deficit transfer Lemma 4.5 of Section 4 combined with the *exact* additivity of torsion over connected components (the function $u - \hat{v}$ is the common torsion function of every component of $\{u > \hat{v}\}$); this is carried out in Proposition 4.10, yielding $\delta_*(\tilde{E}) \leq (\pi^2/\mu_E^2) \delta_*(\Omega) + O(\delta_T^2/\tau^2)$ with no appeal to the collar geometry. Thus the full chain

$$\delta_*(G_0) \leq \delta_*(\tilde{E}) \leq \frac{\pi^2}{\mu_E^2} \delta_*(\Omega) + O\left(\frac{\delta_T^2}{\tau^2}\right)$$

combines Lemma 7.4 (first inequality, this section) with Lemma 4.5/Proposition 4.10 (second inequality, Section 4); the two play distinct roles and rest on distinct hypotheses.

Remark 7.9 (Why the deficit decreases — the mechanism). The single inequality (107) is the entire content. Cutting a cap of measure g costs at most λg in torsion (*linear* in g , with a small slope $\lambda = 2\tilde{\eta} \rightarrow 0$ under (98)), but it shrinks the equal-area comparison ball, lowering its torsion $T(B_G) = |G|^2 / (8\pi)$ by $\approx \frac{2T(B_E)}{V}g = \frac{V}{4\pi}g$, i.e. by a slope $\approx \frac{2T}{V}$ after accounting for the deficit factor. Because the unperturbed torsion density T/V of a near-disc is an *absolute* constant of order 1 while the collar slope $\lambda = 2\tilde{\eta}$ vanishes, the denominator $T(B_G)$ drops faster than the numerator $T(G)$, and $\delta_* = 1 - T/T(B)$ decreases. There is no square-root loss and no constant that degrades as $\delta_T \rightarrow 0$: the deficit algebra is exact and the margin in (101) only widens. The exactness of the torsion-loss *decomposition* (90) is likewise unconditional; the only inexact step is the localisation (98) of the harmonic boundary-layer term, isolated in Remark 7.2.

8. PERTURBATIVE STABILITY OF NEARLY SPHERICAL SETS

This section proves, in full and self-containedly, the perturbative torsional stability of small-amplitude radial graphs consumed by the reduction of Section 9. The result is a Fuglede-type estimate *for the torsion functional*: a planar set whose boundary is a graph of small amplitude over a circle, after the natural volume normalisation, has scale-invariant torsional deficit controlling the square of its Fraenkel asymmetry. We invoke no selection principle, no spectral black box, and no quantitative isoperimetric inequality.

The graph is taken about an *arbitrary* centre z : we do not assume that z is the barycentre, nor do we recentre the set. The only structural information beyond the volume constraint is that the centre be chosen so that the first Fourier mode of the profile — the translation mode — is *quadratically* small, $\|P_1\varphi\|_{L^2} \leq K\|\varphi\|_{L^2}^2$. This is an explicit, checkable condition that the constant and first modes play symmetric, neutral roles, and it is the natural minimal hypothesis: it holds for the barycentre exactly, and we verify in Section 9 that it holds for the L^1 -optimal centre produced by the construction. Recentring the set to enforce $P_1\varphi = 0$ is *not* used, and would be unsafe under the present hypotheses, because the profile is only Lipschitz with slope that may degenerate.

The point requiring care is that the graph produced by Section 6 has *small amplitude but no slope control*: its radial profile is only *measurable*, with no modulus of continuity and no bounded variation available. The classical nearly-spherical estimates (Fuglede [6] and their torsional analogues) assume the boundary is a C^1 - or $W^{2,\infty}$ -small graph; they do not apply here, and it is precisely to manufacture such regularity that a selection principle is otherwise introduced. We therefore give a proof valid under an L^∞ amplitude bound alone. The mechanism is an exact energy identity together with a *slope-free* annular Steklov lower bound that is diagonal in the boundary data; the coercive output is at the L^2 (Fraenkel) scale, which is all that is needed and all that survives at high frequency.

8.1. Statement and standing normalisation. For a measurable $\varphi: S^1 \rightarrow \mathbb{R}$ with $\|\varphi\|_{L^\infty} < 1$, a radius $\hat{r}_G > 0$ and a centre $z \in \mathbb{R}^2$ we write

$$G = G(z, \hat{r}_G, \varphi) := \{z + r\omega(\theta) : \theta \in S^1, 0 \leq r < \hat{r}_G(1 + \varphi(\theta))\}, \quad \omega(\theta) = (\cos \theta, \sin \theta), \quad (109)$$

the radial graph of profile φ . We recall that for a bounded open $E \subset \mathbb{R}^2$ of finite measure with torsion function $u_E \in H_0^1(E)$ ($-\Delta u_E = 1$) and $T(E) = \int_E u_E$,

$$\delta_*(E) = \frac{T(B_E) - T(E)}{T(B_E)}, \quad T(B_E) = \frac{|E|^2}{8\pi}, \quad \mathcal{A}(E) = \inf_{x_0} \frac{|E \triangle B(x_0, r_E)|}{|E|}, \quad r_E = \sqrt{|E|/\pi}, \quad (110)$$

are the scale-invariant torsional deficit and the Fraenkel asymmetry; both are invariant under translations and dilations.

Theorem 8.1 (Radial-graph torsion stability). *Fix any $K < \infty$. There are constants $\eta_G \in (0, \frac{1}{13}]$ and $C_G < \infty$, depending only on K (one may take*

$$\eta_G = \min\left\{\frac{1}{80}, \frac{1}{16(1+K)^2}\right\}, \quad C_G = 3 \quad (111)$$

) such that the following holds. Let $G = G(z, \hat{r}_G, \varphi)$ be a radial graph as in (109) about an arbitrary centre $z \in \mathbb{R}^2$, with $\varphi: S^1 \rightarrow \mathbb{R}$ measurable (so that G is a bounded open star-shaped set about z , taken in its open representative), $\|\varphi\|_{L^\infty} \leq \eta_G$, and

$$\hat{r}_G = \sqrt{|G|/\pi}.$$

Assume the quadratic mode-one bound

$$\|P_1 \varphi\|_{L^2(S^1)} \leq K \|\varphi\|_{L^2(S^1)}^2, \quad (112)$$

where P_1 is the projection onto $\{\cos \theta, \sin \theta\}$ in the polar parametrisation of ∂G about z . Then

$$\mathcal{A}(G)^2 \leq C_G \delta_*(G). \quad (113)$$

The statement is scale-invariant; G need not have area π . It uses no property of the original set Ω and no selection. No slope, perimeter, curvature, barycentre or recentering hypothesis is made: only the volume constraint $\hat{r}_G = \sqrt{|G|/\pi}$, the L^∞ amplitude bound, and the quadratic mode-one bound (112).

Remark 8.2. The hypotheses are on the L^∞ amplitude of φ , the volume constraint, and the quadratic mode-one bound (112): no slope, perimeter, curvature, C^1 or C^2 bound is assumed. The amplitude bound serves only to trap ∂G in a thin annulus; the volume constraint removes the constant mode quantitatively; and (112) removes the first (translation) mode quantitatively, playing exactly the role that a barycentre normalisation would. This is exactly the regime delivered by Proposition 6.6, where the amplitude is $O(\delta_T^{1/8}) \rightarrow 0$.

Remark 8.3 (The mode-one hypothesis and the barycentre). The quadratic mode-one bound (112) holds, with K absolute, whenever z is the barycentre of G : then $\int_{S^1} R^3 \omega d\theta = 0$, and expanding $R^3 = 1 + 3\varphi + 3\varphi^2 + \varphi^3$ together with $\int_{S^1} \omega = 0$ gives $3 \int \varphi \omega = - \int (3\varphi^2 + \varphi^3) \omega$, so $\|P_1 \varphi\|_{L^2} \leq \frac{1}{\sqrt{\pi}}(1 + \frac{1}{3}\|\varphi\|_{L^\infty})\|\varphi\|_{L^2}^2$. Thus Theorem 8.1 contains the barycentre-normalised case as the special instance $K = \frac{1}{\sqrt{\pi}}(1 + \frac{1}{39})$ (using $\|\varphi\|_{L^\infty} \leq \frac{1}{13}$). The point of stating it as the explicit condition (112) is that it can also be verified *at a prescribed centre* — in particular at the L^1 -optimal centre of the construction (Section 9) — without recentering the set, which under an L^∞ amplitude bound alone need not preserve single-valuedness of the radial graph.

Since δ_* and $\mathcal{A}$ are dilation- and translation-invariant, replacing G by $\hat{r}_G^{-1}(G - z)$ we assume throughout the proof

$$z = 0, \quad \hat{r}_G = 1, \quad |G| = \pi, \quad (114)$$

where the last equality uses the hypothesis $\hat{r}_G = \sqrt{|G|/\pi}$. (This rescaling fixes the origin at the prescribed centre z and leaves (112) unchanged, since both sides scale the same way under $\varphi \mapsto \hat{r}_G^{-1}\varphi$ once $\hat{r}_G$ is normalised to 1.) Thus $R := 1 + \varphi$ satisfies the single volume constraint

$$\int_{S^1} (R^2 - 1) d\theta = 0 \quad \left(\text{i.e.} \quad \int_{S^1} (2\varphi + \varphi^2) d\theta = 0 \right), \quad (115)$$

which is just $|G| = \frac{1}{2} \int R^2 = \pi$. We do *not* impose any barycentre constraint; the first Fourier mode of φ is controlled instead through the quadratic mode-one hypothesis (112). We write $u_0(x) := \frac{1}{4}(1 - |x|^2)$ for the disc torsion, $-\Delta u_0 = 1$, $u_0|_{\partial B_1} = 0$, $T(B_1) = \pi/8$. Expand φ and the boundary datum

$$h(\theta) := \frac{R(\theta)^2 - 1}{4} = \frac{2\varphi + \varphi^2}{4} = \frac{\varphi}{2} + \frac{\varphi^2}{4} \quad (116)$$

in real Fourier series, with P_k the projection onto the k -th mode $\{\cos k\theta, \sin k\theta\}$ (for $k = 0$ the constants) and $P_{\geq 2} = \sum_{k \geq 2} P_k$. Recall $\|\psi\|_{L^2(S^1)}^2 = \sum_{k \geq 0} \|P_k \psi\|_{L^2}^2$.

8.2. An exact energy identity. Write the energy functional and the (negative) torsion energy

$$J_E(v) := \frac{1}{2} \int_E |\nabla v|^2 - \int_E v \quad (v \in H_0^1(E)), \quad \mathcal{E}(E) := \min_{v \in H_0^1(E)} J_E(v) = J_E(u_E) = -\frac{1}{2} T(E), \quad (117)$$

the last equality because $\int_E |\nabla u_E|^2 = \int_E u_E = T(E)$. Thus $\mathcal{E}(G) - \mathcal{E}(B_1) = \frac{1}{2}(T(B_1) - T(G))$, and on the normalisation (114), $\delta_*(G) = \frac{8}{\pi}(\frac{\pi}{8} - T(G)) = \frac{16}{\pi}(\mathcal{E}(G) - \mathcal{E}(B_1))$.

Lemma 8.4 (Exact graph identity). *Let G be any bounded open star-shaped (about 0) set with $B_{\rho_i} \subset G \subset B_{\rho_e}$ and measurable radial profile $R = \rho: S^1 \rightarrow [\rho_i, \rho_e]$, normalised by (114). Let $z := u_G - u_0 \in H^1(G)$. Then z is harmonic in G , with $z = h$ as its a.e.-radial boundary trace (h from (116)), and*

$$\mathcal{E}(G) - \mathcal{E}(B_1) = \frac{1}{2} \int_G |\nabla z|^2 dx - \frac{1}{32} \int_{S^1} (R^4 - 1) d\theta. \quad (118)$$

Proof. Both u_G and u_0 solve $-\Delta(\cdot) = 1$, so $z = u_G - u_0$ is harmonic in G . The boundary trace is read off ray by ray: for a.e. θ the restriction of u_G to the radial segment $r \mapsto u_G(r\omega(\theta))$ lies in $H^1(0, \rho(\theta))$, is absolutely continuous, and (being the restriction of an $H_0^1(G)$ function) has radial limit 0 at the endpoint $r = \rho(\theta)$; hence $z = u_G - u_0$ has radial limit $-u_0(\rho\omega) = \frac{1}{4}(\rho^2 - 1) = h(\theta)$ there. We emphasise, however, that the identity (118) below uses *no* boundary regularity: it is proved by a boundary-term-free Green identity valid for any bounded open G .

A weak Green identity. Since $u_0 \in C^\infty(\mathbb{R}^2)$ with $-\Delta u_0 = 1$ and $u_G \in H_0^1(G)$,

$$\int_G \nabla u_0 \cdot \nabla u_G dx = \int_G u_G dx = T(G). \quad (119)$$

Indeed, choose $\phi_j \in C_c^\infty(G)$ with $\phi_j \rightarrow u_G$ in $H_0^1(G)$; integrating by parts on the compact support of ϕ_j (no boundary term), $\int_G \nabla u_0 \cdot \nabla \phi_j = \int_G (-\Delta u_0) \phi_j = \int_G \phi_j$, and passing to the limit in $H_0^1(G)$ (the left side is continuous in $\nabla \phi_j$ because $\nabla u_0 \in L^\infty(G)$, the right side in ϕ_j) yields (119). No boundary regularity of G is invoked.

Polar integrals for the smooth datum. By Fubini in polar coordinates, valid for the measurable profile ρ ,

$$\int_G |\nabla u_0|^2 dx = \int_{S^1} \int_0^\rho \frac{r^2}{4} r dr d\theta = \frac{1}{16} \int_{S^1} \rho^4 d\theta, \quad (120)$$

$$\int_G u_0 dx = \int_{S^1} \int_0^\rho \frac{1-r^2}{4} r dr d\theta = \int_{S^1} \left(\frac{\rho^2}{8} - \frac{\rho^4}{16} \right) d\theta, \quad (121)$$

using $|\nabla u_0|^2 = r^2/4$ and $u_0 = \frac{1}{4}(1-r^2)$.

Expansion of $T(G)$. Write $\nabla u_G = \nabla u_0 + \nabla z$. Then

$$T(G) = \int_G |\nabla u_G|^2 = \int_G |\nabla u_0|^2 + 2 \int_G \nabla u_0 \cdot \nabla z + \int_G |\nabla z|^2.$$

By (119) and (120),

$$\int_G \nabla u_0 \cdot \nabla z = \int_G \nabla u_0 \cdot \nabla u_G - \int_G |\nabla u_0|^2 = T(G) - \frac{1}{16} \int_{S^1} \rho^4 d\theta.$$

Substituting,

$$T(G) = \frac{1}{16} \int_{S^1} \rho^4 + 2 \left(T(G) - \frac{1}{16} \int_{S^1} \rho^4 \right) + \int_G |\nabla z|^2,$$

so that

$$T(G) = \frac{1}{16} \int_{S^1} \rho^4 d\theta - \int_G |\nabla z|^2 dx. \quad (122)$$

(On the disc $\rho \equiv 1$ this reads $T(B_1) = \frac{1}{16} \cdot 2\pi = \pi/8$, as it must.)

Conversion to $\mathcal{E}$. Recall $\mathcal{E}(E) = -\frac{1}{2}T(E)$ and, on the normalisation (114), $\mathcal{E}(B_1) = -\frac{\pi}{16}$. From (122),

$$\mathcal{E}(G) - \mathcal{E}(B_1) = -\frac{1}{2}T(G) + \frac{\pi}{16} = -\frac{1}{32} \int_{S^1} \rho^4 d\theta + \frac{1}{2} \int_G |\nabla z|^2 + \frac{\pi}{16},$$

and since $\frac{\pi}{16} = \frac{1}{32} \int_{S^1} 1 d\theta$, the two $\rho^4/\text{constant}$ terms combine into $-\frac{1}{32} \int_{S^1} (\rho^4 - 1)$. With $R = \rho$ this is exactly (118). $\square$

8.3. A slope-free annular Steklov lower bound. The next lemma is the heart of the matter: it bounds the Dirichlet energy of *any* H^1 function with prescribed outer trace h from below by a diagonal, L^2 -coercive functional of h — with no reference to the slope of ∂G .

Assume G is a bounded open star-shaped set with measurable profile $1-\eta \leq R \leq 1+\eta$, $0 < \eta \leq \frac{1}{4}$, and set $\rho := 1-\eta$ (so that $B_\rho \subset G$). Expand $h = \sum_{k \geq 0} h_k$ into Fourier modes.

Lemma 8.5 (Annular Steklov lower bound). *Let $w \in H^1(G)$ have, for a.e. θ , a radial trace $w(R(\theta)\theta) = h(\theta)$ at the outer endpoint of the ray. Then*

$$\int_G |\nabla w|^2 dx \geq b(\eta) \sum_{k \geq 2} \|h_k\|_{L^2(S^1)}^2, \quad b(\eta) := \frac{2\mu(\eta)}{2 + \mu(\eta)}, \quad \mu(\eta) := \frac{1-\eta}{2\eta}, \quad (123)$$

and $b(\eta) \uparrow 2$ as $\eta \downarrow 0$; in particular $b(\eta) \geq \frac{3}{2}$ whenever $\eta \leq \frac{1}{13}$.

Proof. Let $f(\theta) := w(\rho\theta)$ be the trace of w on the inner circle $\partial B_\rho \subset G$ (it lies in G since $\rho = 1-\eta \leq R$). Split $\int_G |\nabla w|^2 = \int_{B_\rho} |\nabla w|^2 + \int_{G \setminus B_\rho} |\nabla w|^2$.

Inner ball. Among all $H^1(B_\rho)$ functions with trace f on ∂B_ρ the Dirichlet energy is minimised by the harmonic extension, so with $f = \sum_k f_k$,

$$\int_{B_\rho} |\nabla w|^2 \geq \sum_{k \geq 1} k \|f_k\|_{L^2(S^1)}^2 \quad (124)$$

(in the plane the harmonic extension of the mode f_k is $(r/\rho)^k$ times it, with Dirichlet energy $k\|f_k\|_{L^2}^2$, independent of ρ).

Annular shell. By Fubini, for a.e. θ the restriction $r \mapsto w(r\omega(\theta))$ lies in $H^1(\rho, R(\theta))$, is absolutely continuous on $[\rho, R(\theta)]$, equals $f(\theta)$ at $r = \rho$, and has radial limit $h(\theta)$ at the outer endpoint $r = R(\theta)$ by hypothesis; no regularity of the profile R is used. Along the radial segment $r \in [\rho, R(\theta)]$ Cauchy–Schwarz then gives

$$|h(\theta) - f(\theta)|^2 = \left(\int_\rho^{R(\theta)} \partial_r w \, dr \right)^2 \leq \left(\int_\rho^{R(\theta)} |\partial_r w|^2 r \, dr \right) \left(\int_\rho^{R(\theta)} \frac{dr}{r} \right).$$

Since $R \leq 1 + \eta$, $\rho = 1 - \eta$ and $\log \frac{1+t}{1-t} \leq \frac{2t}{1-t}$ for $t \in [0, \frac{1}{4}]$, $\int_\rho^R \frac{dr}{r} = \log \frac{R}{\rho} \leq \log \frac{1+\eta}{1-\eta} \leq \frac{2\eta}{1-\eta} = \mu(\eta)^{-1}$. Hence $\int_\rho^{R(\theta)} |\partial_r w|^2 r \, dr \geq \mu(\eta) |h(\theta) - f(\theta)|^2$, and integrating in θ ,

$$\int_{G \setminus B_\rho} |\nabla w|^2 \geq \mu(\eta) \|h - f\|_{L^2(S^1)}^2 = \mu(\eta) \sum_{k \geq 0} \|h_k - f_k\|_{L^2}^2. \quad (125)$$

Mode-by-mode minimisation. Adding (124) and (125) (and dropping the $k = 0$ shell term) gives the lower bound $\sum_{k \geq 1} (k\|f_k\|^2 + \mu\|h_k - f_k\|^2)$. Minimising $k|f_k|^2 + \mu|h_k - f_k|^2$ over f_k yields $\frac{k\mu}{k+\mu}|h_k|^2$. The coefficient $\frac{k\mu}{k+\mu}$ is increasing in k , so for $k \geq 2$ it is at least its value at $k = 2$, namely $\frac{2\mu}{2+\mu} = b(\eta)$. This is (123). Finally $b(\eta) \geq \frac{3}{2} \iff \mu \geq 6 \iff \frac{1-\eta}{2\eta} \geq 6 \iff \eta \leq \frac{1}{13}$. $\square$

8.4. Removing the neutral modes.

Lemma 8.6 (Neutral modes). *Assume $\|\varphi\|_{L^\infty} \leq \frac{1}{13}$, the volume constraint (115), and the mode-one hypothesis (112) with parameter K . Then, with $C_0 := \frac{1}{2\sqrt{2\pi}} + K$,*

$$\|P_0\varphi\|_{L^2} + \|P_1\varphi\|_{L^2} \leq C_0 \|\varphi\|_{L^2}^2. \quad (126)$$

Consequently, if $\|\varphi\|_{L^2} \leq \varepsilon_1 := 1/(\sqrt{32} C_0)$, then $\|P_{\geq 2}\varphi\|_{L^2}^2 \geq \frac{15}{16} \|\varphi\|_{L^2}^2$.

Proof. The volume constraint $\int_{S^1} (2\varphi + \varphi^2) = 0$ gives $2 \int \varphi = - \int \varphi^2$, so the mean obeys $|\bar{\varphi}| = \frac{1}{2\pi} \int \varphi = \frac{1}{4\pi} \int \varphi^2$ and $\|P_0\varphi\|_{L^2} = \sqrt{2\pi} |\bar{\varphi}| = \frac{1}{2\sqrt{2\pi}} \|\varphi\|_{L^2}^2$. The first-mode term is bounded directly by the hypothesis (112), $\|P_1\varphi\|_{L^2} \leq K \|\varphi\|_{L^2}^2$. Adding the two bounds gives (126) with $C_0 = \frac{1}{2\sqrt{2\pi}} + K$. Then $\|P_0\varphi\|^2 + \|P_1\varphi\|^2 \leq 2C_0^2 \|\varphi\|_{L^2}^4 \leq \frac{1}{16} \|\varphi\|_{L^2}^2$ once $\|\varphi\|_{L^2}^2 \leq \varepsilon_1^2 = 1/(32C_0^2)$, and the last claim follows from $\|P_{\geq 2}\varphi\|^2 = \|\varphi\|^2 - \|P_0\varphi\|^2 - \|P_1\varphi\|^2$. $\square$

Lemma 8.7 (Boundary datum and moment). *If $\|\varphi\|_{L^\infty} \leq \eta \leq \frac{1}{13}$ and (115) holds, then*

$$\int_{S^1} (R^4 - 1) \, d\theta \leq (4 + 4\eta + \eta^2) \|\varphi\|_{L^2}^2, \quad (127)$$

$$\|P_{\geq 2}h\|_{L^2}^2 \geq \frac{1}{4} \|P_{\geq 2}\varphi\|_{L^2}^2 - \frac{1}{4} \eta \|\varphi\|_{L^2}^2. \quad (128)$$

Proof. Expand $R^4 - 1 = 4\varphi + 6\varphi^2 + 4\varphi^3 + \varphi^4$. By (115), $4 \int \varphi = -2 \int \varphi^2$, so

$$\int_{S^1} (R^4 - 1) = 4 \int \varphi^2 + \int (4\varphi^3 + \varphi^4) \leq (4 + 4\eta + \eta^2) \|\varphi\|_{L^2}^2,$$

using $\int |4\varphi^3 + \varphi^4| \leq (4\eta + \eta^2) \int \varphi^2$; this is (127). For (128), from (116), $P_{\geq 2}h = \frac{1}{2}P_{\geq 2}\varphi + \frac{1}{4}P_{\geq 2}(\varphi^2)$ with $\|P_{\geq 2}(\varphi^2)\|_{L^2} \leq \|\varphi^2\|_{L^2} \leq \|\varphi\|_{L^\infty} \|\varphi\|_{L^2} \leq \eta \|\varphi\|_{L^2}$. Hence, by the triangle inequality, $\|P_{\geq 2}h\|_{L^2} \geq \frac{1}{2}\|P_{\geq 2}\varphi\|_{L^2} - \frac{1}{4}\eta\|\varphi\|_{L^2}$, and both terms are nonnegative; squaring and discarding the nonnegative square term,

$$\|P_{\geq 2}h\|_{L^2}^2 \geq \frac{1}{4}\|P_{\geq 2}\varphi\|_{L^2}^2 - 2 \cdot \frac{1}{2} \cdot \frac{1}{4}\eta\|P_{\geq 2}\varphi\|_{L^2}\|\varphi\|_{L^2} \geq \frac{1}{4}\|P_{\geq 2}\varphi\|_{L^2}^2 - \frac{1}{4}\eta\|\varphi\|_{L^2}^2,$$

using $\|P_{\geq 2}\varphi\|_{L^2} \leq \|\varphi\|_{L^2}$ in the last step. $\square$

8.5. Proof of Theorem 8.1. Fix K and set $\eta_G := \min\{\frac{1}{80}, \frac{1}{16(1+K)^2}\}$ as in (111). We prove (113) for every φ satisfying the hypotheses of the theorem, i.e. the volume constraint (115), the mode-one bound (112), and

$$\|\varphi\|_{L^\infty} \leq \eta_G \leq \frac{1}{80}. \quad (129)$$

We first check that Lemmas 8.6 and 8.7 apply with $\eta = \eta_G \leq \frac{1}{13}$. From (129), $\|\varphi\|_{L^2} \leq \sqrt{2\pi} \|\varphi\|_{L^\infty} \leq \sqrt{2\pi} \eta_G$. With $C_0 = \frac{1}{2\sqrt{2\pi}} + K < 1 + K$ and $\varepsilon_1 = 1/(\sqrt{32}C_0)$ from Lemma 8.6, the smallness $\|\varphi\|_{L^2} \leq \varepsilon_1$ needed there holds because

$$\sqrt{2\pi} \eta_G \cdot \sqrt{32} C_0 < \sqrt{64\pi} (1 + K) \eta_G \leq \sqrt{64\pi} (1 + K) \frac{1}{16(1 + K)^2} = \frac{\sqrt{64\pi}}{16(1 + K)} < 1,$$

using $\sqrt{64\pi} < 16$ and $1 + K \geq 1$. Thus $\|\varphi\|_{L^2} < \varepsilon_1$ and Lemma 8.6 gives $\|P_{\geq 2}\varphi\|^2 \geq \frac{15}{16}\|\varphi\|^2$, while Lemma 8.7 applies with $\eta = \eta_G$. In the application, Proposition 6.6 delivers $\|\varphi\|_{L^\infty} \leq C\delta_T^{1/8} \rightarrow 0$, so (129) holds for δ_T small, and Section 9 supplies a centre for which (112) holds.

By Lemma 8.5 applied to $w = z$ (legitimate: z is harmonic with outer trace h , and $1 - \eta_G \leq R \leq 1 + \eta_G$ with $\eta_G \leq \frac{1}{13}$, so $b(\eta_G) \geq \frac{3}{2}$), then (128) and the $\|P_{\geq 2}\varphi\|^2 \geq \frac{15}{16}\|\varphi\|^2$ of Lemma 8.6,

$$\int_G |\nabla z|^2 \geq \frac{3}{2} \|P_{\geq 2}h\|_{L^2}^2 \geq \frac{3}{2} \left(\frac{1}{4} \cdot \frac{15}{16} - \frac{1}{4}\eta_G \right) \|\varphi\|_{L^2}^2 = \left(\frac{45}{128} - \frac{3}{8}\eta_G \right) \|\varphi\|_{L^2}^2. \quad (130)$$

Combining (130) and the moment bound (127) in the exact identity (118),

$$\begin{aligned} \mathcal{E}(G) - \mathcal{E}(B_1) &\geq \frac{1}{2} \left(\frac{45}{128} - \frac{3}{8}\eta_G \right) \|\varphi\|_{L^2}^2 - \frac{1}{32} (4 + 4\eta_G + \eta_G^2) \|\varphi\|_{L^2}^2 \\ &= \left(\frac{45}{256} - \frac{1}{8} - \frac{5}{16}\eta_G - \frac{1}{32}\eta_G^2 \right) \|\varphi\|_{L^2}^2 = \left(\frac{13}{256} - \frac{5}{16}\eta_G - \frac{1}{32}\eta_G^2 \right) \|\varphi\|_{L^2}^2. \end{aligned} \quad (131)$$

The bracket is decreasing in η_G , so its minimum over $\eta_G \leq \frac{1}{80}$ is attained at $\eta_G = \frac{1}{80}$, where it equals $\frac{13}{256} - \frac{5}{1280} - \frac{1}{204800} = 0.04687 \geq 0.046 =: c_0$; hence for every admissible $\eta_G \leq \frac{1}{80}$,

$$\mathcal{E}(G) - \mathcal{E}(B_1) \geq c_0 \|\varphi\|_{L^2}^2, \quad c_0 = 0.046, \quad (132)$$

and therefore, by $\delta_*(G) = \frac{16}{\pi}(\mathcal{E}(G) - \mathcal{E}(B_1))$,

$$\delta_*(G) \geq \frac{16c_0}{\pi} \|\varphi\|_{L^2}^2 = \frac{0.736}{\pi} \|\varphi\|_{L^2}^2 \geq 0.234 \|\varphi\|_{L^2}^2. \quad (133)$$

Finally we bound the asymmetry. Comparing G with B_1 ,

$$|G \triangle B_1| = \int_{S^1} \left| \int_0^R r dr - \int_0^1 r dr \right| d\theta = \frac{1}{2} \int_{S^1} |R^2 - 1| d\theta = \frac{1}{2} \int_{S^1} |2\varphi + \varphi^2| d\theta \leq (1 + \frac{\eta_G}{2}) \int_{S^1} |\varphi| d\theta,$$

so by Cauchy–Schwarz $|G \triangle B_1| \leq (1 + \frac{\eta_G}{2}) \sqrt{2\pi} \|\varphi\|_{L^2}$, and since $\mathcal{A}(G) \leq |G \triangle B_1|/\pi$,

$$\mathcal{A}(G)^2 \leq \frac{2(1 + \eta_G/2)^2}{\pi} \|\varphi\|_{L^2}^2 \leq \frac{2(1 + 1/160)^2}{\pi} \|\varphi\|_{L^2}^2 \leq 0.645 \|\varphi\|_{L^2}^2, \quad (134)$$

where in the last two steps we used $\eta_G \leq \frac{1}{80}$. Combining (134) with (133),

$$\mathcal{A}(G)^2 \leq 0.645 \|\varphi\|_{L^2}^2 \leq \frac{0.645}{0.234} \delta_*(G) = 2.76 \delta_*(G) \leq 3 \delta_*(G),$$

which is (113) with $C_G = 3$. □

Remark 8.8 (Sharpness and what is used). The exponent 2 in (113) is sharp: for $\varphi = a \cos 2\theta$ (a small ellipse) one has $\mathcal{A}(G)^2 \sim c_1 a^2$ and $\delta_*(G) \sim c_2 a^2$ as $a \rightarrow 0$ with $c_1, c_2 > 0$, so $\mathcal{A}(G)^2/\delta_*(G)$ tends to a finite positive limit and no exponent larger than 2 can hold. The proof uses only: the exact energy identity Lemma 8.4 (integration by parts and the radial-graph area element); the Dirichlet principle and a one-dimensional Cauchy–Schwarz along radii (Lemma 8.5); the harmonicity of the disc modes $r^k e^{ik\theta}$; Parseval; the volume constraint; and the quadratic mode-one bound (112). There is *no* quantitative isoperimetric inequality, *no* selection or symmetrisation, *no* second shape derivative, *no* barycentre recentring of the set, and crucially *no* slope or curvature control on ∂G — only the L^∞ amplitude bound. The coercive output is at the L^2 (Fraenkel) scale; the stronger $\dot{H}^{1/2}$ bound is false at this level of generality (high-frequency ripples) but is not needed. The constants (111) are not optimised; the threshold $\eta \leq 1/13$ for $b(\eta) \geq 3/2$ in Lemma 8.5 is the only place the amplitude enters the coercivity, and any smaller absolute η_G is admissible, which is all Section 9 requires since the inset amplitude is $O(\delta_T^{1/8}) \rightarrow 0$.

8.6. The translation mode: reduction of the mode-one hypothesis. We record what the hypothesis (112) is really controlling, and how far one can go without it. The asymmetry side requires no mode-one assumption at all: optimising the comparison disc over translations removes the first Fourier mode automatically, up to an error of the natural quadratic order.

Lemma 8.9 (Translation-reduced asymmetry). *Let $G = G(0, 1, \varphi)$ be normalised as in (114), with φ measurable, $\|\varphi\|_{L^\infty} \leq \eta_G \leq \frac{1}{13}$, and the volume constraint (115). Let $q \in \mathbb{R}^2$ be the first Fourier coefficient vector of φ , i.e. $\omega(\theta) \cdot q = P_1 \varphi(\theta)$. Then $B(q, 1)$ is star-shaped about 0 and*

$$\mathcal{A}(G) \leq \frac{|G \triangle B(q, 1)|}{\pi} \leq C(\|P_{\geq 2} \varphi\|_{L^2} + \|\varphi\|_{L^2}^2), \quad \text{hence} \quad \mathcal{A}(G)^2 \leq C(\|P_{\geq 2} \varphi\|_{L^2}^2 + \|\varphi\|_{L^2}^4), \quad (135)$$

with C absolute and no mode-one hypothesis.

Proof. Since $|q| = \|P_1 \varphi\|_{L^2}/\sqrt{\pi} \leq \|\varphi\|_{L^2}/\sqrt{\pi} \leq \sqrt{2} \eta_G < 1$, the origin lies in $B(q, 1)$, which is therefore star-shaped about 0 with polar radius $\rho_q(\theta)$ given by $|\rho_q \omega - q| = 1$, i.e. $\rho_q^2 - 2(\omega \cdot q) \rho_q + |q|^2 = 1$, so $\rho_q^2 = 1 - |q|^2 + 2(\omega \cdot q) \rho_q$ and $\rho_q = (\omega \cdot q) + \sqrt{1 - |q|^2 + (\omega \cdot q)^2} = 1 + \omega \cdot q + O(|q|^2)$. Hence

$$\rho_q^2 = 1 - |q|^2 + 2(\omega \cdot q) + 2(\omega \cdot q)^2 + r, \quad |r| \leq C|q|^3,$$

with C absolute for $|q| \leq \sqrt{2}/13$. Both G and $B(q, 1)$ are star-shaped about 0 with measurable radial profiles $R = 1 + \varphi$ and ρ_q , so the polar area formula gives

$$|G \triangle B(q, 1)| = \frac{1}{2} \int_{S^1} |R^2 - \rho_q^2| d\theta, \quad R^2 - \rho_q^2 = 2(\varphi - \omega \cdot q) + \varphi^2 + |q|^2 - 2(\omega \cdot q)^2 - r.$$

Therefore, by the triangle inequality and $\omega \cdot q = P_1 \varphi$,

$$|G \triangle B(q, 1)| \leq \int_{S^1} |\varphi - P_1 \varphi| d\theta + \frac{1}{2} \int_{S^1} (\varphi^2 + |q|^2 + 2(\omega \cdot q)^2) d\theta + \frac{1}{2} \int_{S^1} |r| d\theta.$$

For the first term, $\varphi - P_1 \varphi = P_0 \varphi + P_{\geq 2} \varphi$, so by Cauchy–Schwarz $\int_{S^1} |\varphi - P_1 \varphi| \leq \sqrt{2\pi} (\|P_0 \varphi\|_{L^2} + \|P_{\geq 2} \varphi\|_{L^2})$; and the volume constraint gives $\|P_0 \varphi\|_{L^2} = \sqrt{2\pi} |\bar{\varphi}| = \frac{1}{\sqrt{2\pi}} \|\varphi\|_{L^2}^2$ (as in Lemma 8.6). For the second term, $\int_{S^1} \varphi^2 = \|\varphi\|_{L^2}^2$, $\int_{S^1} |q|^2 = 2\pi |q|^2$ and $\int_{S^1} (\omega \cdot q)^2 = \pi |q|^2$, so it equals $\frac{1}{2} \|\varphi\|_{L^2}^2 + 2\pi |q|^2 = \frac{1}{2} \|\varphi\|_{L^2}^2 + 2\|P_1 \varphi\|_{L^2}^2$. The third term is $\leq \frac{1}{2} \cdot 2\pi C |q|^3 \leq C \|\varphi\|_{L^2}^3$. Collecting, and using $\|P_1 \varphi\|_{L^2}^2 \leq \|\varphi\|_{L^2}^2$ and $\|\varphi\|_{L^2}^3 \leq \eta_G \|\varphi\|_{L^2}^2$,

$$|G \triangle B(q, 1)| \leq \sqrt{2\pi} \|P_{\geq 2} \varphi\|_{L^2} + C \|\varphi\|_{L^2}^2.$$

Dividing by π (and $\mathcal{A}(G) \leq |G \triangle B(q, 1)|/|G| = |G \triangle B(q, 1)|/\pi$) gives the first bound in (135); squaring and using $2\|P_{\geq 2} \varphi\| \|\varphi\|^2 \leq \|P_{\geq 2} \varphi\|^2 + \|\varphi\|^4$ gives the second. $\square$

Remark 8.10 (What remains: the translation mode is a quartic direction). Lemma 8.9 isolates the exact role of (112). The deficit lower bound coerces the modes $k \geq 2$: keeping the $k = 1$ term in Lemma 8.5 (coefficient $c_1(\eta) = \mu/(1 + \mu) = 1 - O(\eta)$) and matching it against the mode-one part of the moment term (127) produces, for *any* centre and with no mode-one hypothesis, the cancellation

$$\delta_*(G) \geq c \|P_{\geq 2} \varphi\|_{L^2}^2 - C \eta \|\varphi\|_{L^2}^2, \quad (136)$$

with c, C absolute. Comparing (136) with the asymmetry bound (135), the estimate $\mathcal{A}(G)^2 \leq C_G \delta_*(G)$ would follow from the single *quartic translation bound*

$$\|P_1 \varphi\|_{L^2}^4 \leq C \delta_*(G) \quad (\text{equivalently } \delta_*(G) \geq c \|P_1 \varphi\|_{L^2}^4), \quad (137)$$

which is exactly the statement that translating the disc is a direction of *vanishing first variation and strictly positive second variation* of the torsional deficit. The hypothesis (112), $\|P_1 \varphi\|_{L^2} \leq K \|\varphi\|_{L^2}^2$, implies (137) a fortiori (it makes $\|\varphi\|_{L^2}^2 \leq C \|P_{\geq 2} \varphi\|_{L^2}^2$, absorbing the $-C\eta \|\varphi\|^2$ loss in (136)), which is why we adopt it. We emphasise that (137) is *not* delivered by the slope-free energy method of this section: the translation contributes to δ_* only at order $\|P_1 \varphi\|^4 = O(\eta^2) \|P_1 \varphi\|^2$, which lies below the $O(\eta) \|P_1 \varphi\|^2$ loss of the one-sided annular Steklov bound Lemma 8.5 and of the cubic remainder in (127); extracting it would require an exact (rather than one-sided) second-order computation of the harmonic energy under the domain perturbation. This is the precise content of the residual recorded as (154) in Section 9.

9. PROOF OF THE MAIN THEOREM

We now assemble the constructions of §§4–7 with the perturbative nearly-spherical estimate of §8. Throughout, $\Omega \subset \mathbb{R}^2$ is bounded open with $|\Omega| = \pi$, $u \in H_0^1(\Omega)$ is its torsion function, $\delta_T = \delta_T(\Omega) = \frac{\pi}{8} - T(\Omega) \geq 0$ is the Saint–Venant deficit, and $\delta_*(\Omega) = \frac{8}{\pi} \delta_T$ is the scale-invariant relative torsion deficit. We write C for an absolute constant whose value may change from line to

line, and η_G, C_G for the constants of the radial-graph theorem Theorem 8.1, which depend only on the absolute mode-one constant K of (154) below and are therefore themselves absolute. The threshold $\delta_\star > 0$ is absolute and may be shrunk finitely many times.

The section has two results. Theorem 9.1 reduces a near-extremal Ω to a measurable small-amplitude radial graph G that inherits the torsion deficit and is close to Ω in measure; this is the entire payload of §§4–7. Theorem 9.2 then feeds G into Theorem 8.1 and transfers the resulting asymmetry bound back to Ω by the triangle inequality.

9.1. The reduction. The construction selects an inset level $\hat{v} \sim \sqrt{\delta_T}$, passes to the trapping annulus, cuts the radial folds to make an honest graph, and optimises the free scale τ at $\tau = \sqrt{\delta_T}$. We carry τ symbolic through the estimates and set $\tau = \sqrt{\delta_T}$ only at the end, so that the balance producing the exponent $\frac{1}{2}$ is visible.

Theorem 9.1 (Sharp reduction to a graph inset). *There is an absolute $\delta_\star > 0$ such that the following holds. Let $\Omega \subset \mathbb{R}^2$ be bounded open with $|\Omega| = \pi$ and $\delta_T \leq \delta_\star$, and put $\tau := \sqrt{\delta_T}$. Then there exist a point $z \in \mathbb{R}^2$ and a bounded open star-shaped radial graph*

$$G = \{z + r\omega : 0 \leq r < \rho(\omega), \omega \in S^1\}, \quad \rho: S^1 \rightarrow (0, \infty) \text{ measurable,}$$

with $\hat{r}_G := \sqrt{|G|/\pi}$, such that

- (i) (small amplitude) $\left\| \frac{\rho}{\hat{r}_G} - 1 \right\|_{L^\infty(S^1)} \leq C \delta_T^{1/8} \leq \eta_G$;
- (ii) (controlled deficit) $\delta_*(G) \leq C \delta_*(\Omega)$;
- (iii) (closeness) $|\Omega \Delta G| \leq C \sqrt{\delta_T}$.

Every constant is absolute. The construction uses only the two level deficits D_H, D_I (through Theorems 3.4 and 5.10 and Lemmas 3.9, 3.11 and 6.1); it invokes no selection principle, no transport, and no Serrin/Magnanini–Poggesi symmetry estimate.

Proof. Throughout we abbreviate $\delta_{\text{iso}} = \delta_{\text{iso}}(\tilde{E})$, $\text{Exc} = P(\tilde{E}) - 2\pi\hat{r}$, $a = |\tilde{E} \Delta B_{\hat{r}}(z)|$ and $\eta = \rho_e - \rho_i$, with the notation of Definition 4.9 and Theorem 5.10.

Step 0: the inset. Apply Lemma 4.6 with the given $\tau \in (0, \frac{1}{16})$ (legitimate since $\tau = \sqrt{\delta_T} \leq \sqrt{\delta_\star} < \frac{1}{16}$ once $\delta_\star < \frac{1}{256}$): there is a regular value $\hat{v} \in (\tau, 2\tau)$ with

$$D_H(\hat{v}) + D_I(\hat{v}) \leq A_0 \frac{\delta_T}{\tau}, \quad \mu_B(\hat{v}) - \mu(\hat{v}) \leq A_0 \frac{\delta_T}{\tau}, \quad (138)$$

A_0 absolute. Let $\tilde{E}_0$ be the connected inset of Definition 4.9 (the component of $\{u > \hat{v}\}$ of largest measure); by Lemma 4.3 it is connected and has no interior holes, and $\mu_E = |\tilde{E}_0|$, $\hat{r} = \sqrt{\mu_E/\pi}$. By Lemmas 4.7 and 4.8,

$$|\Omega \Delta \tilde{E}_0| \leq 8\pi\tau + A_0 \frac{\delta_T}{\tau} + \frac{A_0^2}{16\pi^3} \frac{\delta_T^2}{\tau^2}. \quad (139)$$

Since $\mu_E = |\tilde{E}_0| = \pi - |\Omega \Delta \tilde{E}_0|$ (as $\tilde{E}_0 \subset \Omega$), (139) gives $\mu_E \geq \pi - (8\pi\tau + A_0\delta_T/\tau + A_0^2\delta_T^2/(16\pi^3\tau^2))$; for $\delta_\star$ small (so that the right-hand bound is $\leq \pi/2$, which is exactly the standing hypothesis (29) of Section 4 at $\tau = \sqrt{\delta_T}$, with the higher-order spillover term absorbed) we obtain $\mu_E \geq \pi/2$, hence $\hat{r} = \sqrt{\mu_E/\pi} \geq 1/\sqrt{2}$ and, for the annulus produced below, $\rho_i \geq \frac{1}{2}$ after possibly shrinking $\delta_\star$.

Step 0': hole-filling. Pass from $\tilde{E}_0$ to its saturation $\tilde{E} := E^h$ of Definition 4.11, with all bounded complementary components filled in (Remark 5.1). By Lemma 4.12(a) the set $\tilde{E}$ is bounded, open,

connected and *globally hole-free* — $\mathbb{R}^2 \setminus \widetilde{E}$ connected — *unconditionally*, with no hypothesis on $\partial\Omega$; this discharges the exterior-connectedness convention Lemma 5.2 required by the trapping annulus and the inscribed-ball lemma. The filled mass obeys $H := |\widetilde{E} \setminus \widetilde{E}_0| \leq D_I(\widetilde{E}_0)^2/(64\pi^2\mu_E) \leq C(\delta_T/\tau)^2$ (Lemma 4.12(c),(d)), so at $\tau = \sqrt{\delta_T}$, $H \leq C\delta_T$, of the same higher order as the spillover. By Lemma 4.12(d) all inset scales are preserved with the same rates: $\delta_{\text{iso}}(\widetilde{E}) \leq \delta_{\text{iso}}(\widetilde{E}_0)$, $\text{Exc}(\widetilde{E}) \leq \text{Exc}(\widetilde{E}_0) + CH$, $a(\widetilde{E}) \leq a(\widetilde{E}_0) + CH$, $\delta_*(\widetilde{E}) \leq \delta_*(\widetilde{E}_0) + CH/\mu_E$, and the closeness deteriorates only by H :

$$|\Omega \Delta \widetilde{E}| \leq |\Omega \Delta \widetilde{E}_0| + H \leq 8\pi\tau + A_0 \frac{\delta_T}{\tau} + \frac{A_0^2}{16\pi^3} \frac{\delta_T^2}{\tau^2} + CH. \quad (140)$$

From here on $\widetilde{E}$ denotes the hole-filled inset, $\mu_E := |\widetilde{E}|$ (now $\mu_E + H$ of the previous step, still $\geq \pi/2$), $\hat{r} := \sqrt{\mu_E/\pi}$, and we run Steps 1–4 on it; all of Sections 5 to 7 apply to it directly (Remark 5.1).

Step 1: the scales (carrying τ symbolic). From the inset control (33) of Proposition 4.10 (which includes the spillover term $4\pi S$ and absorbs it),

$$\delta_{\text{iso}} \leq C \frac{\delta_T}{\tau}, \quad \text{Exc} = 2\pi\hat{r} \delta_{\text{iso}} \leq C \frac{\delta_T}{\tau}, \quad (141)$$

using $\hat{r} \leq 1$. The sharp quantitative isoperimetric inequality Theorem 2.11, in the form $a = |\widetilde{E} \Delta B_{\hat{r}}(z)| \leq C_F \sqrt{\delta_{\text{iso}}} |\widetilde{E}|$, gives

$$a \leq C \sqrt{\delta_{\text{iso}}} \leq C \left(\frac{\delta_T}{\tau} \right)^{1/2}. \quad (142)$$

The trapping annulus Theorem 5.10, applied to the hole-filled inset $\widetilde{E} = E^h$ (globally hole-free unconditionally by Lemma 4.12(a), Remark 5.1), yields $B_{\rho_i}(z) \subset \widetilde{E} \subset B_{\rho_e}(z)$ with

$$\eta = \rho_e - \rho_i \leq C (\text{Exc} + a)^{1/2}. \quad (143)$$

By Corollary 6.5 the radial-graph defect of $\widetilde{E}$ about z obeys

$$\text{gd}(\widetilde{E}) \leq C \eta (\text{Exc} + a\eta). \quad (144)$$

Step 2: optimise $\tau = \sqrt{\delta_T}$ and re-derive every exponent. Set $\tau = \sqrt{\delta_T}$, so $\delta_T/\tau = \sqrt{\delta_T}$. We re-derive each scale independently from (141)–(144):

$$\delta_{\text{iso}} \leq C \frac{\delta_T}{\tau} = C \delta_T^{1/2}, \quad \text{Exc} \leq C \delta_T^{1/2}, \quad (145)$$

$$a \leq C \left(\frac{\delta_T}{\tau} \right)^{1/2} = C \delta_T^{1/4}, \quad (146)$$

$$\eta \leq C (\text{Exc} + a)^{1/2} \leq C a^{1/2} \leq C \delta_T^{1/8}, \quad (147)$$

where in (147) we used that $a \sim \delta_T^{1/4}$ dominates $\text{Exc} \sim \delta_T^{1/2}$ as $\delta_T \rightarrow 0$, so $\text{Exc} + a \leq 2a \leq C \delta_T^{1/4}$ and $(\text{Exc} + a)^{1/2} \leq C \delta_T^{1/8}$. For the graph defect, (144) gives

$$\text{gd}(\widetilde{E}) \leq C \eta (\text{Exc} + a\eta) \leq C (\delta_T^{1/8} \cdot \delta_T^{1/2} + \delta_T^{1/4} \cdot \delta_T^{1/4}) = C (\delta_T^{5/8} + \delta_T^{1/2}) \leq C \delta_T^{1/2}, \quad (148)$$

the dominant term being $a\eta \cdot \eta$ with $a\eta^2 = \delta_T^{1/4} \cdot \delta_T^{1/4} = \delta_T^{1/2}$. As a cross-check against the heuristic $\text{gd} \sim \delta_T/\tau$: at $\tau = \sqrt{\delta_T}$ one has $\delta_T/\tau = \delta_T^{1/2}$, matching (148). Likewise the closeness of the hole-filled

inset (140) reads

$$\left| \Omega \Delta \tilde{E} \right| \leq 8\pi \delta_T^{1/2} + A_0 \delta_T^{1/2} + \frac{A_0^2}{16\pi^3} \delta_T + CH \leq C \delta_T^{1/2}, \quad (149)$$

i.e. both contributions $\tau = \delta_T^{1/2}$ (the level height) and $\delta_T/\tau = \delta_T^{1/2}$ (the level smoothing) coincide at the optimum, while the spillover $\delta_T^2/\tau^2 = \delta_T$ and the filled mass $H \leq C\delta_T$ are genuinely higher order. This is precisely the balance $\tau \sim \delta_T/\tau$, solved by $\tau = \sqrt{\delta_T}$; any other power of τ would leave one of the two terms larger than $\delta_T^{1/2}$.

Step 3: realisation as the measurable core graph G . We take $G := G_0$ the measurable open star-shaped core graph of Proposition 6.6, with profile $\rho := \rho_0: S^1 \rightarrow [\rho_i, \rho_e]$ measurable. No mollification or continuity is required: the nearly-spherical estimate Theorem 8.1 admits measurable profiles. By Proposition 6.6(iii) and Corollary 6.5,

$$\left| \tilde{E} \Delta G \right| = \left| \tilde{E} \setminus G_0 \right| \leq C \eta (\text{Exc} + a\eta) \leq C \delta_T^{1/2}, \quad (150)$$

the last step being (148), and $\|\rho/\hat{r}_G - 1\|_{L^\infty} \leq 2\eta$ by Proposition 6.6(ii) with $\hat{r}_G = \hat{r}_{G_0} = \sqrt{|G|/\pi} \in [\rho_i, \rho_e]$.

Step 4: verification of (i), (ii), (iii).

(i) *Amplitude.* By Proposition 6.6(ii) the profile $\rho = \rho_0$ obeys $\|\rho/\hat{r}_G - 1\|_\infty \leq 2\eta$ with $\hat{r}_G = \sqrt{|G|/\pi} \in [\rho_i, \rho_e]$, and by (147) $\eta \leq C\delta_T^{1/8}$, so $\|\rho/\hat{r}_G - 1\|_\infty \leq C\delta_T^{1/8}$. Shrinking $\delta_\star$ so that $C\delta_\star^{1/8} \leq \eta_G$ gives $\|\rho/\hat{r}_G - 1\|_\infty \leq \eta_G$, which is (i). Here it is essential that $\eta \sim \delta_T^{1/8}$ is an *absolute* smallness (decaying with δ_T), so the hypothesis of Theorem 8.1 is met for δ_T small, *regardless* of the cruder L^1 -roundness $a \sim \delta_T^{1/4}$.

(iii) *Closeness.* By (149) and (150), with $\tilde{E} = E^h$ so that $\left| \Omega \Delta \tilde{E} \right|$ already includes the filled mass $H \leq C\delta_T$ ((140)),

$$\left| \Omega \Delta G \right| \leq \left| \Omega \Delta \tilde{E}_0 \right| + H + \left| \tilde{E} \Delta G \right| = \left| \Omega \Delta \tilde{E} \right| + \left| \tilde{E} \Delta G \right| \leq C \delta_T^{1/2} + C \delta_T^{1/2} = C \sqrt{\delta_T},$$

which is (iii); the filled mass $H \leq C\delta_T$ is negligible.

(ii) *Deficit.* Here $G = G_0 = \{z + r\omega : r < \rho_0(\omega)\}$ is the measurable core graph of Corollary 6.5 and Proposition 6.6, so $\rho_0 \in [\rho_i, \rho_e]$, $B_{\rho_i}(z) \subset G_0 \subset \tilde{E} \subset B_{\rho_e}(z)$, and $\tilde{E} \setminus G_0$ (the radial caps) lies in the collar $\{\rho_i < |x - z| < \rho_e\}$. The torsion-loss decomposition (90) of Lemma 7.1 holds unconditionally,

$$T(\tilde{E}) - T(G_0) = \int_{\tilde{E} \setminus G_0} u_{\tilde{E}} + \int_{G_0} h \leq \tilde{\eta} g + \int_{G_0} h, \quad g = \left| \tilde{E} \setminus G_0 \right|,$$

with $\tilde{\eta} = \frac{1}{4}(\rho_e^2 - \rho_i^2) \leq \eta \rho_e \leq C\delta_T^{1/8}$ and the cap term $\int_{\tilde{E} \setminus G_0} u_{\tilde{E}} \leq \tilde{\eta} g$ rigorous. The torsion density is an absolute constant bounded below,

$$\frac{T(\tilde{E})}{\left| \tilde{E} \right|} = (1 - \delta_*(\tilde{E})) \frac{\left| \tilde{E} \right|}{8\pi} \geq \left(1 - \frac{1}{2}\right) \frac{\pi/2}{8\pi} = \frac{1}{32},$$

using $\delta_*(\tilde{E}) \leq \frac{1}{2}$ (from Step 5 below, equivalently Proposition 4.10 (34)) and $\left| \tilde{E} \right| \geq \pi/2$. Under the boundary-layer control (98) of Remark 7.2 — the single residual analytic point, namely the localisation $\int_{G_0} h \leq \tilde{\eta} g$ of the harmonic boundary-layer term to the cap region — the loss-slope

hypothesis (101) holds with $\lambda = 2\tilde{\eta} \leq \frac{1}{32} \leq T(\tilde{E})/|\tilde{E}|$ (Remarks 7.5 and 7.6), so Lemma 7.4 applies to $G = G_0 \subset \tilde{E}$ and gives directly, with no further regularisation step (since $G = G_0$),

$$\delta_*(G) = \delta_*(G_0) \leq \delta_*(\tilde{E}). \quad (151)$$

This is the only place in the reduction where a residual analytic point is invoked; everything else — the exact decomposition, the cap term, and the deficit algebra — is unconditional.

Step 5: transfer of the deficit from Ω to $\tilde{E}$. The deficit of the raw component $\tilde{E}_0$ is controlled by the inset control (34) of Proposition 4.10, $\delta_*(\tilde{E}_0) \leq (\pi^2/\mu_E^2) \delta_*(\Omega) \leq C \delta_*(\Omega)$; and the hole-filling raises it only by the higher-order term $CH/\mu_E \leq C\delta_T$ (Lemma 4.12(d), eq. (45)), so

$$\delta_*(\tilde{E}) \leq \delta_*(\tilde{E}_0) + \frac{CH}{\mu_E} \leq \frac{\pi^2}{\mu_E^2} \delta_*(\Omega) + C\delta_T \leq C \delta_*(\Omega). \quad (152)$$

We stress that the transfer to $\tilde{E}_0$ is established in Section 4 *without* appeal to the cutting principle Lemma 7.4: the passage from the super-level set $\{u > \hat{v}\}$ to its largest component $\tilde{E}_0$ is handled there by the exact additivity of torsion over the (disjoint) level components and the Saint-Venant bound on the spillover torsion (Proposition 4.10, eqs. (36)–(37)), both internal to Sections 3 and 4; and the hole-filling step (45) uses only domain monotonicity of torsion and the area bound $H \leq C\delta_T$, again internal to Section 4. In particular the spillover components need *not* lie in the trapping collar of $\tilde{E}$ — so Lemma 7.4, whose hypothesis requires the excised set to sit in a thin concentric collar, is *not* invoked here, and there is no circular dependence on Section 7 (see Remark 4.14). With $\mu_E \geq \pi/2$ (Step 0) the prefactor is $\pi^2/\mu_E^2 \leq 4$, so for δ_* small (152) gives $\delta_*(\tilde{E}) \leq 5\delta_*(\Omega)$; in particular $\delta_*(\tilde{E}) \leq \frac{1}{2}$ once $\delta_*(\Omega) \leq \frac{1}{10}$, which justifies the value used in Step 4. Inserting (152) into (151) and using $\delta_T = \frac{\pi}{8}\delta_*(\Omega) \leq C\delta_*(\Omega)$,

$$\delta_*(G) \leq \delta_*(\tilde{E}) + C\delta_T \leq C\delta_*(\Omega) + C\delta_*(\Omega) = C \delta_*(\Omega),$$

which is (ii). This completes the proof. $\square$

9.2. The main theorem.

Theorem 9.2 (Sharp quantitative Saint-Venant, selection-free). *There is an absolute constant C such that every bounded open set $\Omega \subset \mathbb{R}^2$ with $|\Omega| = \pi$ satisfies*

$$\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega). \quad (153)$$

Proof. Trivial regime. If $\delta_T \geq \delta_*$, then since $|\Omega| = |B| = \pi$ for the optimal disc B we have $|\Omega \triangle B| \leq |\Omega| + |B| = 2\pi$, so $\mathcal{A}(\Omega) \leq 2$ and $\mathcal{A}(\Omega)^2 \leq 4 \leq (4/\delta_*) \delta_T$. Thus (153) holds with $C = 4/\delta_*$ in this regime.

Near-extremal regime. Assume $\delta_T \leq \delta_*$ with δ_* the threshold of Theorem 9.1, and let $G = \{z + r\omega : r < \rho(\omega)\}$ be the radial graph it provides, with $\hat{r}_G = \sqrt{|G|/\pi}$ and amplitude $\|\rho/\hat{r}_G - 1\|_\infty \leq C\delta_T^{1/8} \leq \eta_G$.

Theorem 8.1 now applies directly to the graph G as it is produced by Theorem 9.1, radial about the annulus centre z , with no barycentre recentring: its hypotheses are the volume normalisation $\hat{r}_G = \sqrt{|G|/\pi}$ (built into Theorem 9.1), the L^∞ amplitude bound $\|\rho/\hat{r}_G - 1\|_\infty \leq \eta_G$ (part (i)), and the quadratic mode-one bound (112), i.e. $\|P_1\varphi\|_{L^2} \leq K\|\varphi\|_{L^2}^2$ for the profile $\varphi = \rho/\hat{r}_G - 1$ in polar

coordinates about z . The first two hold by Theorem 9.1. The third is supplied by the choice of centre, as we now record.

Remark 9.3 (The mode-one bound at the L^1 -optimal centre). The centre z is the *exact* L^1 -optimal (Fraenkel) centre of the inset $\tilde{E}$ (Lemma 5.13), an interior point with a definite inscribed ball. Exact L^1 -optimality is a first-order extremality condition on the comparison ball: for the optimal centre the directional derivative of $p \mapsto |\tilde{E} \triangle B_{\hat{r}}(z+p)|$ vanishes, which is the statement that the translation mode of $\partial \tilde{E}$ carries no first-order L^1 -content. This is precisely the mechanism behind the quadratic mode-one bound (112): it forces the first Fourier mode of the radial profile to be of second order in the amplitude, exactly as the barycentre condition does through Remark 8.3, but *at the centre z already in hand*, with no recentring of the set. What remains to make this fully quantitative for the measurable core graph G (as opposed to $\tilde{E}$) is the passage of the first-order extremality from the L^1 -content of $\partial \tilde{E}$ to the L^2 mode-one coefficient of the profile of G , with the collar error $|\tilde{E} \triangle G| \leq C\sqrt{\delta_T}$ of Theorem 9.1 absorbed. We isolate this as the explicit scoped hypothesis:

$$\|P_1\varphi\|_{L^2(S^1)} \leq K \|\varphi\|_{L^2(S^1)}^2 \quad (\varphi = \rho/\hat{r}_G - 1 \text{ about the } L^1\text{-optimal } z), \quad (154)$$

for an absolute K . Under (154) the argument below is unconditional. This replaces the earlier, false-as-stated requirement of a barycentre recentring — which need not preserve single-valuedness of the graph under an L^∞ amplitude bound alone — by a precise condition at the existing centre, and removes the recentring obstruction entirely from Theorem 8.1. By Lemma 8.9 the asymmetry side needs no mode-one hypothesis at all, and by Remark 8.10 the bound (154) is equivalent, for the present purpose, to the quartic translation estimate $\delta_*(G) \geq c\|P_1\varphi\|_{L^2}^4$ (137) — the flatness of the translation direction of the torsional deficit; this is the precise form of the residual.

Asymmetry of G . By Theorem 8.1 (with the mode-one bound (154)) and Theorem 9.1(ii),

$$\mathcal{A}(G)^2 \leq C_G \delta_*(G) \leq C_G \cdot C \delta_*(\Omega) = C \delta_*(\Omega) = C \frac{8}{\pi} \delta_T \leq C \delta_T, \quad (155)$$

so $\mathcal{A}(G) \leq C\sqrt{\delta_T}$.

Transfer to Ω by the triangle inequality. Let $B_G = B(y, \hat{r}_G)$ be the disc of area $|G|$ realising the asymmetry of G , i.e. centred at the optimal y so that $|G \triangle B_G| = \mathcal{A}(G) |G|$, and let $B = B(y, 1)$ be the disc of area π *concentric* with B_G . Since B_G and B are concentric discs of areas $|G|$ and π , their symmetric difference is the annulus of area

$$|B_G \triangle B| = \left| |G| - \pi \right| = \left| |G| - |\Omega| \right| \leq |\Omega \triangle G|. \quad (156)$$

Because $|\Omega| = \pi = |B|$, the disc B is admissible in the definition of $\mathcal{A}(\Omega)$, so by the triangle inequality for the symmetric difference,

$$\begin{aligned} \mathcal{A}(\Omega) &\leq \frac{|\Omega \triangle B|}{\pi} \leq \frac{|\Omega \triangle G| + |G \triangle B_G| + |B_G \triangle B|}{\pi} \\ &\leq \frac{1}{\pi} \left(|\Omega \triangle G| + \mathcal{A}(G) |G| + |\Omega \triangle G| \right), \end{aligned}$$

using (156) and $|G \triangle B_G| = \mathcal{A}(G) |G|$. With $|G| = \pi + O(\sqrt{\delta_T}) \leq 2\pi$, $|\Omega \triangle G| \leq C\sqrt{\delta_T}$ by Theorem 9.1(iii), and $\mathcal{A}(G) \leq C\sqrt{\delta_T}$ by (155),

$$\mathcal{A}(\Omega) \leq \frac{1}{\pi} (C\sqrt{\delta_T} + \mathcal{A}(G) |G| + C\sqrt{\delta_T}) \leq C\sqrt{\delta_T} + C \mathcal{A}(G) \leq C\sqrt{\delta_T}. \quad (157)$$

Squaring (157) gives $\mathcal{A}(\Omega)^2 \leq C \delta_T$, which is (153) in the near-extremal regime. Combining the two regimes proves the theorem. $\square$

Remark 9.4 (What was used, what was not, and the two sharp ingredients). The reduction Theorem 9.1 draws on the two level deficits and nothing else of substance:

- $D_H + D_I$ enters through the level identity $\int_0^m (D_H + D_I) dv = 4\pi\delta_T$ (Theorem 3.4), which powers both the level selection Lemma 4.6 and the deficit transfer Lemma 4.5, and through the pointwise bound $-\mu' \geq 4\pi$ (Lemma 3.11) and the profile identity (Lemma 3.9) that carry closeness;
- D_I enters through the isoperimetric/annulus inputs (Theorems 2.11 and 5.10) and the multi-component spillover bound (Lemma 4.8).

At no point is the Brasco–De Philippis–Velichkov selection principle used, nor any mass–transport (moving–ball, kinetic) argument, nor a Serrin / Magnanini–Poggesi symmetry estimate. The only nonlinear stability input is the *perturbative* nearly–spherical bound Theorem 8.1, applied to a graph whose amplitude is already absolutely small.

Two ingredients make the reduction *sharp*, i.e. produce the exponent $\frac{1}{2}$ rather than $\frac{1}{4}$:

- (a) the fold–content bound Proposition 6.3, in which subtracting the constant $1/\hat{r}$ (legitimate because $|\tilde{E}| = |B_{\hat{r}}(z)|$) and confining $\tilde{E} \triangle B_{\hat{r}}(z)$ to the η –annulus replaces a by $a\eta$, so the graph defect avoids the quantitative–isoperimetric square–root loss; without it the fold content would carry the full $a \sim \sqrt{\delta_{\text{iso}}}$ and cap the result at $\delta_T^{1/4}$ closeness;
- (b) the cutting lemma Lemma 7.4: excising the low–torsion boundary caps to manufacture the graph G *decreases* the scale–invariant deficit δ_* , because the equal–area comparison ball shrinks faster (by $\text{gd}/4$ in $\tilde{\eta}$ –weighted terms) than the torsion is lost.

Finally, the optimisation $\tau = \sqrt{\delta_T}$ is the balance between the two τ –dependent contributions to closeness: the level height $\sim \tau$ grows in τ , while the smoothing/graph defect $\sim \delta_T/\tau$ decays in τ ; equating them, $\tau \sim \delta_T/\tau$, forces $\tau = \sqrt{\delta_T}$, at which both equal $\sqrt{\delta_T}$ and the amplitude is $\eta \sim (\delta_T/\tau)^{1/4} = \delta_T^{1/8}$, the absolute smallness the graph theorem requires.

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